MULTI-TEACHER KNOWLEDGE DISTILLATION

1 OVERVIEW

Four fine-tuned backbones act as teachers. Their logits are averaged and converted into soft targets, and a small student is trained against those targets and the ground-truth labels at once. The question the experiment answers is which combination of teachers, and which student, gives the best accuracy per megabyte.

The best result is **EfficientNetB0 distilled from Custom-ENB5 and ResNet50** at $T = 3$ and $\alpha = 0.5$: 0.9077 test accuracy and 0.9095 macro F1 from 4.38 M parameters. That is above every individual teacher, within 0.3 points of the 1.2 GB fused ensemble it was distilled towards, and 70 times smaller than that ensemble. The same student trained on hard labels alone reaches 0.8332 macro F1, so distillation is worth 7.0 points to it.

2 THE STUDY END TO END

This report covers the last of four phases. The earlier three decided which models deserved to be teachers, and Figure 1 places the distillation step in that sequence: a frozen-backbone baseline, a screening pass that added early stopping and saved weights, a grid search that unfroze the backbones and more than doubled the macro F1, and finally distillation, which beat every individual teacher with a model one seventieth of the fused ensemble's size.

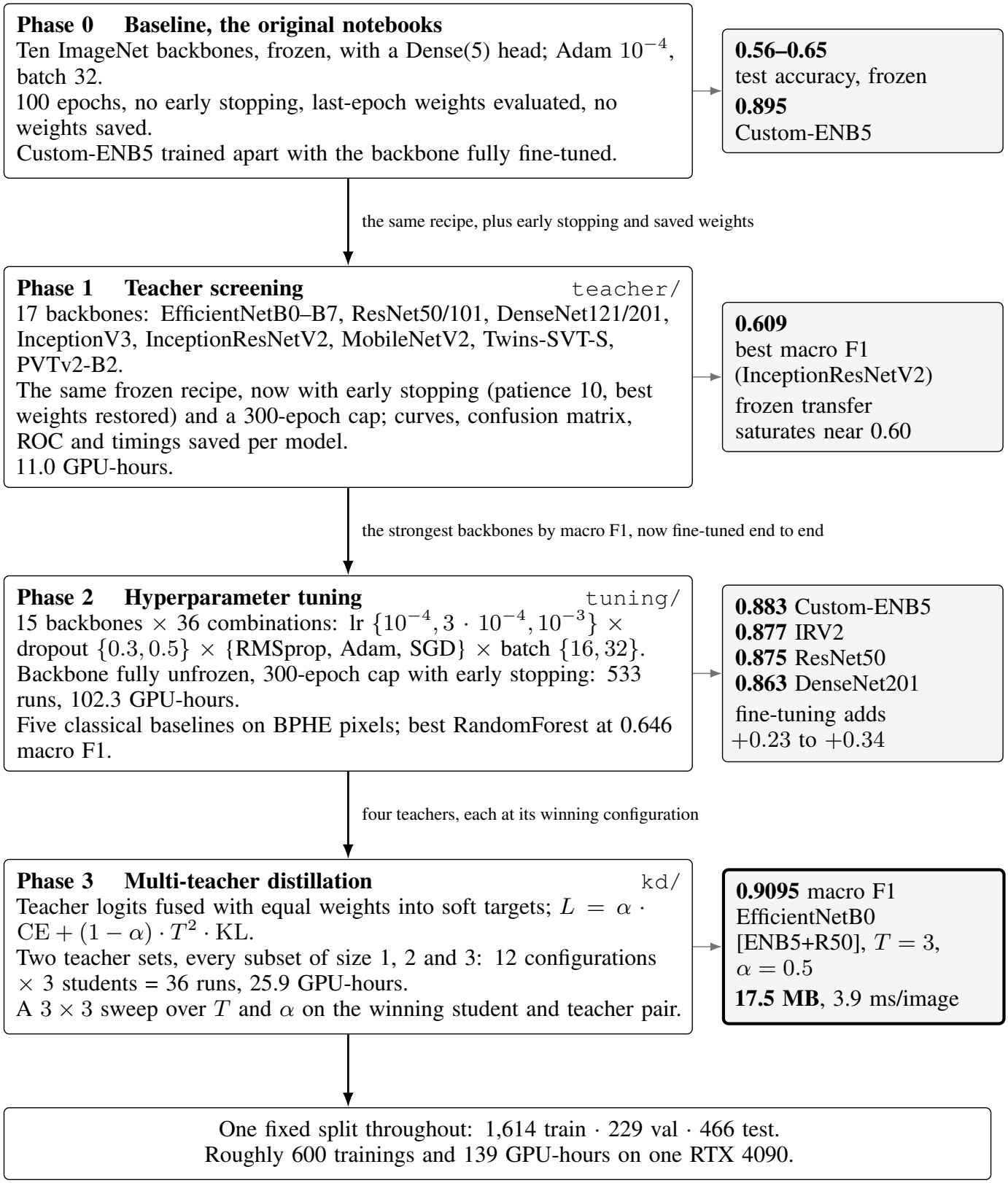


Figure 1: The whole study, from the baseline notebooks to multi-teacher distillation. The right-hand column carries each phase's headline test-set result, so the progression 0.61 to 0.88 to 0.91 macro F1 can be read down the page. Arrows name what each phase passed to the next.

3 EXPERIMENTAL SETUP

Table 1: Common experimental configuration.

Item	Configuration
Data	1,614 training, 229 validation, and 466 test knee radiographs; KL grades 0–4; one fixed split shared by every teacher, student, and control.
Preprocessing	Each backbone's native `preprocess_input` and nothing else, matching the hyperparameter-tuning phase.
Input	RGB images resized to 224×224 for every model. Phase 2 tuned all fifteen backbones at that size, following notebook 3, so the teachers are evaluated here at exactly the resolution they were tuned at.
Training augmentation	Rotation 30°; width/height shift 0.2; shear 0.2; zoom 0.2; horizontal flip; brightness 0.8–1.2; nearest fill.
Teachers	Four backbones, each loaded at its grid-search winning configuration with the backbone fully fine-tuned; frozen during distillation.
Students	MobileNetV2, EfficientNetB0, EfficientNetB2, ImageNet-initialised, with a shared head: global average pooling, Dense(256, swish), dropout 0.3, Dense(5) emitting logits.
Optimisation	Adam, learning rate 10^{-4}, batch size 32, maximum 300 epochs, seed 42.
Early stopping	Validation loss, patience 10, restored best-epoch weights. Validation loss is hard-label cross-entropy for every run, including the control, so model selection asks the same question everywhere.
Evaluation	Accuracy; weighted and macro precision, recall, and F1; confusion matrix; one-vs-rest ROC and AUC; parameter count, on-disk size, inference latency, and wall-clock training time.
Execution	One fresh subprocess per run; `metrics.json` written last, so an interrupted job resumes at the first unfinished run. 36 runs, 25.9 GPU-hours on an RTX 4090.
Software	Python 3.12.3; TensorFlow 2.20.0; Keras 3.12.0; NumPy 2.2.6; OpenCV 4.12.0.

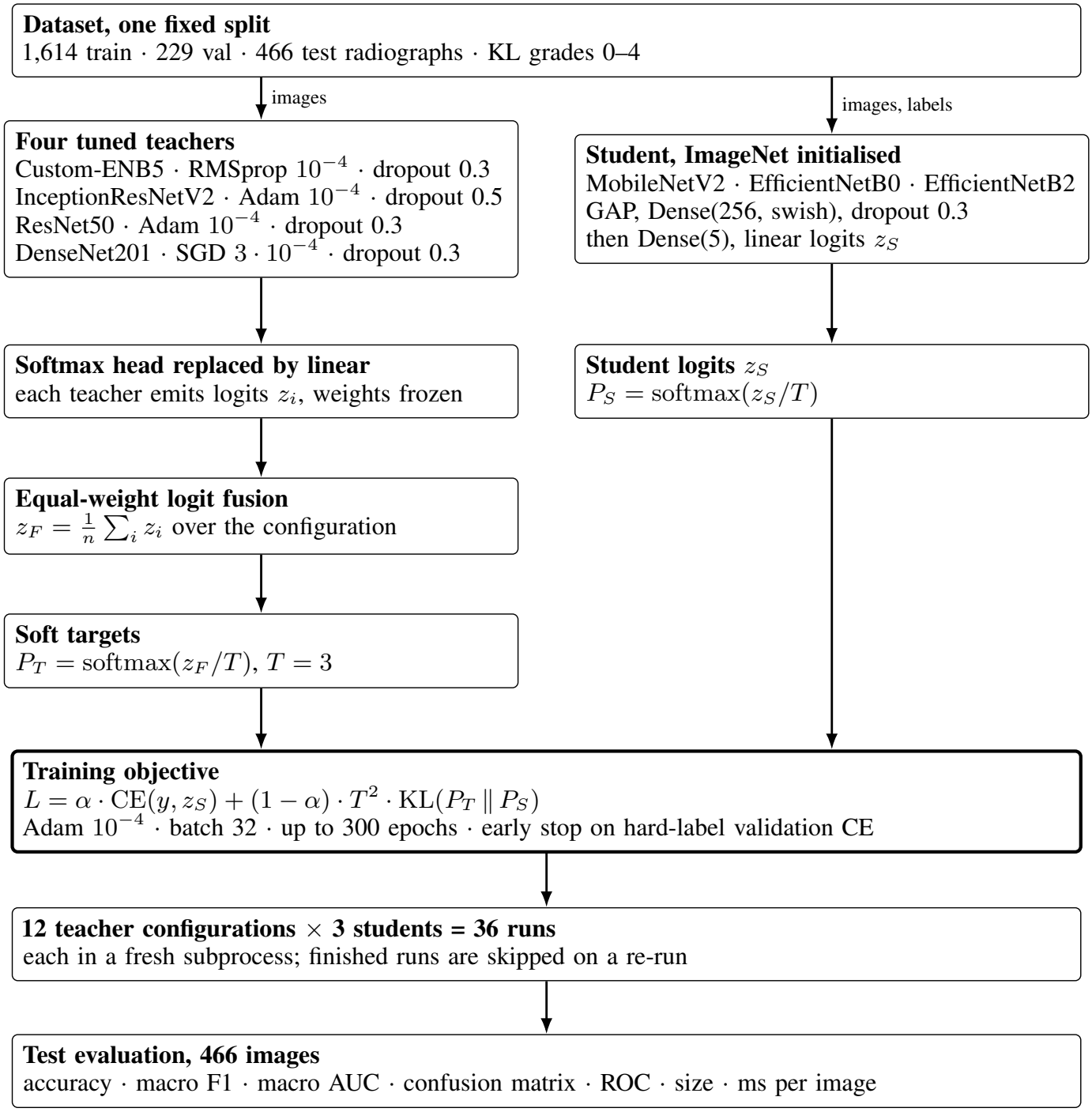


Figure 2: The distillation pipeline. The two paths meet only at the loss: teachers are never updated, they run a forward pass on each batch, their logits are averaged, and the student pays for disagreeing with that average as well as for getting the label wrong.

4 Distillation Objective

Fusion happens in logit space. Each teacher's final softmax layer is replaced by an identical linear layer, so teacher i emits logits z_i rather than probabilities. For a configuration holding n teachers,

$$z_F = \sum_{i=1}^{n} w_i z_i, \qquad w_i = \tfrac{1}{n}, \tag{1}$$

$$L_{\mathrm{KD}} = T^2 \cdot \mathrm{KL}\big(\mathrm{softmax}(z_F/T) \,\big\|\, \mathrm{softmax}(z_S/T)\big)\,, \tag{2}$$

$$L = \alpha \cdot \mathrm{CE}(y, z_S) + (1-\alpha) \cdot L_{\mathrm{KD}}. \tag{3}$$

Term by term: z_F is the fused teacher opinion, averaged with equal weights because no teacher was measurably better than the others; T flattens both distributions so that the relative size of the wrong-class probabilities, which is where a teacher's knowledge of grade similarity lives, survives the softmax; the factor T^2 restores the gradient scale that dividing the logits by T removed, keeping the two terms comparable as T rises; α is the weight on the ground truth, so $\alpha = 0.3$ means 30% hard labels and 70% teachers. The main grid uses $T = 3$, $\alpha = 0.3$; Section 7 varies both.

Both teacher sets share Custom-ENB5 and InceptionResNetV2 and differ in the third member. Every non-empty subset of each set is a configuration; the three shared subsets are trained once and counted for both.

Table 2: The 12 configurations. "Control" is the same student and recipe with hard labels only.

Configuration	Teachers	Count	Set
none	hard labels only	0	control
D201	DenseNet201	1	A
ENB5	Custom-ENB5	1	A+B
ENB5+D201	Custom-ENB5, DenseNet201	2	A
ENB5+IRV2	Custom-ENB5, InceptionResNetV2	2	A+B
ENB5+IRV2+D201	Custom-ENB5, InceptionResNetV2, DenseNet201	3	A
ENB5+IRV2+R50	Custom-ENB5, InceptionResNetV2, ResNet50	3	B
ENB5+R50	Custom-ENB5, ResNet50	2	B
IRV2	InceptionResNetV2	1	A+B
IRV2+D201	InceptionResNetV2, DenseNet201	2	A
IRV2+R50	InceptionResNetV2, ResNet50	2	B
R50	ResNet50	1	B

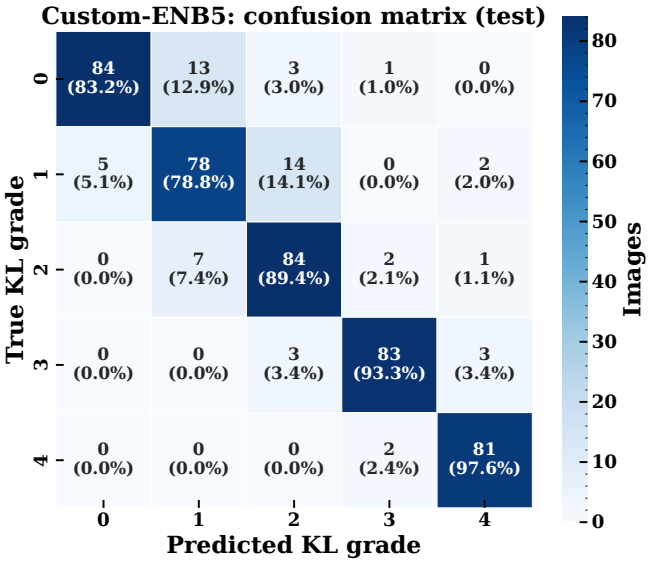
5 Teachers and Fused Ensembles

The four teachers sit within two points of each other and fusing them beats all of them. Set B gains more from fusion than set A, which is also why set B produced the better students: a student can only inherit what its teachers agree on.

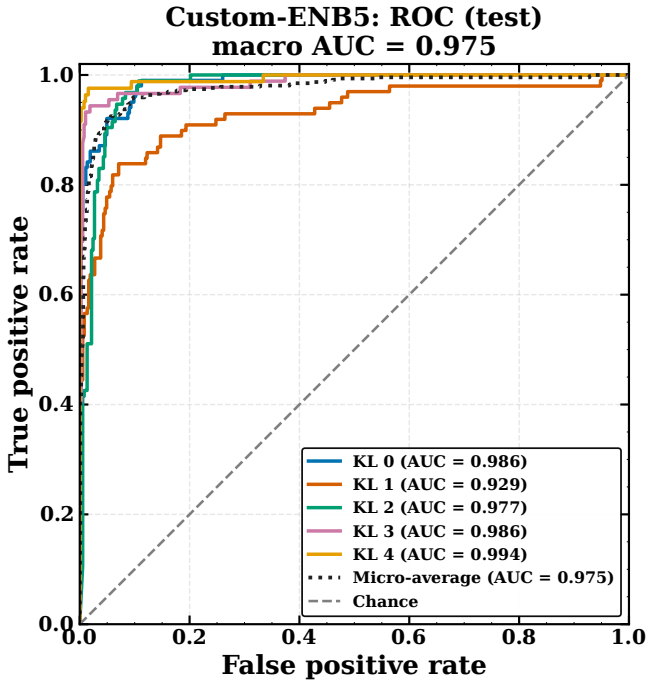
Table 3: Teachers at their tuned settings, and the equal-weight logit fusion of each set, on the test set.

Model	Accuracy	Macro F1	Weighted F1	Macro AUC	Size (MB)	ms/image
Custom-ENB5	0.8798	0.8831	0.8799	0.9746	247.0	7.8
InceptionResNetV2	0.8734	0.8767	0.8741	0.9736	660.4	10.1
ResNet50	0.8712	0.8750	0.8718	0.9728	312.4	3.0
DenseNet201	0.8627	0.8633	0.8592	0.9737	168.6	9.0
Fused set A (ENB5+IRV2+D201)	0.8948	0.8974	0.8946	0.9813	1076.0	26.8
Fused set B (ENB5+IRV2+R50)	0.9099	0.9121	0.9102	0.9806	1219.8	20.9

5.1 CUSTOM-ENB5



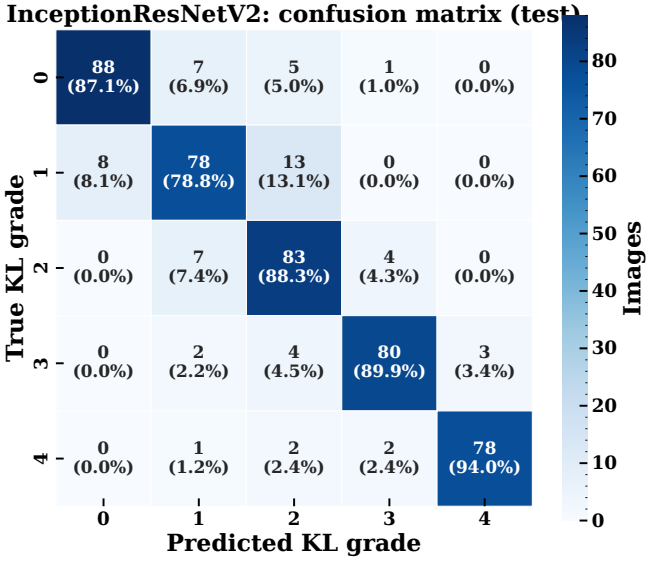
(a) Test confusion matrix.



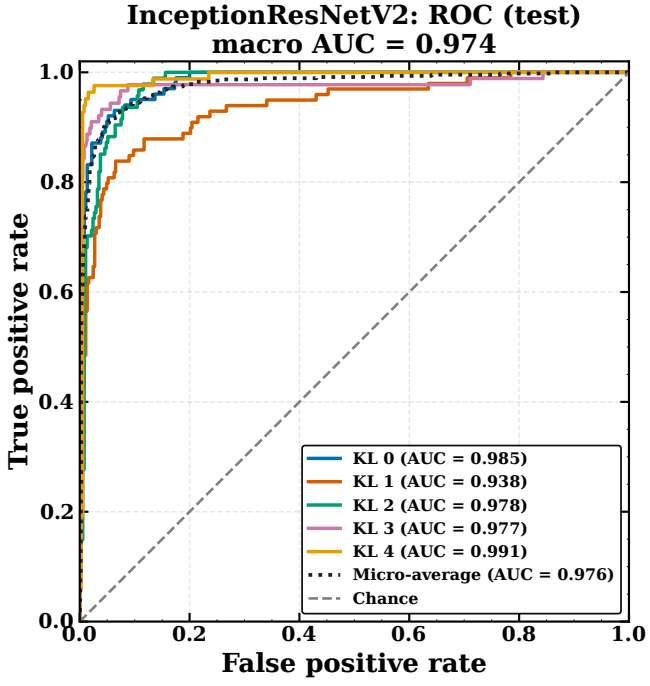
(b) One-vs-rest ROC curves.

Figure 3: Custom-ENB5: test-set results.

5.2 INCEPTIONRESNETV2



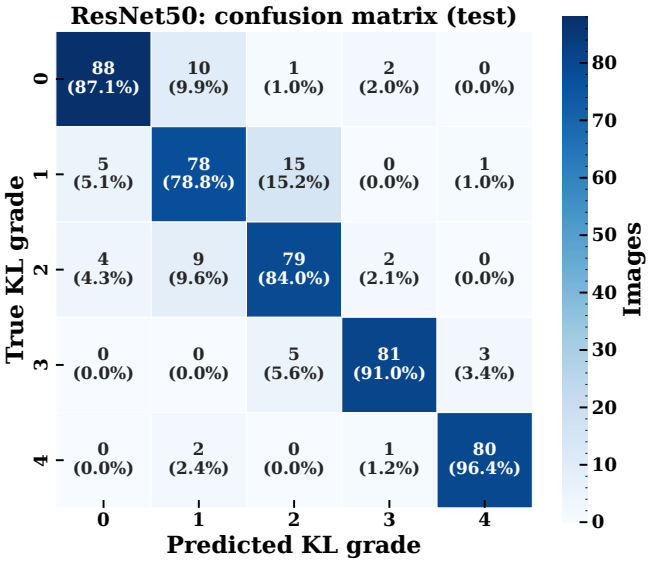
(a) Test confusion matrix.



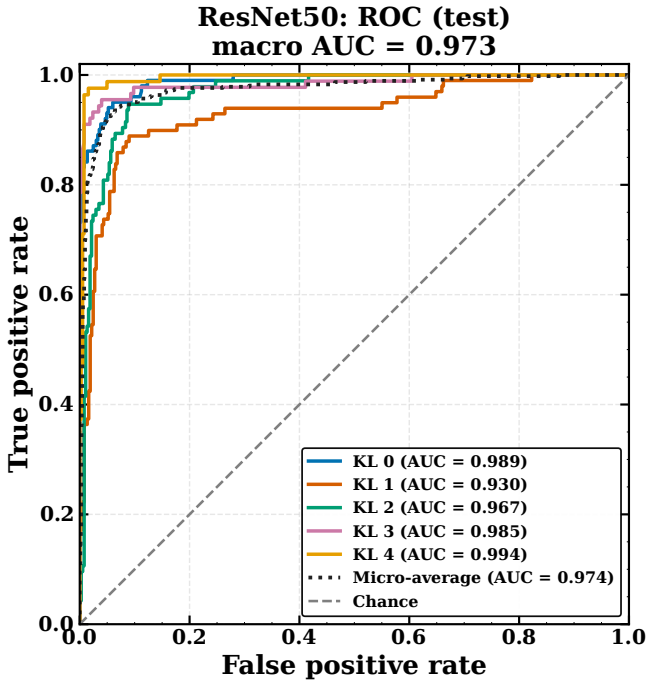
(b) One-vs-rest ROC curves.

Figure 4: InceptionResNetV2: test-set results.

5.3 ResNet50



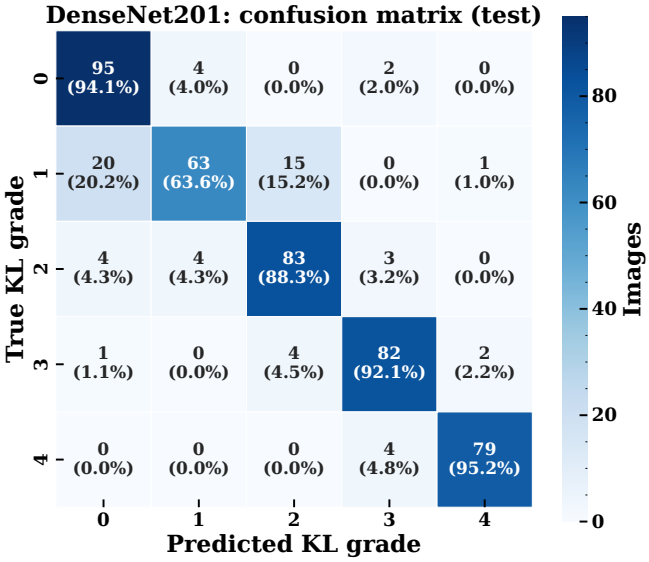
(a) Test confusion matrix.



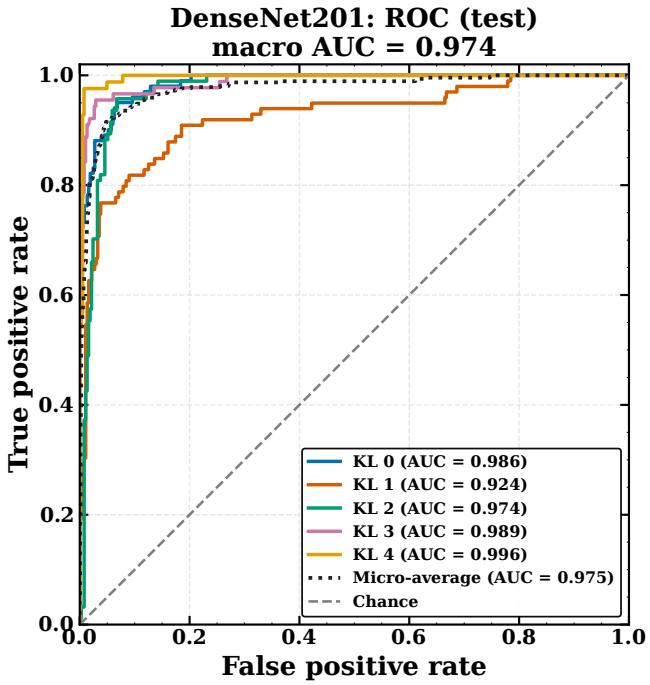
(b) One-vs-rest ROC curves.

Figure 5: ResNet50: test-set results.

5.4 DenseNet201



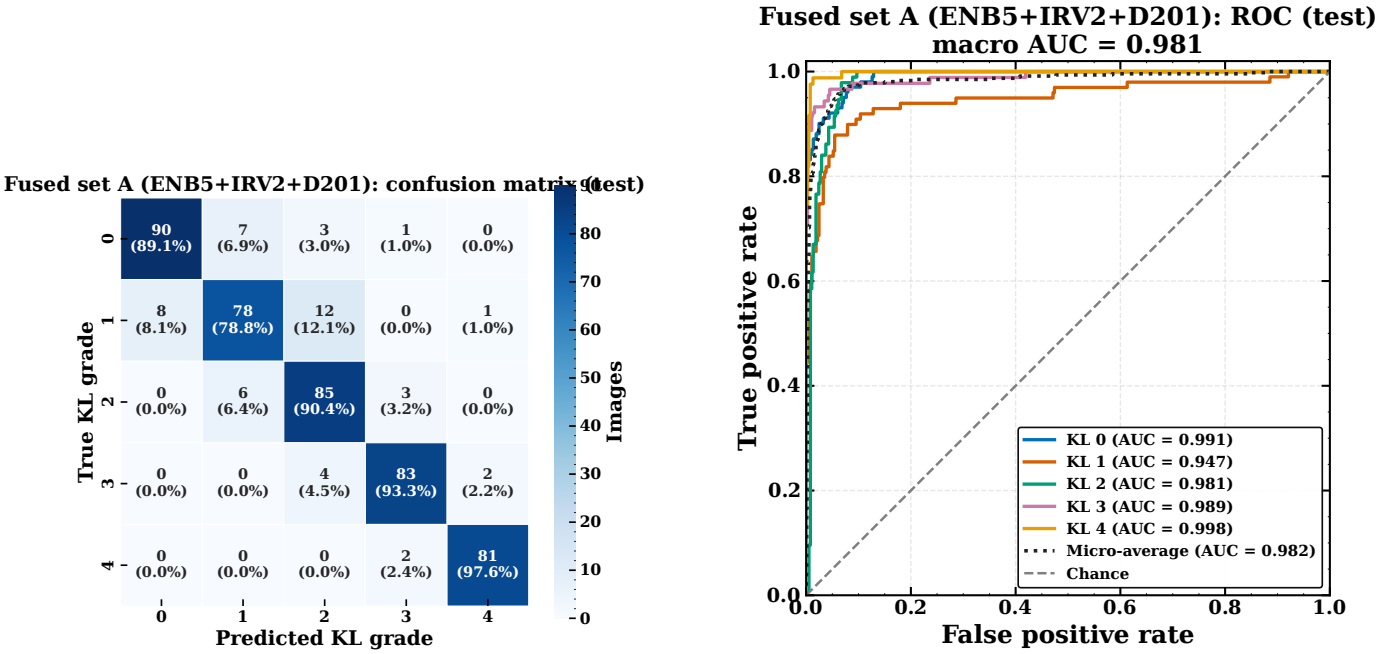
(a) Test confusion matrix.



(b) One-vs-rest ROC curves.

Figure 6: DenseNet201: test-set results.

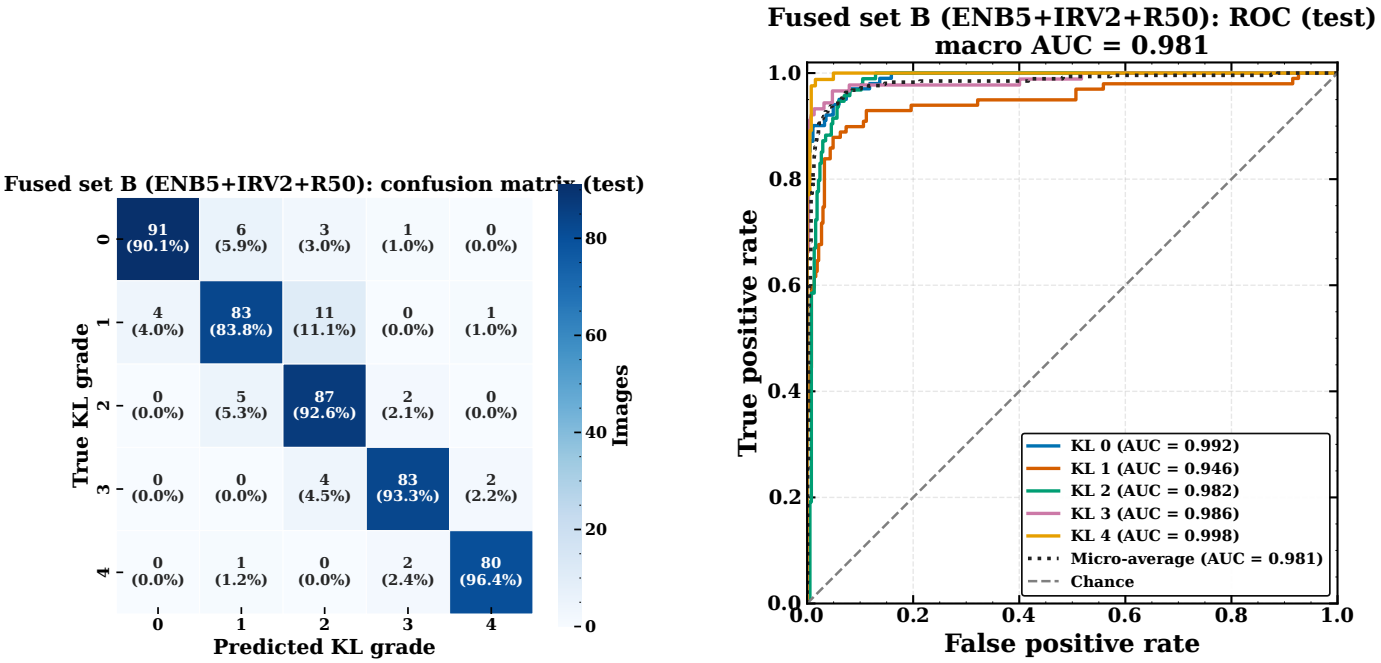
5.5 Fused set A



(a) Test confusion matrix. **(b)** One-vs-rest ROC curves.

Figure 7: Fused set A (ENB5+IRV2+D201): test-set results.

5.6 Fused set B



(a) Test confusion matrix. **(b)** One-vs-rest ROC curves.

Figure 8: Fused set B (ENB5+IRV2+R50): test-set results.

6 Distillation Results

Every student gains from distillation, and the smaller its own ceiling, the more it gains. The eight runs voided by the stopping-rule artifact of Section 8 are excluded from every aggregate below and marked in Table 7.

Table 4: Control against the best distilled configuration for each student, macro F1.

Student	Params (M)	Control	Best configuration	Best	Gain
EfficientNetB0	4.38	0.8332	ENB5+R50	**0.9030**	+7.0
EfficientNetB2	8.13	0.8584	R50	**0.8999**	+4.2
MobileNetV2	2.59	0.8546	ENB5+IRV2	**0.8758**	+2.1

Table 5: Best configuration within each teacher set, macro F1. MobileNetV2's winner belongs to both sets.

Student	Best in set A	Macro F1	Best in set B	Macro F1
EfficientNetB0	ENB5+IRV2+D201	0.8990	ENB5+R50	0.9030
EfficientNetB2	D201	0.8926	R50	0.8999
MobileNetV2	ENB5+IRV2	0.8758	ENB5+IRV2	0.8758

Table 6: Averaging over the healthy distilled runs: more teachers is not better.

Teachers in the configuration	Runs	Mean macro F1	Best macro F1
1	9	0.8789	0.8999
2	14	0.8774	0.9030
3	6	0.8775	0.8990

Three observations follow. First, swapping DenseNet201 for ResNet50 as the third teacher is worth $+0.4$ points to EfficientNetB0 and $+0.7$ to EfficientNetB2. Second, the number of teachers barely moves the mean: equal-weight averaging dilutes a strong teacher with weaker ones as readily as it combines them, and the single best run uses two teachers while EfficientNetB2's best uses one. Third, every control stopped early — MobileNetV2 at 155 epochs, both EfficientNets at 83 and 84 — while nearly every distilled run used all 300 epochs with its best epoch at or next to the cap. The soft targets keep supplying gradient long after hard labels stop helping, so these numbers are a floor rather than a converged result.

Table 7: Every distillation run at $T = 3$, $\alpha = 0.3$. Runs marked † were voided by the stopping-rule artifact of Section 8 and are excluded from all aggregates.

Student	Configuration	Set	Ep./best	Val acc.	Test acc.	Macro P	Macro R	Macro F1	AUC	Time
MobileNetV2	none	–	155/145	0.7594	0.8519	0.8589	0.8567	0.8546	0.9648	25 m
MobileNetV2	D201	A	300/300	0.8286	0.8476	0.8524	0.8494	0.8498	0.9681	49 m
MobileNetV2	ENB5†	A+B	11/1	0.2314	0.2060	0.0704	0.1909	0.0790	0.5973	3 m
MobileNetV2	ENB5+D201	A	300/300	0.8167	0.8433	0.8614	0.8475	0.8502	0.9686	49 m
MobileNetV2	ENB5+IRV2	A+B	300/300	0.8268	0.8734	0.8750	0.8769	0.8758	0.9746	49 m
MobileNetV2	ENB5+IRV2+D201	A	300/299	0.8225	0.8605	0.8691	0.8629	0.8647	0.9697	50 m
MobileNetV2	ENB5+IRV2+R50	B	300/300	0.8318	0.8455	0.8550	0.8474	0.8499	0.9703	50 m
MobileNetV2	ENB5+R50	B	300/300	0.8261	0.8691	0.8734	0.8710	0.8718	0.9731	49 m
MobileNetV2	IRV2	A+B	300/300	0.8148	0.8541	0.8588	0.8563	0.8565	0.9709	49 m
MobileNetV2	IRV2+D201	A	300/300	0.8210	0.8412	0.8644	0.8470	0.8487	0.9707	49 m
MobileNetV2	IRV2+R50	B	300/300	0.8270	0.8712	0.8737	0.8751	0.8732	0.9748	49 m
MobileNetV2	R50†	B	11/1	0.2402	0.2232	0.1165	0.2067	0.1000	0.6374	2 m
EfficientNetB0	none	–	83/73	0.7875	0.8305	0.8365	0.8325	0.8332	0.9691	14 m
EfficientNetB0	D201	A	300/300	0.8323	0.8884	0.8897	0.8911	0.8895	0.9731	50 m
EfficientNetB0	ENB5†	A+B	11/1	0.2489	0.2232	0.1352	0.2074	0.1009	0.6701	3 m
EfficientNetB0	ENB5+D201†	A	11/1	0.2489	0.2146	0.1319	0.1993	0.0953	0.6278	4 m
EfficientNetB0	ENB5+IRV2	A+B	300/300	0.8603	0.8863	0.8910	0.8892	0.8898	0.9774	50 m
EfficientNetB0	ENB5+IRV2+D201	A	300/300	0.8601	0.8970	0.8994	0.8986	0.8990	0.9750	55 m
EfficientNetB0	ENB5+IRV2+R50	B	300/299	0.8679	0.8798	0.8831	0.8827	0.8822	0.9735	54 m
EfficientNetB0	ENB5+R50	B	300/300	0.8578	0.9013	0.9028	0.9039	0.9030	0.9769	54 m
EfficientNetB0	IRV2	A+B	300/300	0.8526	0.8712	0.8753	0.8728	0.8734	0.9765	49 m
EfficientNetB0	IRV2+D201	A	300/300	0.8616	0.8670	0.8703	0.8694	0.8683	0.9743	51 m
EfficientNetB0	IRV2+R50	B	300/300	0.8518	0.8820	0.8871	0.8837	0.8844	0.9766	54 m
EfficientNetB0	R50	B	300/300	0.8618	0.8691	0.8758	0.8718	0.8733	0.9722	53 m
EfficientNetB2	none	–	84/74	0.8250	0.8541	0.8629	0.8586	0.8584	0.9676	16 m
EfficientNetB2	D201	A	300/300	0.8758	0.8906	0.8924	0.8944	0.8926	0.9756	53 m
EfficientNetB2	ENB5	A+B	300/300	0.8797	0.8884	0.8920	0.8918	0.8917	0.9737	53 m
EfficientNetB2	ENB5+D201	A	286/276	0.8763	0.8884	0.8903	0.8913	0.8901	0.9756	51 m
EfficientNetB2	ENB5+IRV2	A+B	300/298	0.8751	0.8798	0.8823	0.8814	0.8817	0.9752	53 m
EfficientNetB2	ENB5+IRV2+D201	A	300/300	0.8808	0.8734	0.8757	0.8765	0.8754	0.9742	54 m
EfficientNetB2	ENB5+IRV2+R50	B	300/295	0.8730	0.8906	0.8940	0.8938	0.8936	0.9763	52 m
EfficientNetB2	ENB5+R50	B	300/300	0.8831	0.8820	0.8852	0.8859	0.8839	0.9747	51 m
EfficientNetB2	IRV2	A+B	286/276	0.8731	0.8820	0.8834	0.8851	0.8839	0.9743	50 m
EfficientNetB2	IRV2+D201	A	300/299	0.8800	0.8798	0.8836	0.8827	0.8821	0.9781	52 m
EfficientNetB2	IRV2+R50	B	300/300	0.8772	0.8777	0.8830	0.8792	0.8805	0.9768	51 m
EfficientNetB2	R50	B	300/300	0.8644	0.8970	0.9014	0.9000	0.8999	0.9734	51 m

7 Temperature and Alpha

A 3×3 sweep around the winning run, EfficientNetB0 distilled from ENB5+R50, checks that $T = 3$ and $\alpha = 0.3$ were not simply lucky. The $T = 3$, $\alpha = 0.3$ point is the main-grid run itself. Four points were voided by the same stopping-rule artifact.

Table 8: Temperature and alpha sweep on EfficientNetB0 [ENB5+R50]. † marks voided runs.

T	α	Epochs/best	Test accuracy	Macro F1	Macro AUC
2	0.1	300/299	0.8734	0.8772	0.9719
2	0.3	300/300	0.8970	0.8994	0.9754
2	0.5	300/299	0.8584	0.8633	0.9745
3†	0.1	11/1	0.2210	0.0913	0.7017
3	0.3	300/300	0.9013	0.9030	0.9769
3	0.5	300/300	0.9077	**0.9095**	0.9770
5†	0.1	11/1	0.2167	0.0835	0.6931
5†	0.3	11/1	0.2167	0.0834	0.6929
5†	0.5	11/1	0.2189	0.0873	0.6939

The best setting is $T = 3$ with $\alpha = 0.5$, an even split between the teachers and the ground truth, which lifts the headline run by a further 0.65 points. On the surviving evidence α matters more than T, and leaning too far towards the teachers ($\alpha = 0.1$ at $T = 2$) costs about two points. Four of the five missing points sit at $T = 5$ or $\alpha = 0.1$, so the sweep cannot yet rule on high temperature.

7.1 THE BEST MODEL: EFFICIENTNETB0 [ENB5+R50], $T = 3$, $\alpha = 0.5$

Run summary

Teachers	Custom-ENB5 + ResNet50 (set B)
Temperature / alpha	$T = 3$, $\alpha = 0.5$
Parameters / size	4.38 M / 17.5 MB
Epochs / best	300 / 300
Validation / test accuracy	0.8637 / 0.9077
Macro / weighted F1	0.9095 / 0.9075
Macro AUC	0.9770
Training time	50.3 min

Test-set classification report

Class	Precision	Recall	F1-score	Support
0	0.9307	0.9307	0.9307	101
1	0.8632	0.8283	0.8454	99
2	0.8557	0.8830	0.8691	94
3	0.9432	0.9326	0.9379	89
4	0.9529	0.9759	0.9643	83
Accuracy	–	–	0.9077	466
Macro average	0.9091	0.9101	0.9095	466
Weighted average	0.9076	0.9077	0.9075	466

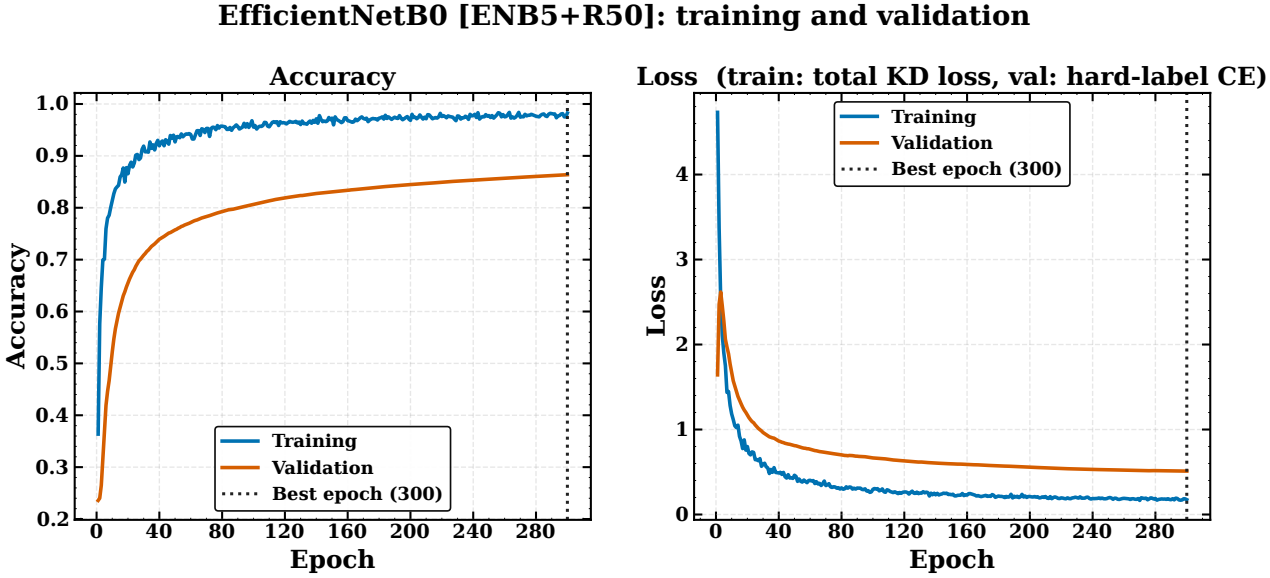
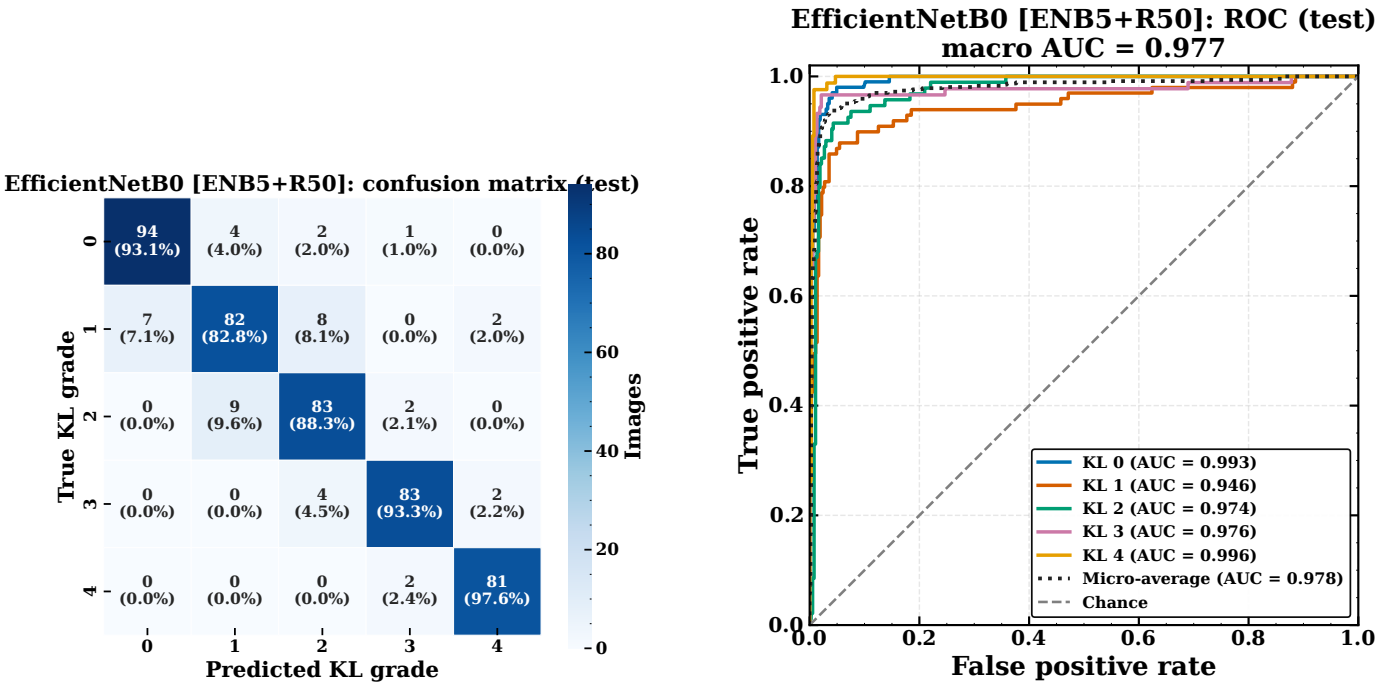


Figure 9: EfficientNetB0 [ENB5+R50], $T = 3$, $\alpha = 0.5$: training and validation curves. Training loss is the combined objective; validation loss is hard-label cross-entropy.



(a) Test confusion matrix.

(b) One-vs-rest ROC curves.

Figure 10: EfficientNetB0 [ENB5+R50], $T = 3$, $\alpha = 0.5$: test-set results.

8 A Stopping Rule that Voided Eight Runs

Eight trainings — four distillation runs and four sweep points — report accuracy near 0.21 and macro F1 near 0.09, which looks like catastrophic failure. It is not: their training accuracy was already 0.83 and still climbing when they stopped.

Validation loss is hard-label cross-entropy, and an untrained student predicts almost uniformly, so its epoch-1 validation loss sits near $\ln 5 = 1.609$ — low, but only because the model is undecided. The student then becomes confident faster than it becomes accurate, validation loss climbs to about 3.0, and needs 15 to 25 epochs to fall back under that first value. With patience counted from epoch 1, these runs reached ten epochs without improvement at epoch 11, stopped, and `restore_best_weights` handed back the undecided epoch-1 weights. The affected runs are EfficientNetB0 [ENB5], EfficientNetB0 [ENB5+D201], MobileNetV2 [ENB5], MobileNetV2 [R50], and the sweep points at $T = 3$, $\alpha = 0.1$ and all three at $T = 5$.

The fix is a warm-up: `start_from_epoch=25` makes the early epochs ineligible as the best and keeps them out of the patience count. It was verified with patience 1 and a warm-up of 5, where the run went its full length and selected no warm-up epoch. Because a finished `metrics.json` marks a run as done, the eight result folders must be deleted before re-running, which costs roughly six GPU-hours. No conclusion in this report depends on them: they can only add to the results, and one of them, MobileNetV2 [R50], may well raise MobileNetV2's modest gain.

9 Per-Run Results

Every run at $T = 3$, $\alpha = 0.3$, grouped by student, in the order of Table 7.

9.1 MobileNetV2

9.1.1 MobileNetV2 [none]

Run summary

Teachers	none (control)
Temperature / alpha	–
Parameters / size	2.59 M / 10.4 MB
Epochs / best	155 / 145
Validation / test accuracy	0.7594 / 0.8519
Macro / weighted F1	0.8546 / 0.8497
Macro AUC	0.9648
Training time	24.5 min

Test-set classification report

Class	Precision	Recall	F1-score	Support
0	0.7607	0.8812	0.8165	101
1	0.8101	0.6465	0.7191	99
2	0.8416	0.9043	0.8718	94
3	0.9518	0.8876	0.9186	89
4	0.9302	0.9639	0.9467	83
Accuracy	–	–	0.8519	466
Macro average	0.8589	0.8567	0.8546	466
Weighted average	0.8542	0.8519	0.8497	466

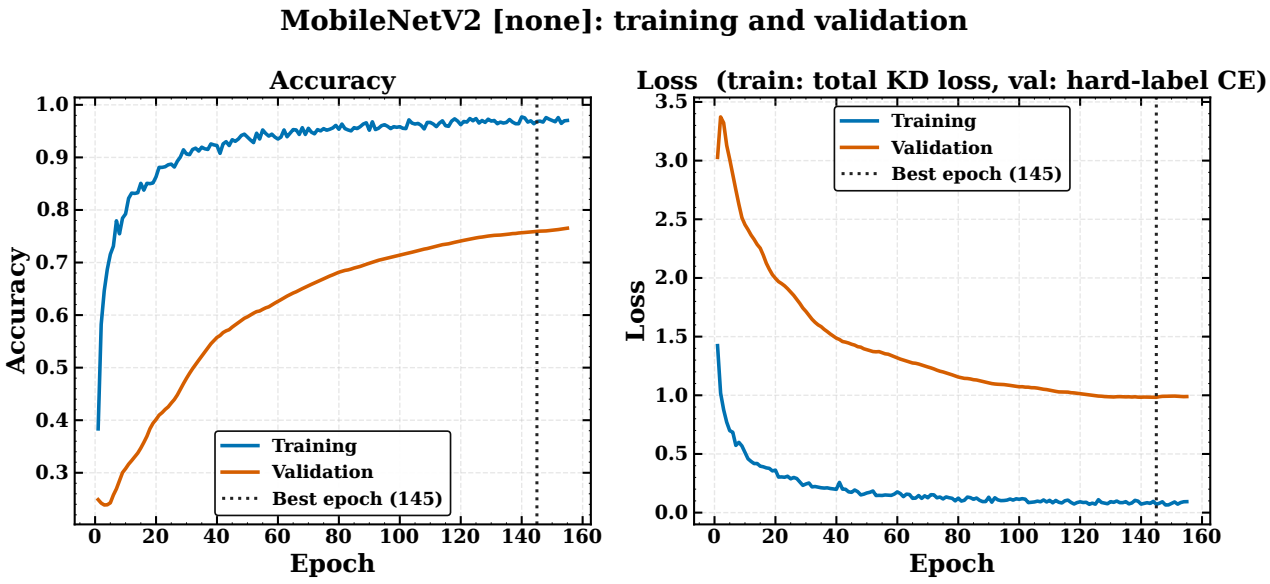
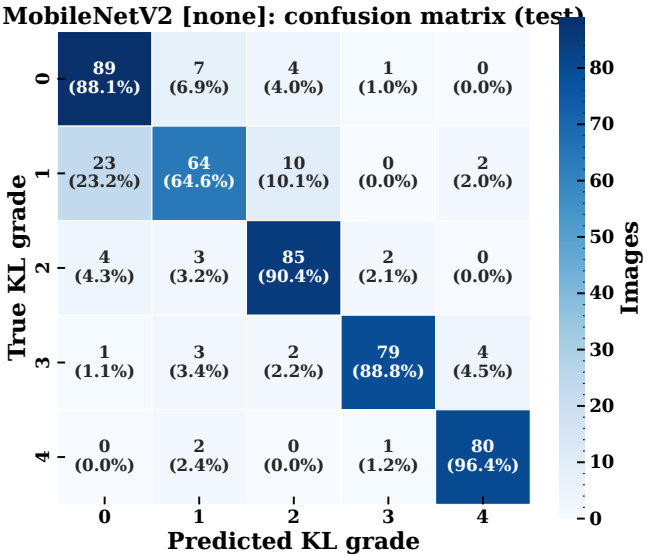
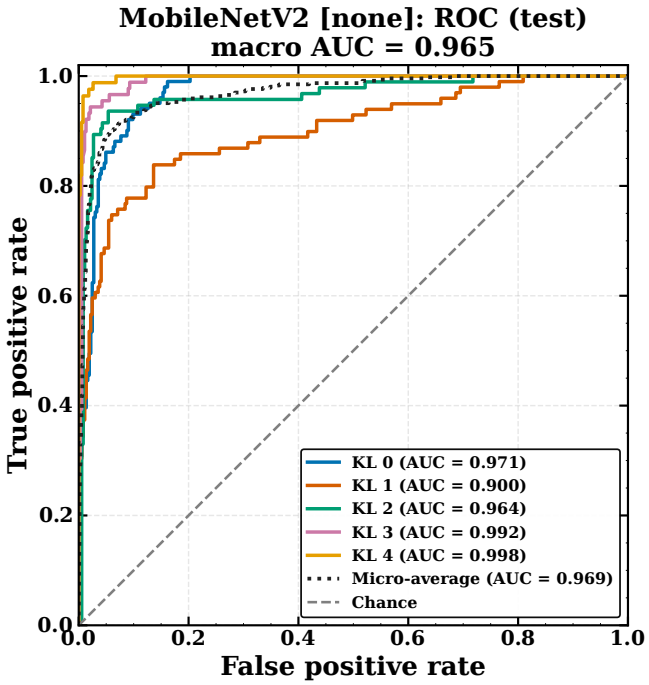


Figure 11: MobileNetV2 [none]: training and validation curves. Training loss is the combined objective; validation loss is hard-label cross-entropy.



(a) Test confusion matrix.



(b) One-vs-rest ROC curves.

Figure 12: MobileNetV2 [none]: test-set results.

9.1.2 MobileNetV2 [D201]

Run summary	
Teachers	DenseNet201
Temperature / alpha	$T = 3$, $\alpha = 0.3$
Parameters / size	2.59 M / 10.4 MB
Epochs / best	300 / 300
Validation / test accuracy	0.8286 / 0.8476
Macro / weighted F1	0.8498 / 0.8473
Macro AUC	0.9681
Training time	49.2 min

Test-set classification report

Class	Precision	Recall	F1-score	Support
0	0.8598	0.9109	0.8846	101
1	0.7451	0.7677	0.7562	99
2	0.8452	0.7553	0.7978	94
3	0.8632	0.9213	0.8913	89
4	0.9487	0.8916	0.9193	83
Accuracy	–	–	0.8476	466
Macro average	0.8524	0.8494	0.8498	466
Weighted average	0.8490	0.8476	0.8473	466

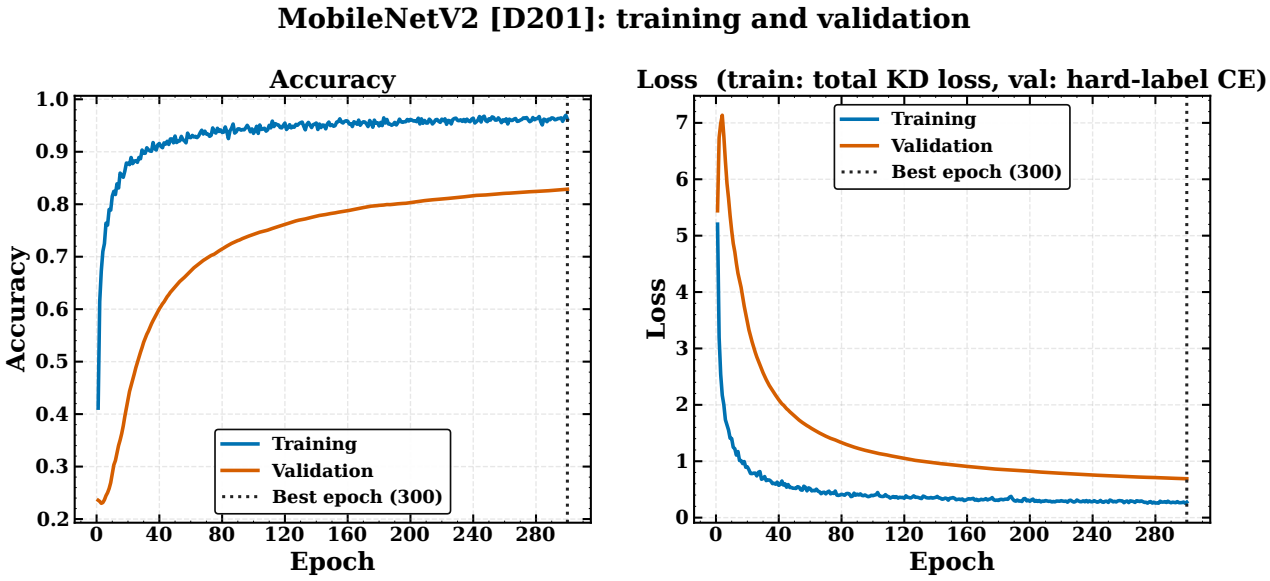
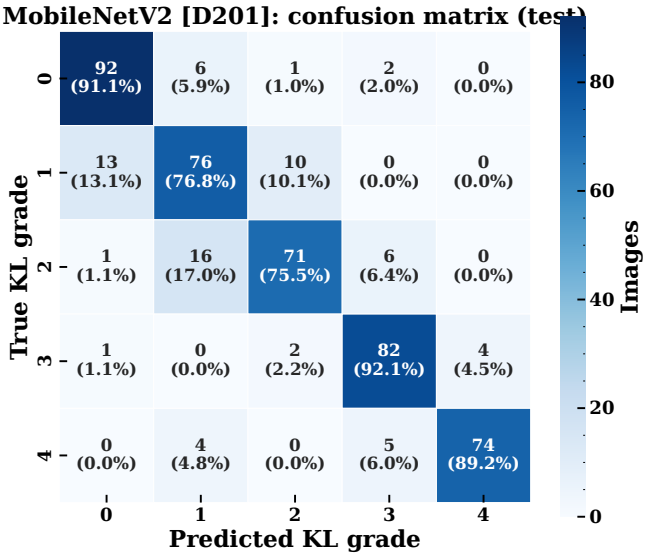
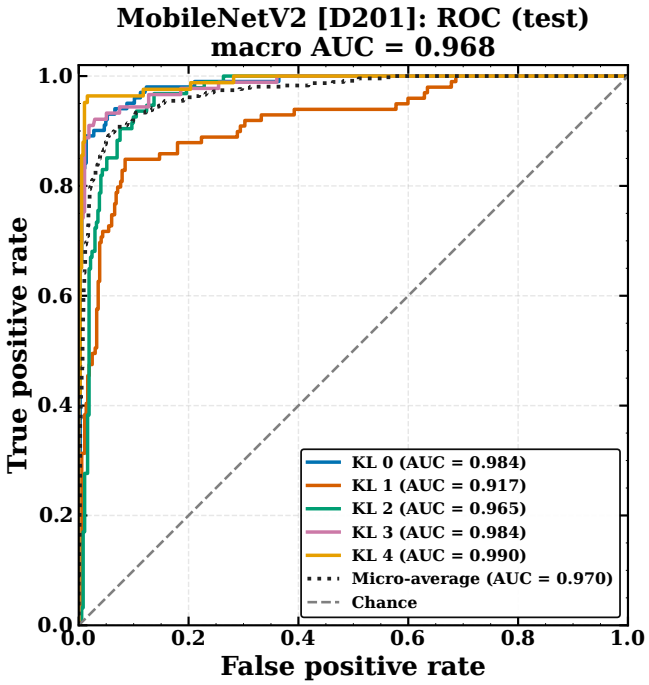


Figure 13: MobileNetV2 [D201]: training and validation curves. Training loss is the combined objective; validation loss is hard-label cross-entropy.



(a) Test confusion matrix.



(b) One-vs-rest ROC curves.

Figure 14: MobileNetV2 [D201]: test-set results.

9.1.3 MobileNetV2 [ENB5]

Voided by the stopping-rule artifact of Section 8.

Run summary

Teachers	Custom-ENB5
Temperature / alpha	$T = 3$, $\alpha = 0.3$
Parameters / size	2.59 M / 10.4 MB
Epochs / best	11 / 1
Validation / test accuracy	0.2314 / 0.2060
Macro / weighted F1	0.0790 / 0.0843
Macro AUC	0.5973
Training time	2.7 min

Test-set classification report

Class	Precision	Recall	F1-score	Support
0	0.2090	0.9208	0.3407	101
1	0.0000	0.0000	0.0000	99
2	0.0000	0.0000	0.0000	94
3	0.1429	0.0337	0.0545	89
4	0.0000	0.0000	0.0000	83
Accuracy	–	–	0.2060	466
Macro average	0.0704	0.1909	0.0790	466
Weighted average	0.0726	0.2060	0.0843	466

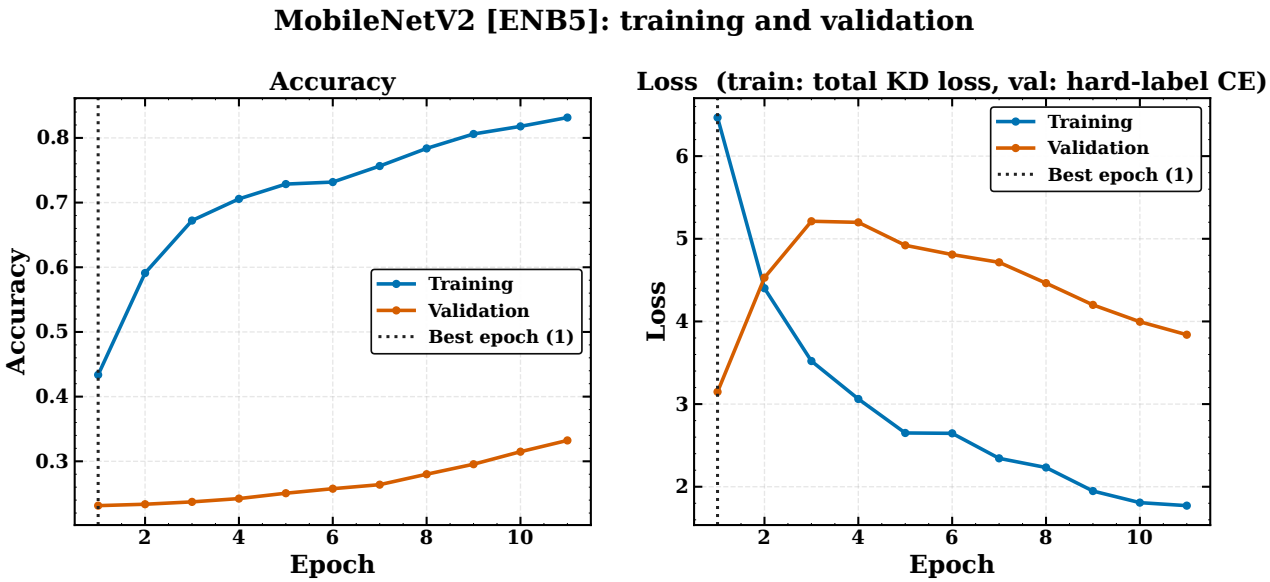
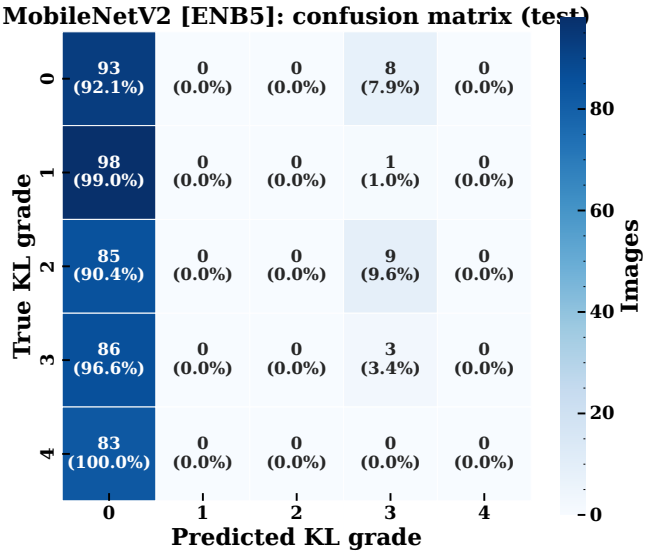
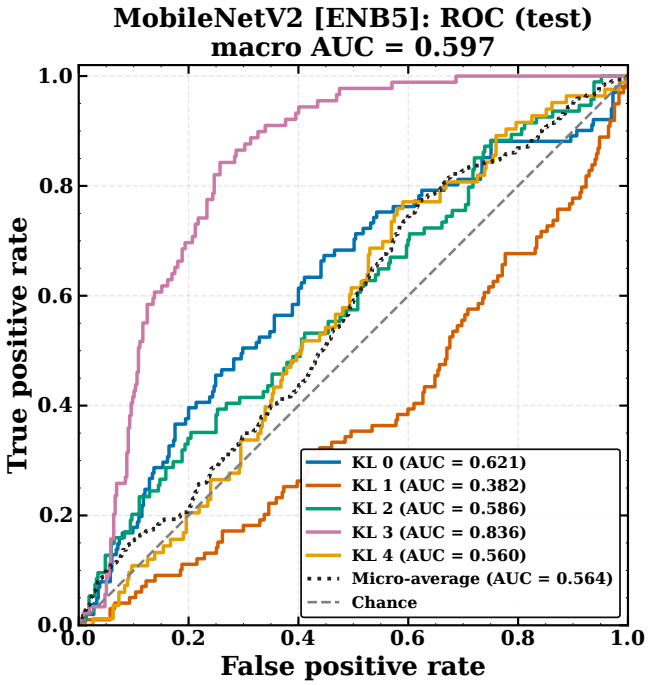


Figure 15: MobileNetV2 [ENB5]: training and validation curves. Training loss is the combined objective; validation loss is hard-label cross-entropy.



(a) Test confusion matrix.



(b) One-vs-rest ROC curves.

Figure 16: MobileNetV2 [ENB5]: test-set results.

9.1.4 MobileNetV2 [ENB5+D201]

Run summary

Teachers	Custom-ENB5, DenseNet201
Temperature / alpha	$T = 3$, $\alpha = 0.3$
Parameters / size	2.59 M / 10.4 MB
Epochs / best	300 / 300
Validation / test accuracy	0.8167 / 0.8433
Macro / weighted F1	0.8502 / 0.8460
Macro AUC	0.9686
Training time	49.4 min

Test-set classification report

Class	Precision	Recall	F1-score	Support
0	0.9390	0.7624	0.8415	101
1	0.6667	0.8485	0.7467	99
2	0.8295	0.7766	0.8022	94
3	0.9205	0.9101	0.9153	89
4	0.9512	0.9398	0.9455	83
Accuracy	–	–	0.8433	466
Macro average	0.8614	0.8475	0.8502	466
Weighted average	0.8577	0.8433	0.8460	466

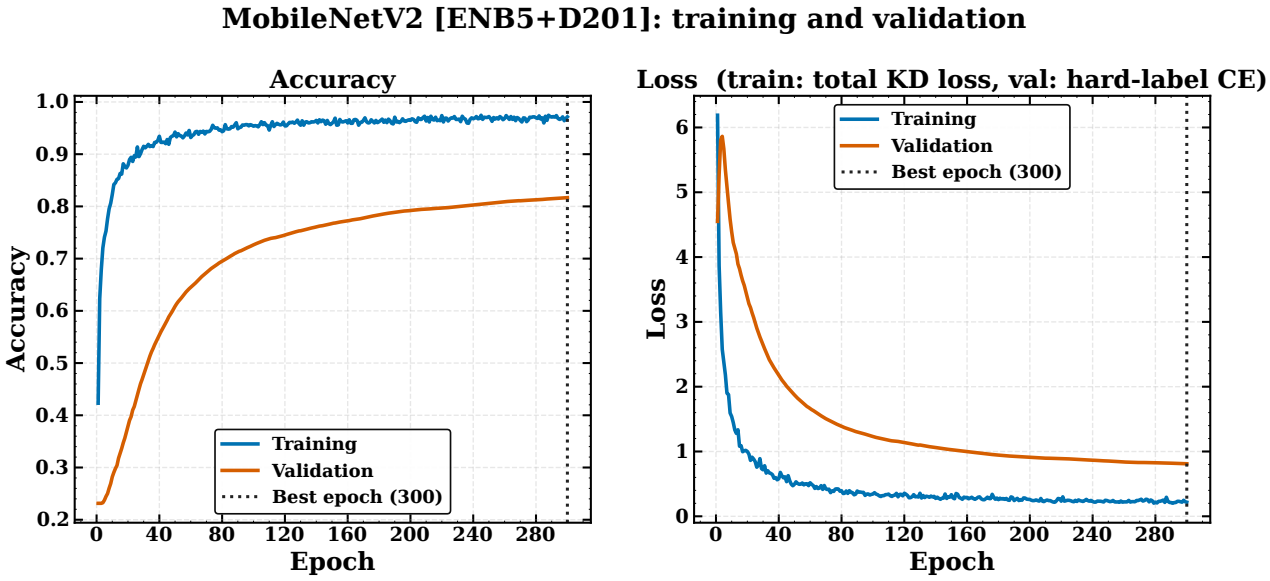
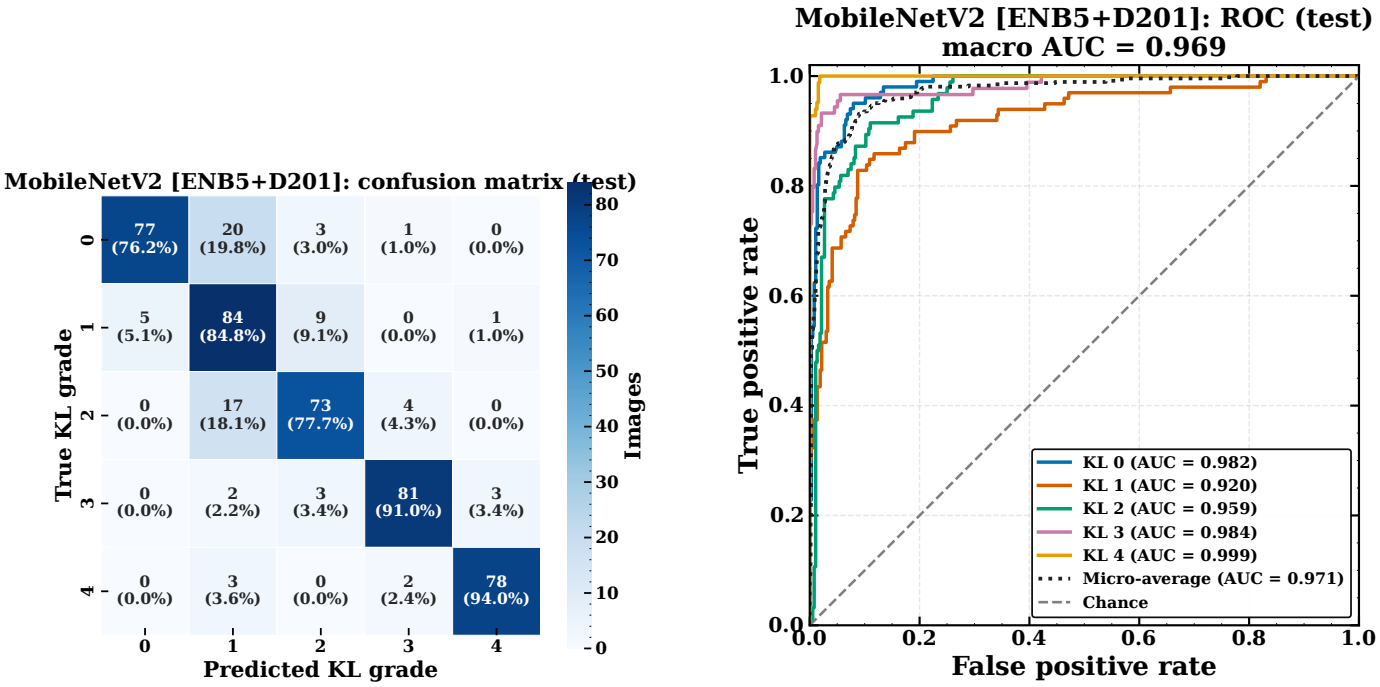


Figure 17: MobileNetV2 [ENB5+D201]: training and validation curves. Training loss is the combined objective; validation loss is hard-label cross-entropy.



(a) Test confusion matrix.

(b) One-vs-rest ROC curves.

Figure 18: MobileNetV2 [ENB5+D201]: test-set results.

9.1.5 MobileNetV2 [ENB5+IRV2]

Run summary

Teachers	Custom-ENB5, Inception-ResNetV2
Temperature / alpha	$T = 3$, $\alpha = 0.3$
Parameters / size	2.59 M / 10.4 MB
Epochs / best	300 / 300
Validation / test accuracy	0.8268 / 0.8734
Macro / weighted F1	0.8758 / 0.8728
Macro AUC	0.9746
Training time	49.3 min

Test-set classification report

Class	Precision	Recall	F1-score	Support
0	0.8824	0.8911	0.8867	101
1	0.7938	0.7778	0.7857	99
2	0.8462	0.8191	0.8324	94
3	0.9222	0.9326	0.9274	89
4	0.9302	0.9639	0.9467	83
Accuracy	–	–	0.8734	466
Macro average	0.8750	0.8769	0.8758	466
Weighted average	0.8724	0.8734	0.8728	466

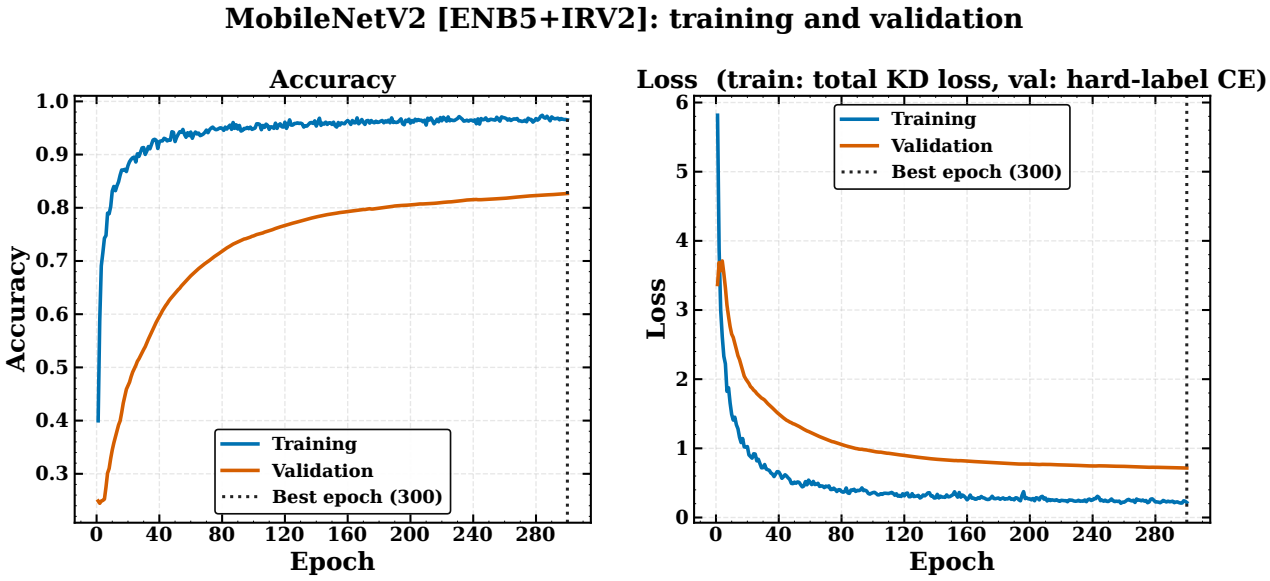
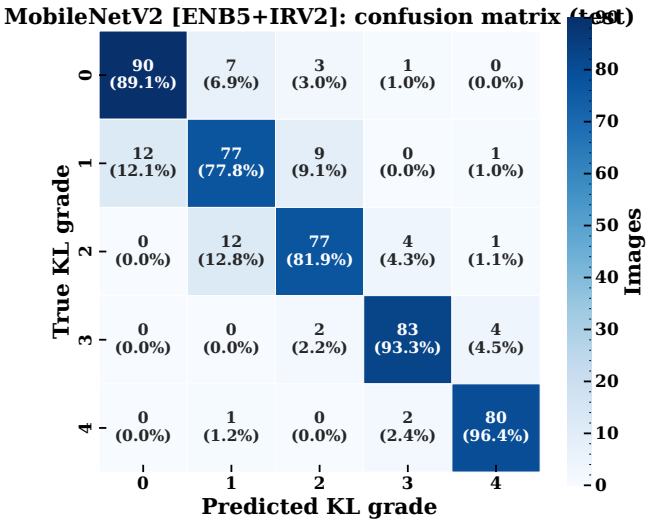
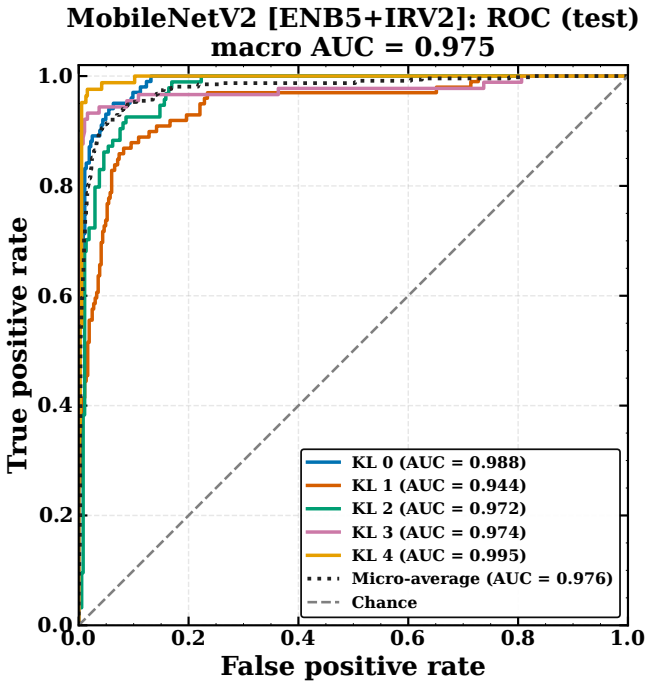


Figure 19: MobileNetV2 [ENB5+IRV2]: training and validation curves. Training loss is the combined objective; validation loss is hard-label cross-entropy.



(a) Test confusion matrix.



(b) One-vs-rest ROC curves.

Figure 20: MobileNetV2 [ENB5+IRV2]: test-set results.

9.1.6 MobileNetV2 [ENB5+IRV2+D201]

Run summary

Teachers	Custom-ENB5, Inception-ResNetV2, DenseNet201
Temperature / alpha	$T = 3$, $\alpha = 0.3$
Parameters / size	2.59 M / 10.4 MB
Epochs / best	300 / 299
Validation / test accuracy	0.8225 / 0.8605
Macro / weighted F1	0.8647 / 0.8619
Macro AUC	0.9697
Training time	50.5 min

Test-set classification report

Class	Precision	Recall	F1-score	Support
0	0.8958	0.8515	0.8731	101
1	0.7193	0.8283	0.7700	99
2	0.8824	0.7979	0.8380	94
3	0.9213	0.9213	0.9213	89
4	0.9268	0.9157	0.9212	83
Accuracy	–	–	0.8605	466
Macro average	0.8691	0.8629	0.8647	466
Weighted average	0.8660	0.8605	0.8619	466

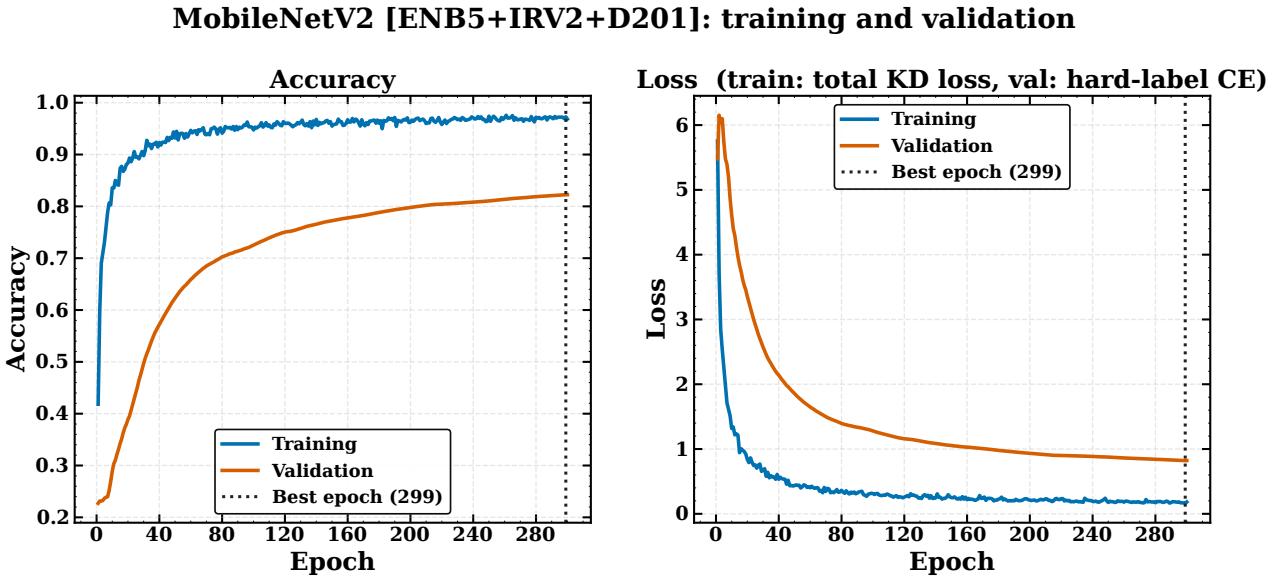
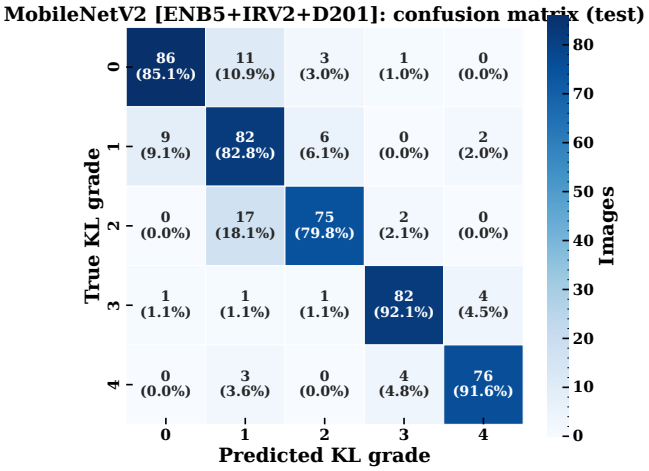
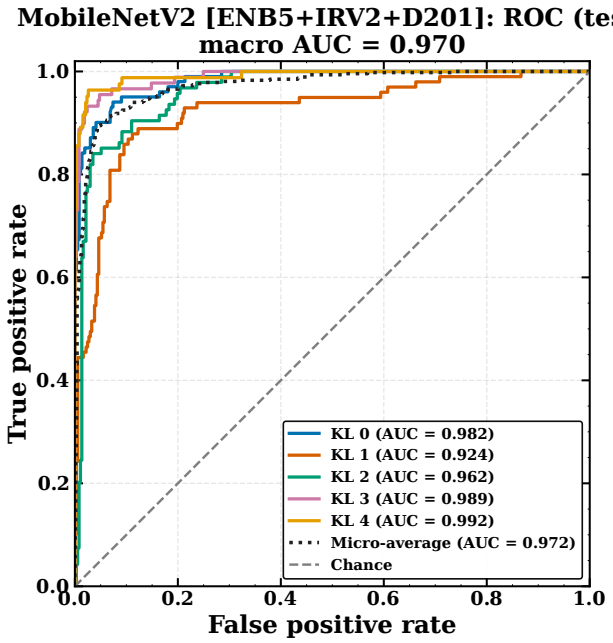


Figure 21: MobileNetV2 [ENB5+IRV2+D201]: training and validation curves. Training loss is the combined objective; validation loss is hard-label cross-entropy.



(a) Test confusion matrix.



(b) One-vs-rest ROC curves.

Figure 22: MobileNetV2 [ENB5+IRV2+D201]: test-set results.

9.1.7 MobileNetV2 [ENB5+IRV2+R50]

Run summary

Teachers	Custom-ENB5, Inception-ResNetV2, ResNet50
Temperature / alpha	$T = 3$, $\alpha = 0.3$
Parameters / size	2.59 M / 10.4 MB
Epochs / best	300 / 300
Validation / test accuracy	0.8318 / 0.8455
Macro / weighted F1	0.8499 / 0.8463
Macro AUC	0.9703
Training time	49.6 min

Test-set classification report

Class	Precision	Recall	F1-score	Support
0	0.8091	0.8812	0.8436	101
1	0.7308	0.7677	0.7488	99
2	0.8706	0.7872	0.8268	94
3	0.8913	0.9213	0.9061	89
4	0.9733	0.8795	0.9241	83
Accuracy	–	–	0.8455	466
Macro average	0.8550	0.8474	0.8499	466
Weighted average	0.8498	0.8455	0.8463	466

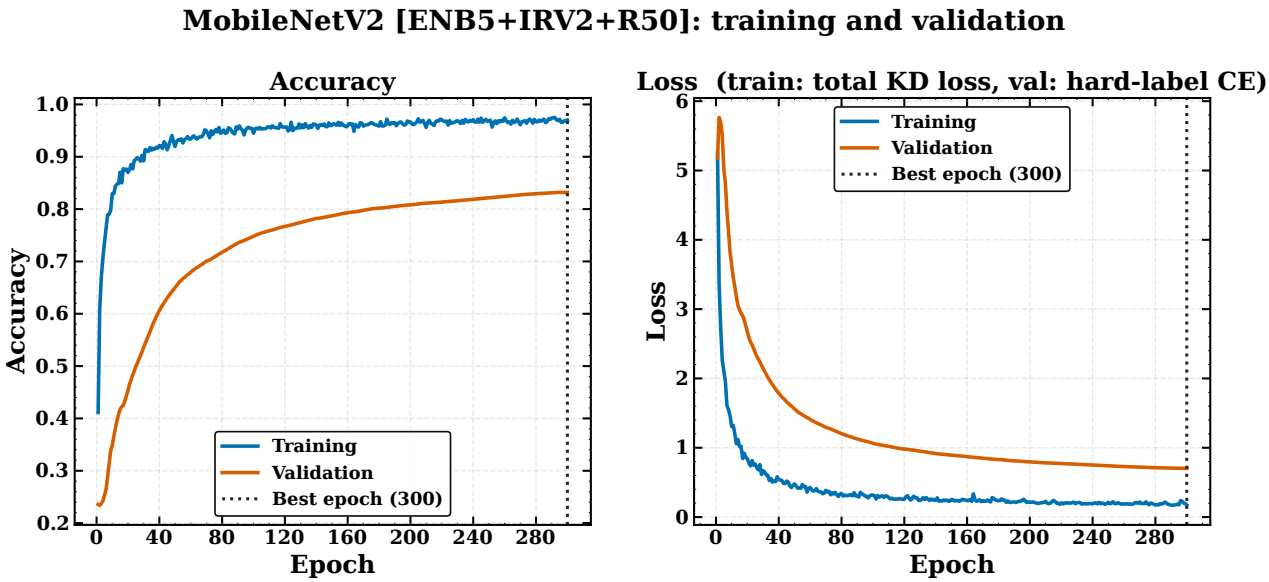
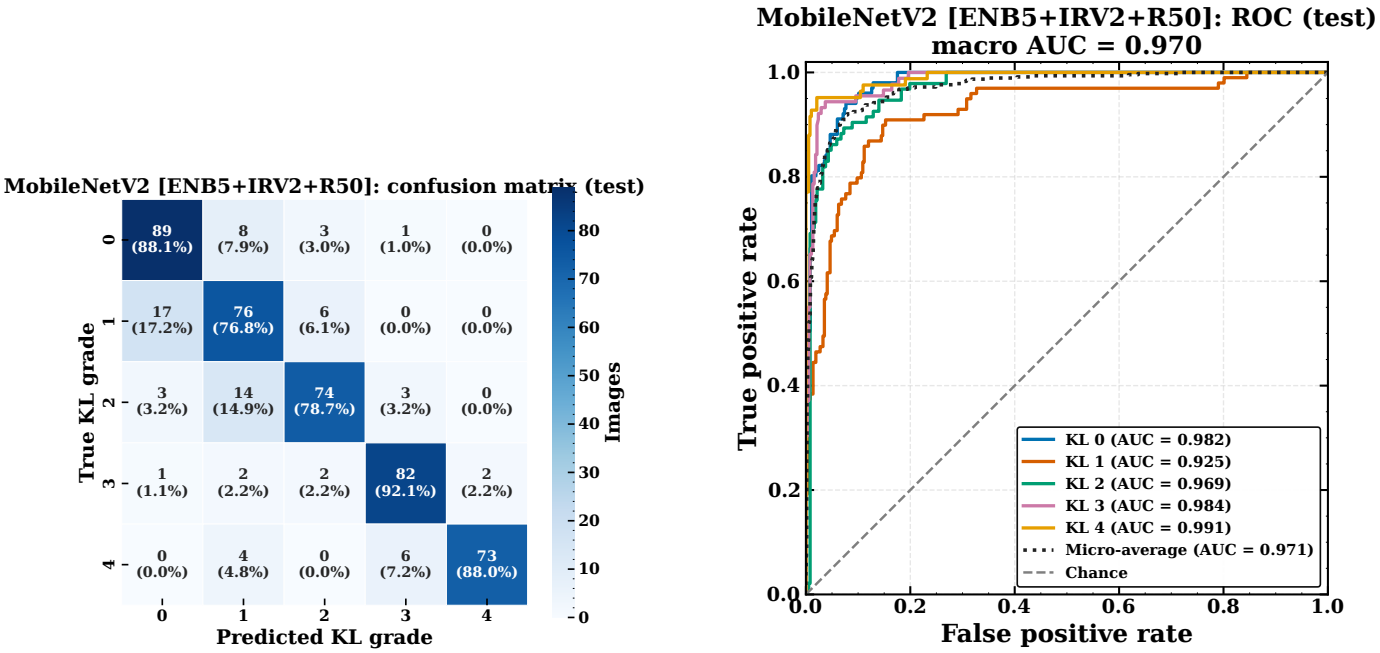


Figure 23: MobileNetV2 [ENB5+IRV2+R50]: training and validation curves. Training loss is the combined objective; validation loss is hard-label cross-entropy.



(a) Test confusion matrix.

(b) One-vs-rest ROC curves.

Figure 24: MobileNetV2 [ENB5+IRV2+R50]: test-set results.

9.1.8 MobileNetV2 [ENB5+R50]

Run summary

Teachers	Custom-ENB5, ResNet50
Temperature / alpha	$T = 3$, $\alpha = 0.3$
Parameters / size	2.59 M / 10.4 MB
Epochs / best	300 / 300
Validation / test accuracy	0.8261 / 0.8691
Macro / weighted F1	0.8718 / 0.8690
Macro AUC	0.9731
Training time	49.4 min

Test-set classification report

Class	Precision	Recall	F1-score	Support
0	0.8598	0.9109	0.8846	101
1	0.7917	0.7677	0.7795	99
2	0.8316	0.8404	0.8360	94
3	0.9222	0.9326	0.9274	89
4	0.9615	0.9036	0.9317	83
Accuracy	–	–	0.8691	466
Macro average	0.8734	0.8710	0.8718	466
Weighted average	0.8697	0.8691	0.8690	466

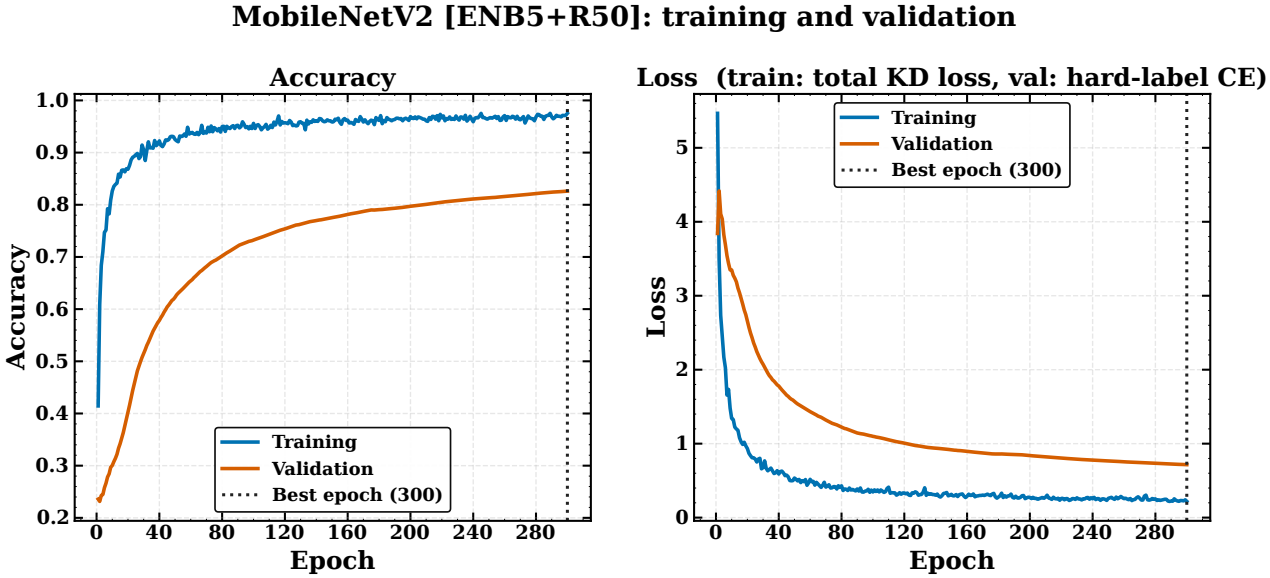
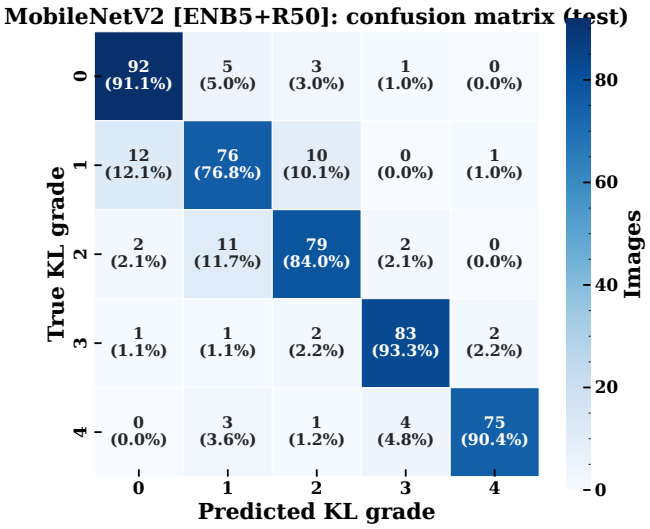
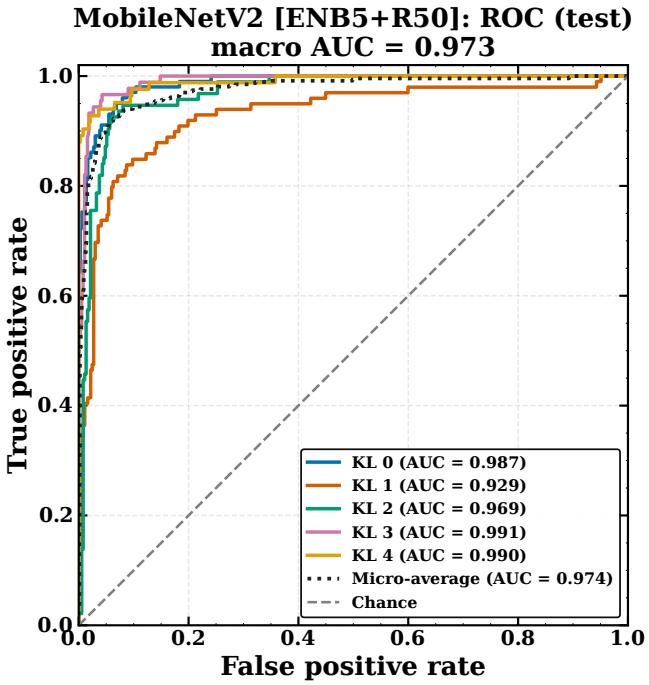


Figure 25: MobileNetV2 [ENB5+R50]: training and validation curves. Training loss is the combined objective; validation loss is hard-label cross-entropy.



(a) Test confusion matrix.



(b) One-vs-rest ROC curves.

Figure 26: MobileNetV2 [ENB5+R50]: test-set results.

9.1.9 MobileNetV2 [IRV2]

Run summary

Teachers	InceptionResNetV2
Temperature / alpha	$T = 3$, $\alpha = 0.3$
Parameters / size	2.59 M / 10.4 MB
Epochs / best	300 / 300
Validation / test accuracy	0.8148 / 0.8541
Macro / weighted F1	0.8565 / 0.8536
Macro AUC	0.9709
Training time	49.0 min

Test-set classification report

Class	Precision	Recall	F1-score	Support
0	0.8491	0.8911	0.8696	101
1	0.8000	0.7273	0.7619	99
2	0.7921	0.8511	0.8205	94
3	0.8925	0.9326	0.9121	89
4	0.9605	0.8795	0.9182	83
Accuracy	–	–	0.8541	466
Macro average	0.8588	0.8563	0.8565	466
Weighted average	0.8553	0.8541	0.8536	466

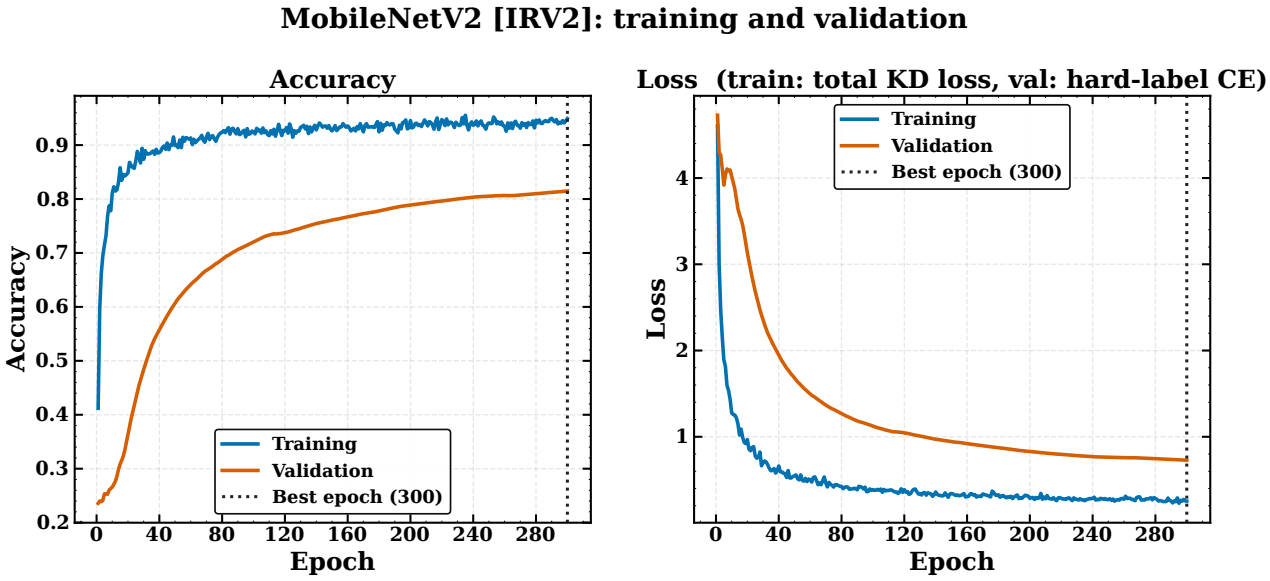
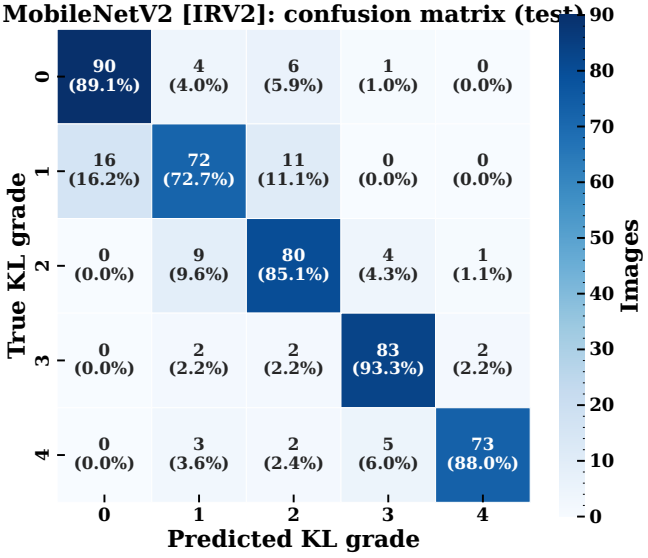
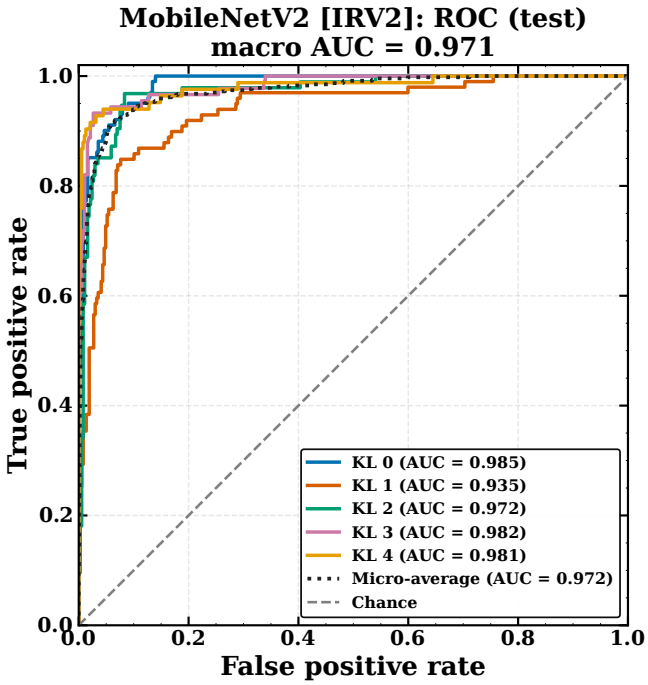


Figure 27: MobileNetV2 [IRV2]: training and validation curves. Training loss is the combined objective; validation loss is hard-label cross-entropy.



(a) Test confusion matrix.



(b) One-vs-rest ROC curves.

Figure 28: MobileNetV2 [IRV2]: test-set results.

9.1.10 MobileNetV2 [IRV2+D201]

Run summary

Teachers	InceptionResNetV2, DenseNet201
Temperature / alpha	$T = 3$, $\alpha = 0.3$
Parameters / size	2.59 M / 10.4 MB
Epochs / best	300 / 300
Validation / test accuracy	0.8210 / 0.8412
Macro / weighted F1	0.8487 / 0.8433
Macro AUC	0.9707
Training time	49.2 min

Test-set classification report

Class	Precision	Recall	F1-score	Support
0	0.9452	0.6832	0.7931	101
1	0.6462	0.8485	0.7336	99
2	0.8462	0.8191	0.8324	94
3	0.9326	0.9326	0.9326	89
4	0.9518	0.9518	0.9518	83
Accuracy	–	–	0.8412	466
Macro average	0.8644	0.8470	0.8487	466
Weighted average	0.8605	0.8412	0.8433	466

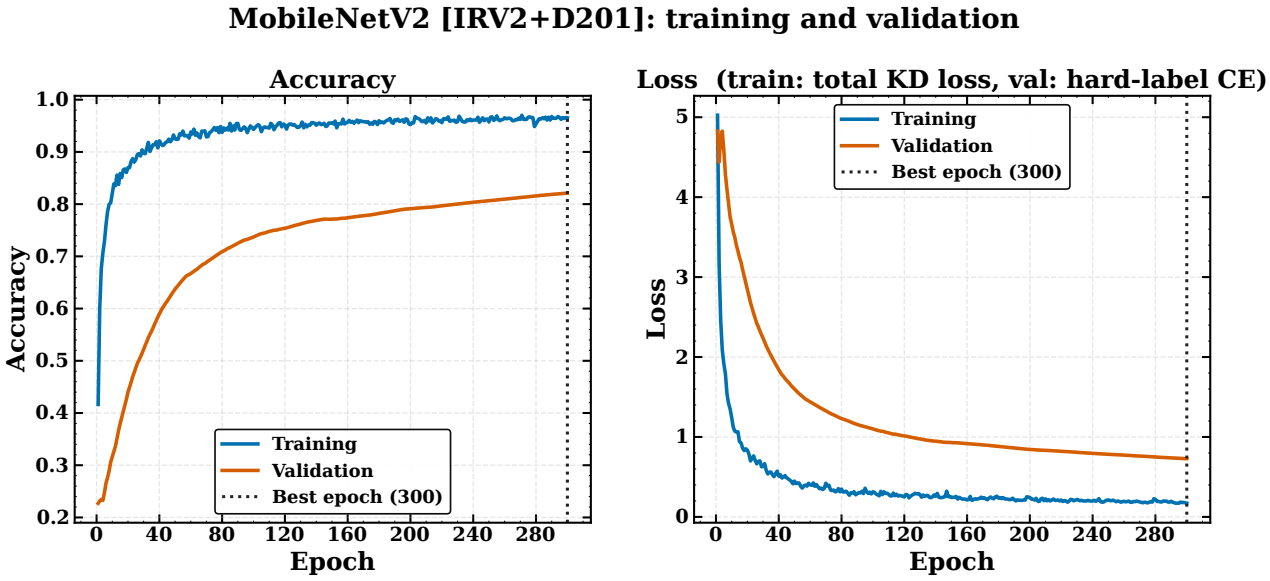
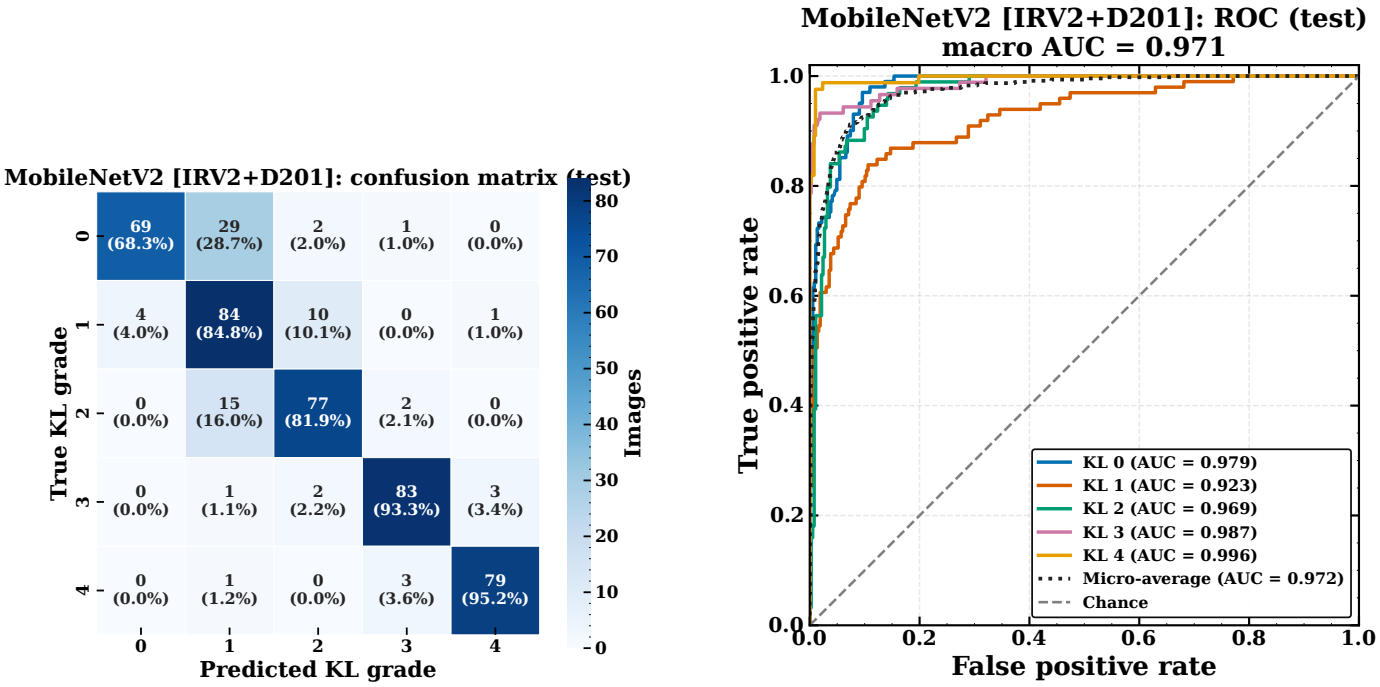


Figure 29: MobileNetV2 [IRV2+D201]: training and validation curves. Training loss is the combined objective; validation loss is hard-label cross-entropy.



(a) Test confusion matrix.

(b) One-vs-rest ROC curves.

Figure 30: MobileNetV2 [IRV2+D201]: test-set results.

9.1.11 MobileNetV2 [IRV2+R50]

Run summary

Teachers	InceptionResNetV2, ResNet50
Temperature / alpha	$T = 3$, $\alpha = 0.3$
Parameters / size	2.59 M / 10.4 MB
Epochs / best	300 / 300
Validation / test accuracy	0.8270 / 0.8712
Macro / weighted F1	0.8732 / 0.8702
Macro AUC	0.9748
Training time	48.8 min

Test-set classification report

Class	Precision	Recall	F1-score	Support
0	0.8713	0.8713	0.8713	101
1	0.8295	0.7374	0.7807	99
2	0.8173	0.9043	0.8586	94
3	0.9412	0.8989	0.9195	89
4	0.9091	0.9639	0.9357	83
Accuracy	–	–	0.8712	466
Macro average	0.8737	0.8751	0.8732	466
Weighted average	0.8716	0.8712	0.8702	466

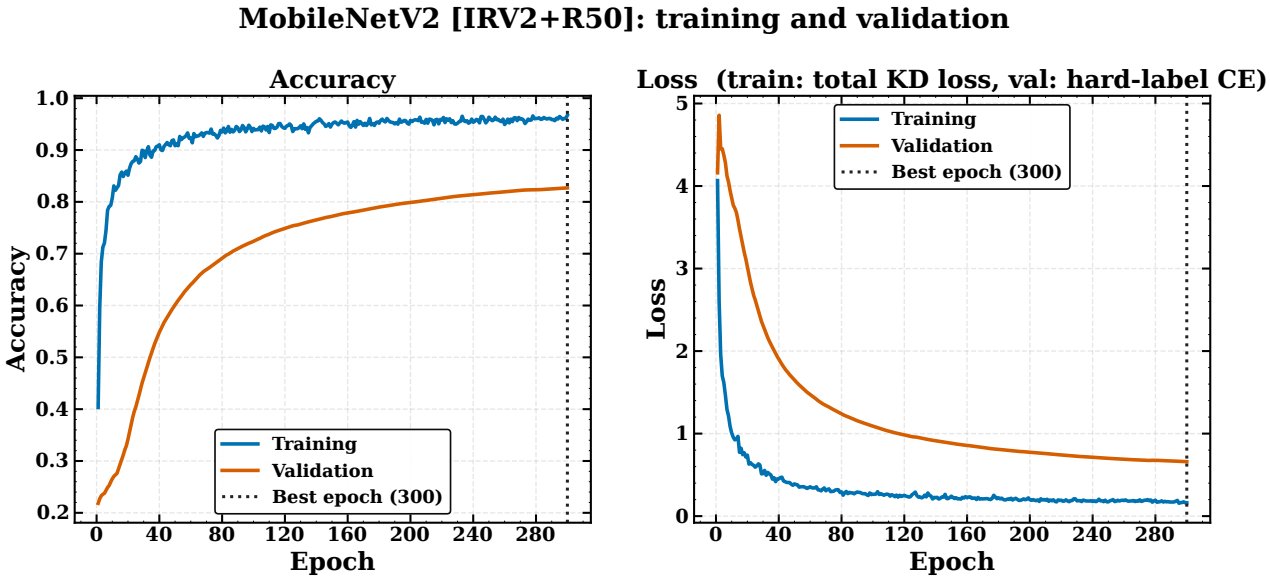
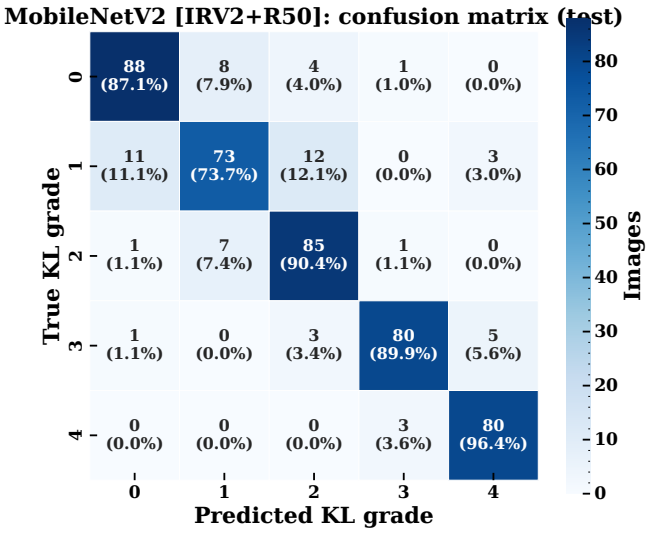
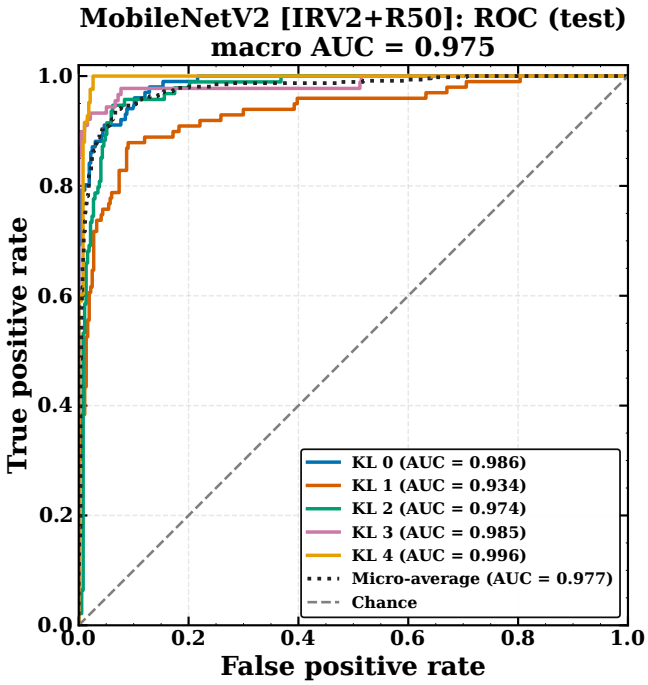


Figure 31: MobileNetV2 [IRV2+R50]: training and validation curves. Training loss is the combined objective; validation loss is hard-label cross-entropy.



(a) Test confusion matrix.



(b) One-vs-rest ROC curves.

Figure 32: MobileNetV2 [IRV2+R50]: test-set results.

9.1.12 MobileNetV2 [R50]

Voided by the stopping-rule artifact of Section 8.

Run summary

Teachers	ResNet50
Temperature / alpha	$T = 3$, $\alpha = 0.3$
Parameters / size	2.59 M / 10.4 MB
Epochs / best	11 / 1
Validation / test accuracy	0.2402 / 0.2232
Macro / weighted F1	0.1000 / 0.1070
Macro AUC	0.6374
Training time	2.4 min

Test-set classification report

Class	Precision	Recall	F1-score	Support
0	0.2270	0.9505	0.3664	101
1	0.2222	0.0606	0.0952	99
2	0.0000	0.0000	0.0000	94
3	0.1333	0.0225	0.0385	89
4	0.0000	0.0000	0.0000	83
Accuracy	–	–	0.2232	466
Macro average	0.1165	0.2067	0.1000	466
Weighted average	0.1219	0.2232	0.1070	466

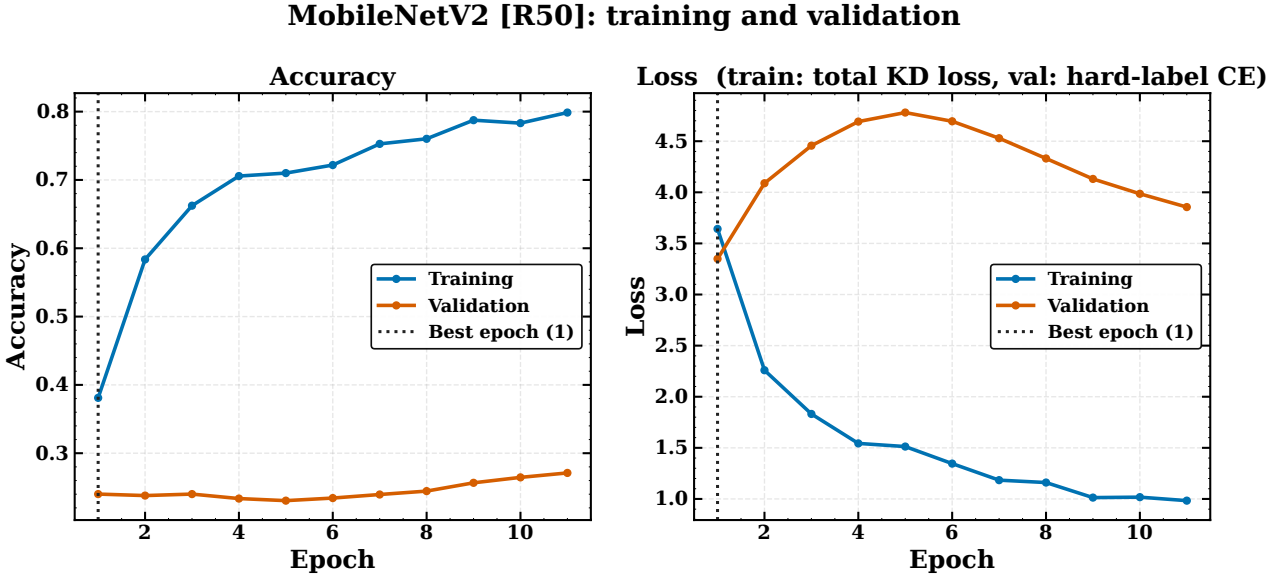
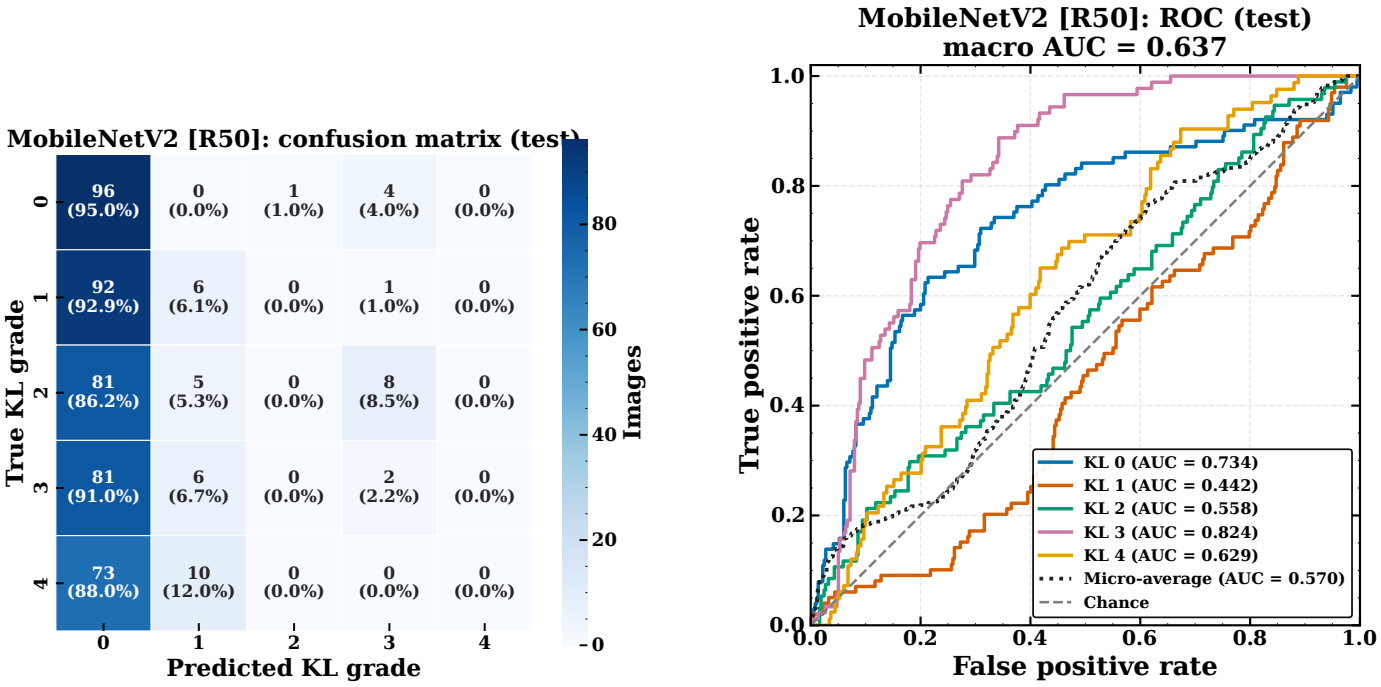


Figure 33: MobileNetV2 [R50]: training and validation curves. Training loss is the combined objective; validation loss is hard-label cross-entropy.



(a) Test confusion matrix.

(b) One-vs-rest ROC curves.

Figure 34: MobileNetV2 [R50]: test-set results.

9.2 EfficientNetB0

9.2.1 EfficientNetB0 [none]

Run summary

Teachers	none (control)
Temperature / alpha	–
Parameters / size	4.38 M / 17.5 MB
Epochs / best	83 / 73
Validation / test accuracy	0.7875 / 0.8305
Macro / weighted F1	0.8332 / 0.8301
Macro AUC	0.9691
Training time	13.8 min

Test-set classification report

Class	Precision	Recall	F1-score	Support
0	0.8198	0.9010	0.8585	101
1	0.7184	0.7475	0.7327	99
2	0.8519	0.7340	0.7886	94
3	0.8667	0.8764	0.8715	89
4	0.9259	0.9036	0.9146	83
Accuracy	–	–	0.8305	466
Macro average	0.8365	0.8325	0.8332	466
Weighted average	0.8326	0.8305	0.8301	466

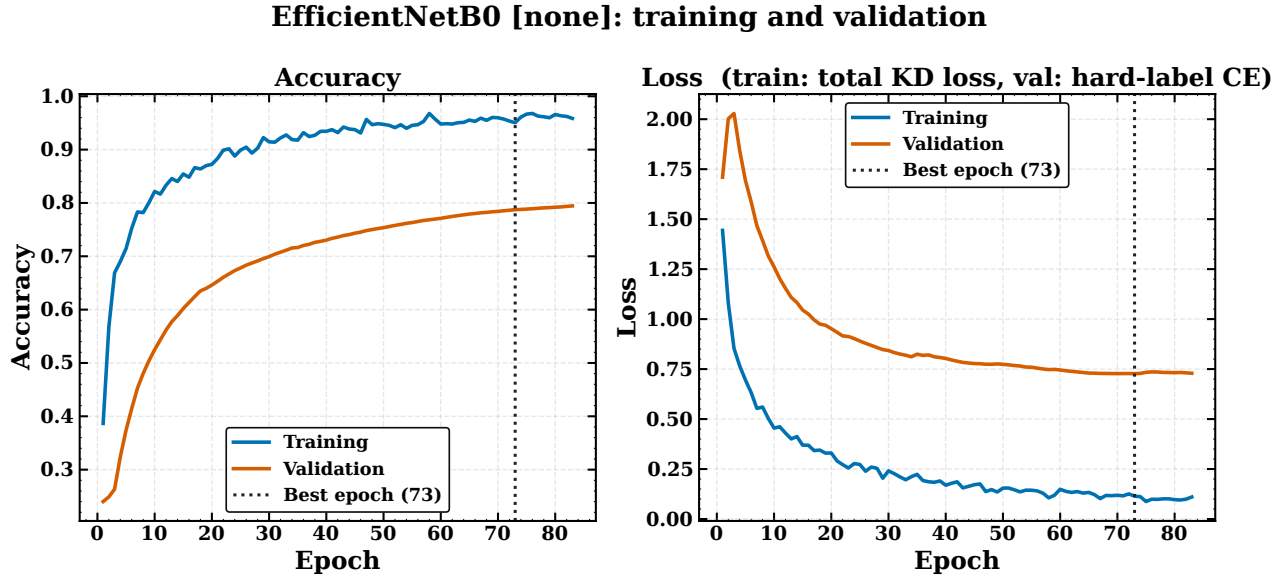
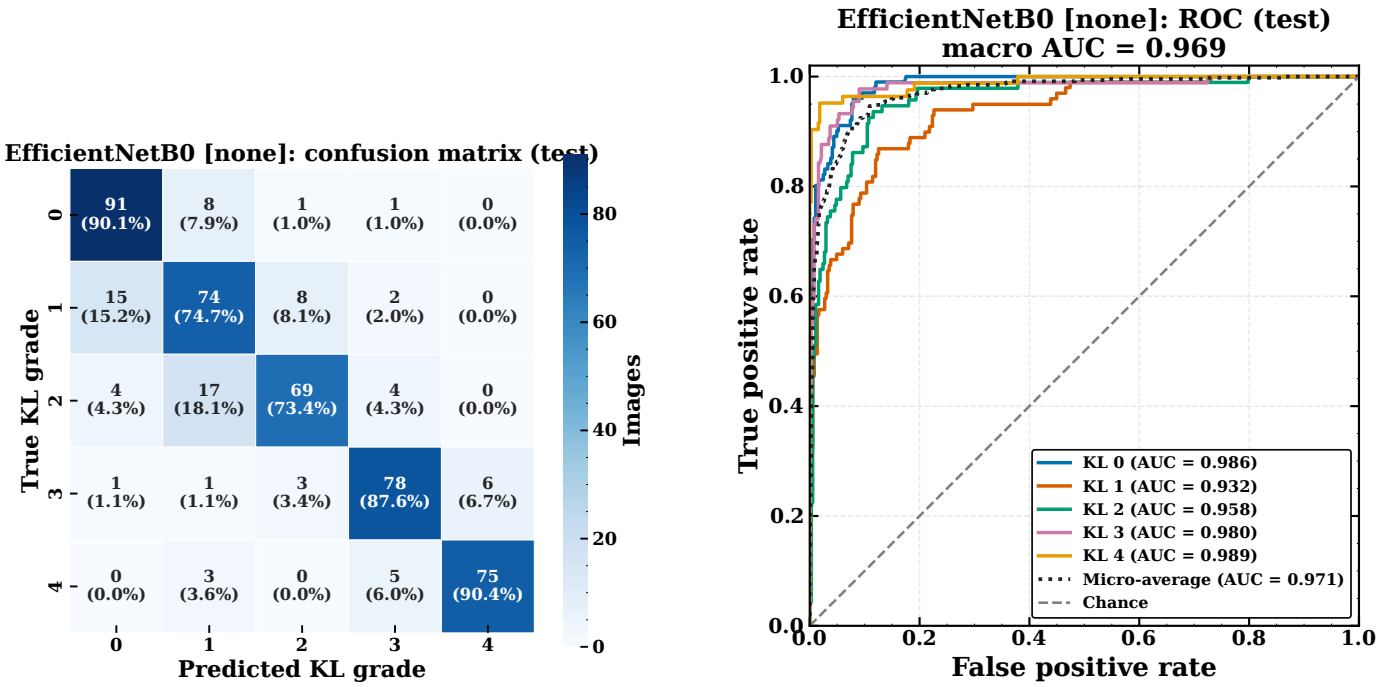


Figure 35: EfficientNetB0 [none]: training and validation curves. Training loss is the combined objective; validation loss is hard-label cross-entropy.



(a) Test confusion matrix.

(b) One-vs-rest ROC curves.

Figure 36: EfficientNetB0 [none]: test-set results.

9.2.2 EfficientNetB0 [D201]

Run summary

Teachers	DenseNet201
Temperature / alpha	$T = 3, \alpha = 0.3$
Parameters / size	4.38 M / 17.5 MB
Epochs / best	300 / 300
Validation / test accuracy	0.8323 / 0.8884
Macro / weighted F1	0.8895 / 0.8871
Macro AUC	0.9731
Training time	49.6 min

Test-set classification report

Class	Precision	Recall	F1-score	Support
0	0.8879	0.9406	0.9135	101
1	0.8621	0.7576	0.8065	99
2	0.8438	0.8617	0.8526	94
3	0.9032	0.9438	0.9231	89
4	0.9518	0.9518	0.9518	83
Accuracy	–	–	0.8884	466
Macro average	0.8897	0.8911	0.8895	466
Weighted average	0.8878	0.8884	0.8871	466

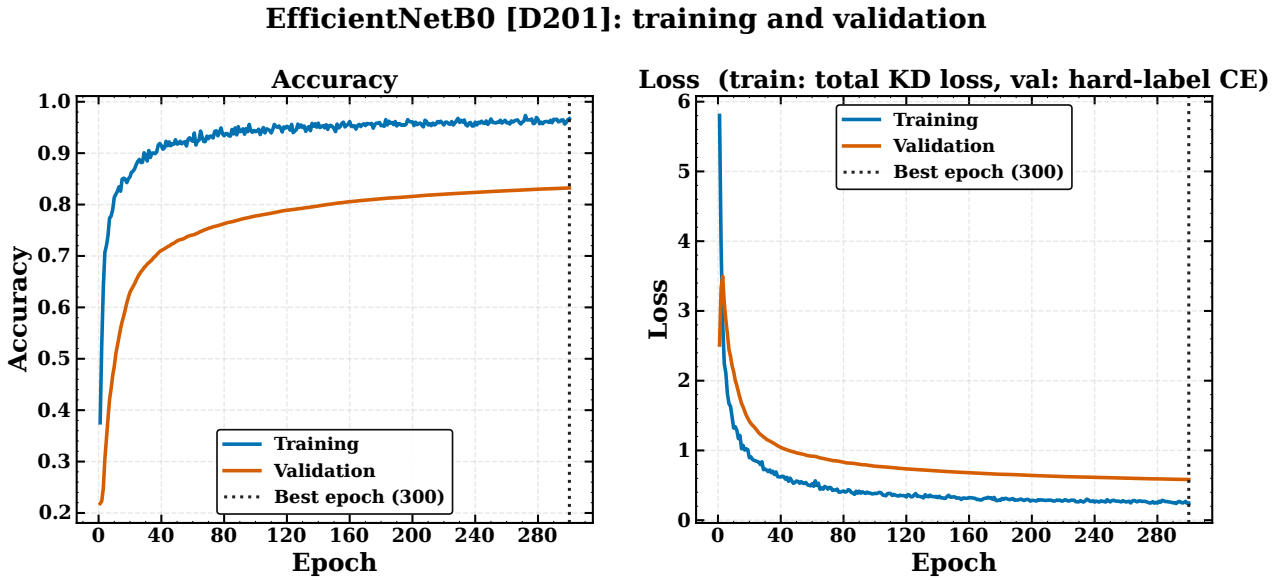
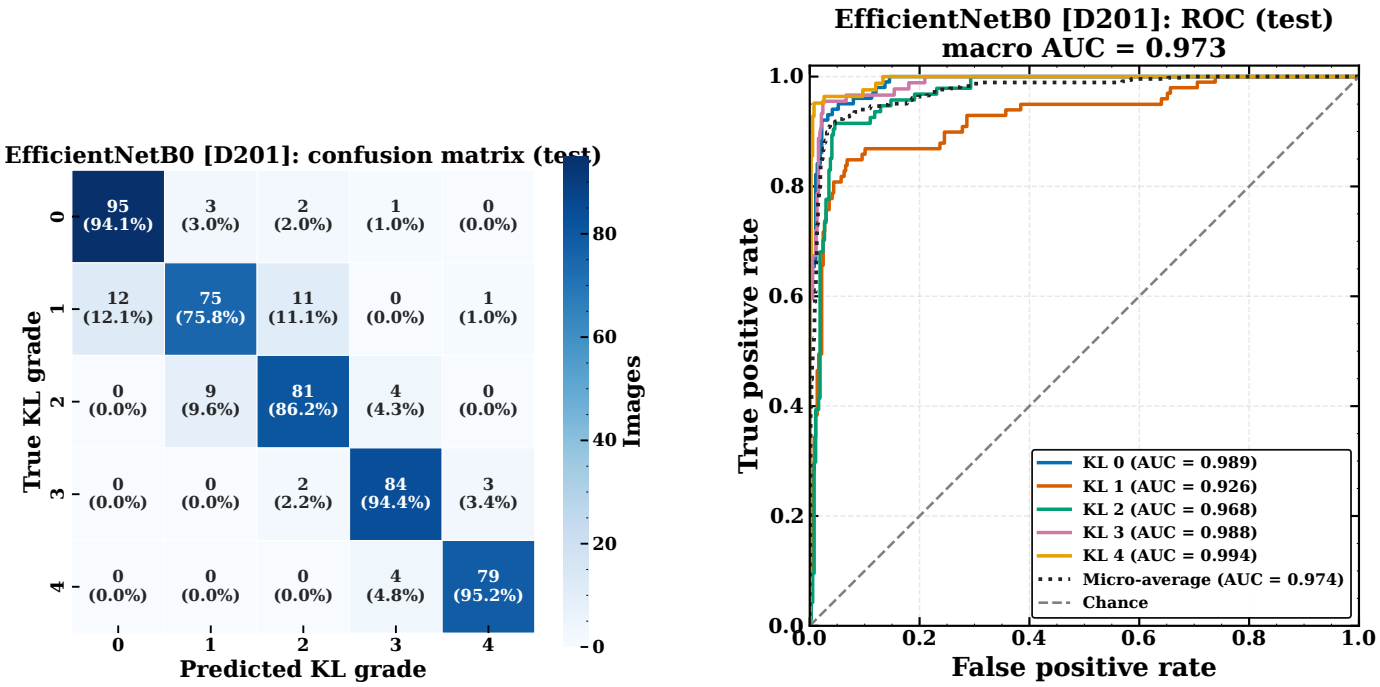


Figure 37: EfficientNetB0 [D201]: training and validation curves. Training loss is the combined objective; validation loss is hard-label cross-entropy.



(a) Test confusion matrix.

(b) One-vs-rest ROC curves.

Figure 38: EfficientNetB0 [D201]: test-set results.

9.2.3 EfficientNetB0 [ENB5]

Voided by the stopping-rule artifact of Section 8.

Run summary

Teachers	Custom-ENB5
Temperature / alpha	$T = 3, \alpha = 0.3$
Parameters / size	4.38 M / 17.5 MB
Epochs / best	11 / 1
Validation / test accuracy	0.2489 / 0.2232
Macro / weighted F1	0.1009 / 0.1068
Macro AUC	0.6701
Training time	3.1 min

Test-set classification report

Class	Precision	Recall	F1-score	Support
0	0.2222	0.9505	0.3602	101
1	0.0000	0.0000	0.0000	99
2	0.3000	0.0638	0.1053	94
3	0.1538	0.0225	0.0392	89
4	0.0000	0.0000	0.0000	83
Accuracy	–	–	0.2232	466
Macro average	0.1352	0.2074	0.1009	466
Weighted average	0.1381	0.2232	0.1068	466

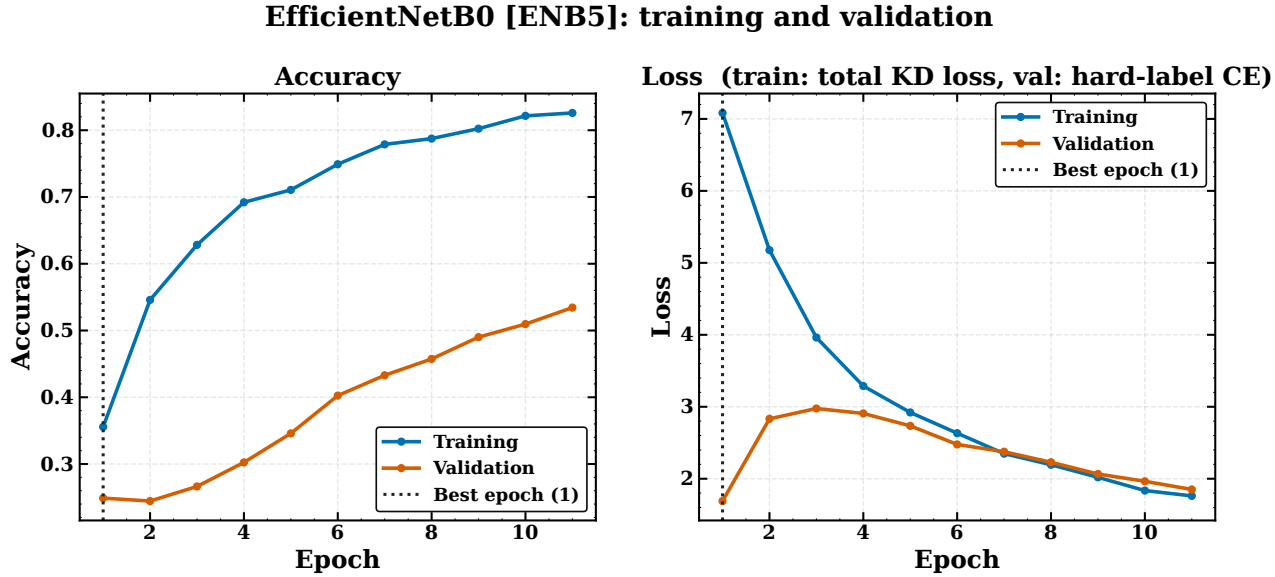
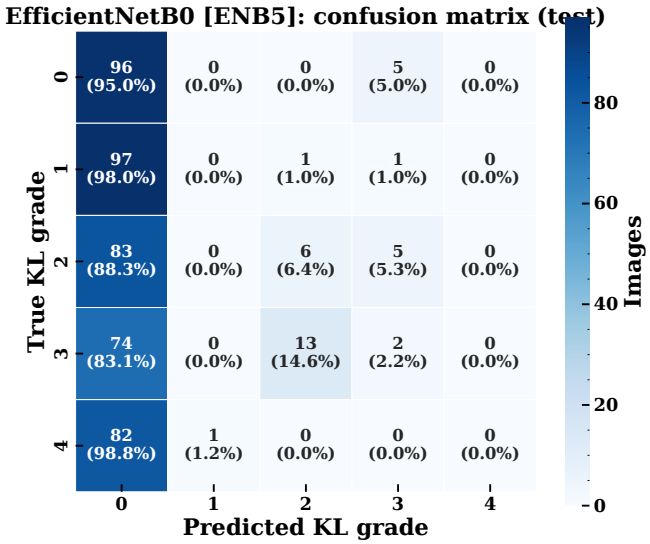
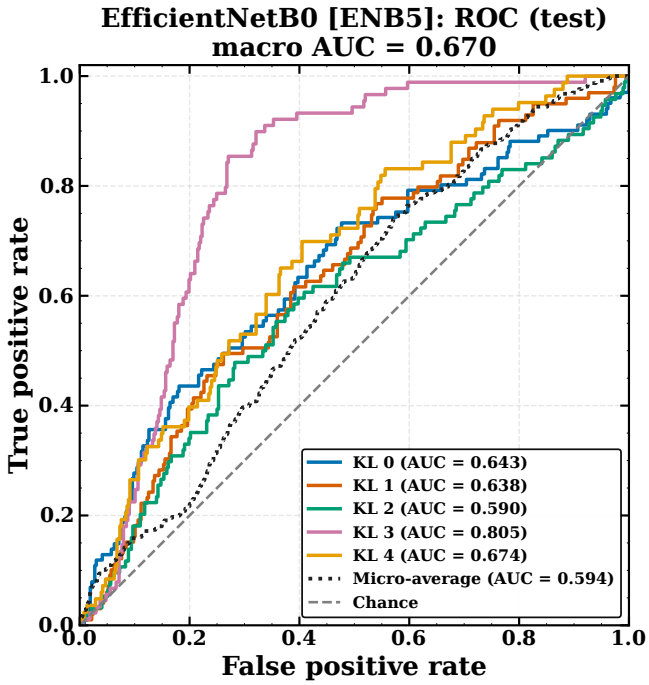


Figure 39: EfficientNetB0 [ENB5]: training and validation curves. Training loss is the combined objective; validation loss is hard-label cross-entropy.



(a) Test confusion matrix.



(b) One-vs-rest ROC curves.

Figure 40: EfficientNetB0 [ENB5]: test-set results.

9.2.4 EfficientNetB0 [ENB5+D201]

Voided by the stopping-rule artifact of Section 8.

Run summary

Teachers	Custom-ENB5, DenseNet201
Temperature / alpha	$T = 3$, $\alpha = 0.3$
Parameters / size	4.38 M / 17.5 MB
Epochs / best	11 / 1
Validation / test accuracy	0.2489 / 0.2146
Macro / weighted F1	0.0953 / 0.1009
Macro AUC	0.6278
Training time	4.3 min

Test-set classification report

Class	Precision	Recall	F1-score	Support
0	0.2138	0.9208	0.3470	101
1	0.0000	0.0000	0.0000	99
2	0.3125	0.0532	0.0909	94
3	0.1333	0.0225	0.0385	89
4	0.0000	0.0000	0.0000	83
Accuracy	–	–	0.2146	466
Macro average	0.1319	0.1993	0.0953	466
Weighted average	0.1348	0.2146	0.1009	466

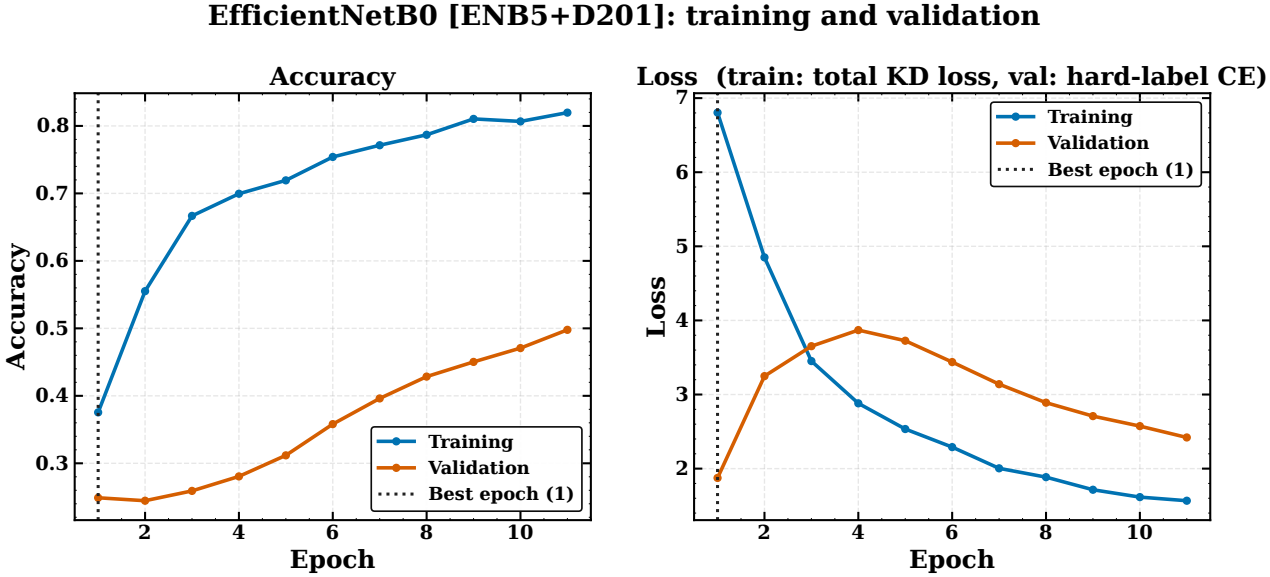
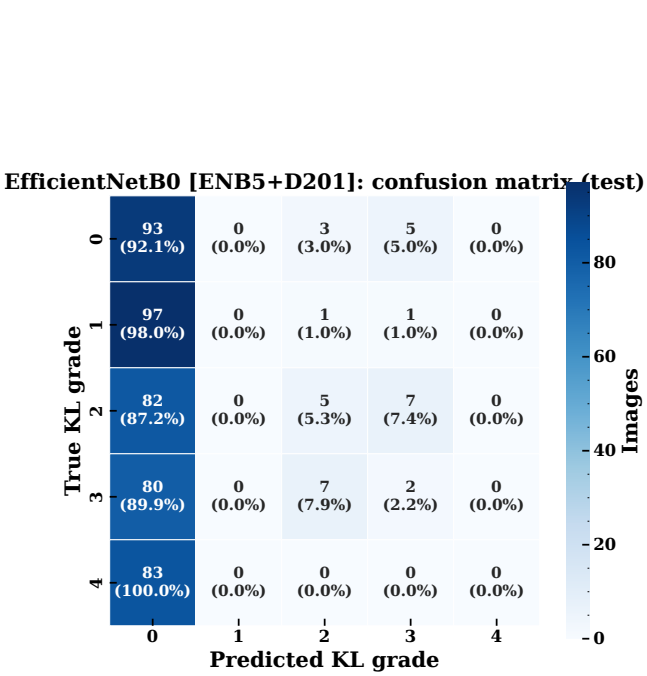
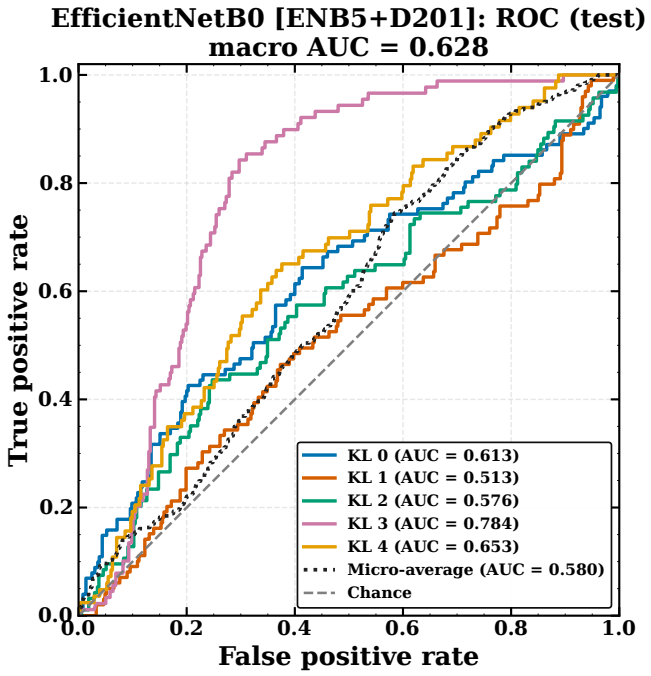


Figure 41: EfficientNetB0 [ENB5+D201]: training and validation curves. Training loss is the combined objective; validation loss is hard-label cross-entropy.



(a) Test confusion matrix.



(b) One-vs-rest ROC curves.

Figure 42: EfficientNetB0 [ENB5+D201]: test-set results.

9.2.5 EfficientNetB0 [ENB5+IRV2]

Run summary

Teachers	Custom-ENB5, Inception-ResNetV2
Temperature / alpha	$T = 3$, $\alpha = 0.3$
Parameters / size	4.38 M / 17.5 MB
Epochs / best	300 / 300
Validation / test accuracy	0.8603 / 0.8863
Macro / weighted F1	0.8898 / 0.8868
Macro AUC	0.9774
Training time	49.9 min

Test-set classification report

Class	Precision	Recall	F1-score	Support
0	0.9158	0.8614	0.8878	101
1	0.8000	0.8485	0.8235	99
2	0.8421	0.8511	0.8466	94
3	0.9213	0.9213	0.9213	89
4	0.9756	0.9639	0.9697	83
Accuracy	–	–	0.8863	466
Macro average	0.8910	0.8892	0.8898	466
Weighted average	0.8880	0.8863	0.8868	466

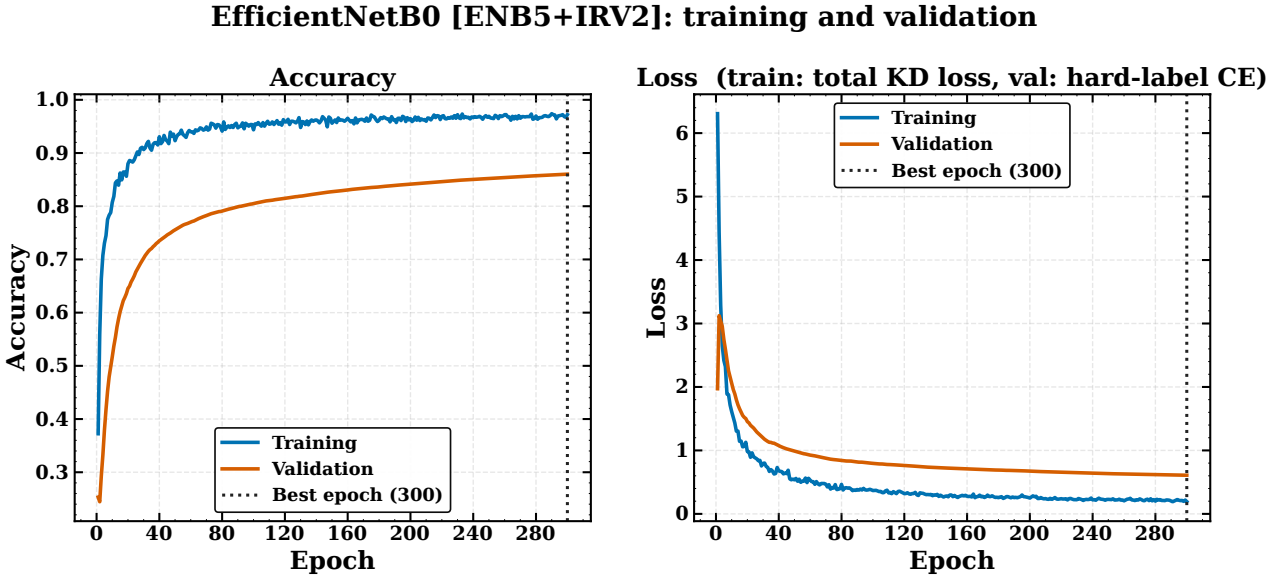
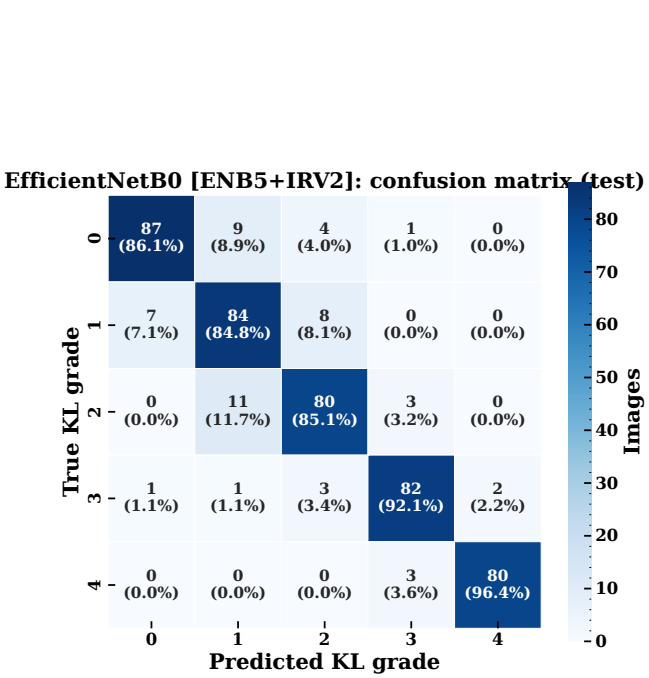
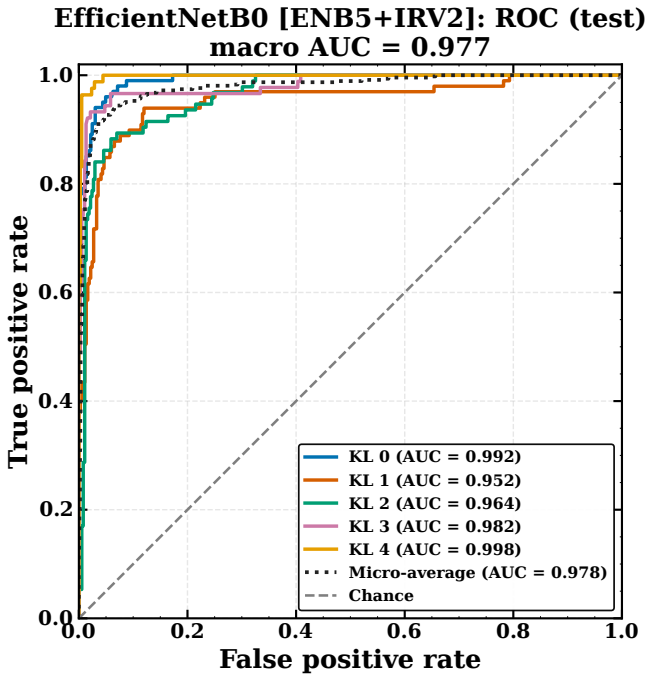


Figure 43: EfficientNetB0 [ENB5+IRV2]: training and validation curves. Training loss is the combined objective; validation loss is hard-label cross-entropy.



(a) Test confusion matrix.



(b) One-vs-rest ROC curves.

Figure 44: EfficientNetB0 [ENB5+IRV2]: test-set results.

9.2.6 EfficientNetB0 [ENB5+IRV2+D201]

Run summary

Teachers	Custom-ENB5, Inception-ResNetV2, DenseNet201
Temperature / alpha	$T = 3$, $\alpha = 0.3$
Parameters / size	4.38 M / 17.5 MB
Epochs / best	300 / 300
Validation / test accuracy	0.8601 / 0.8970
Macro / weighted F1	0.8990 / 0.8972
Macro AUC	0.9750
Training time	55.4 min

Test-set classification report

Class	Precision	Recall	F1-score	Support
0	0.9300	0.9208	0.9254	101
1	0.8469	0.8384	0.8426	99
2	0.8351	0.8617	0.8482	94
3	0.9222	0.9326	0.9274	89
4	0.9630	0.9398	0.9512	83
Accuracy	–	–	0.8970	466
Macro average	0.8994	0.8986	0.8990	466
Weighted average	0.8976	0.8970	0.8972	466

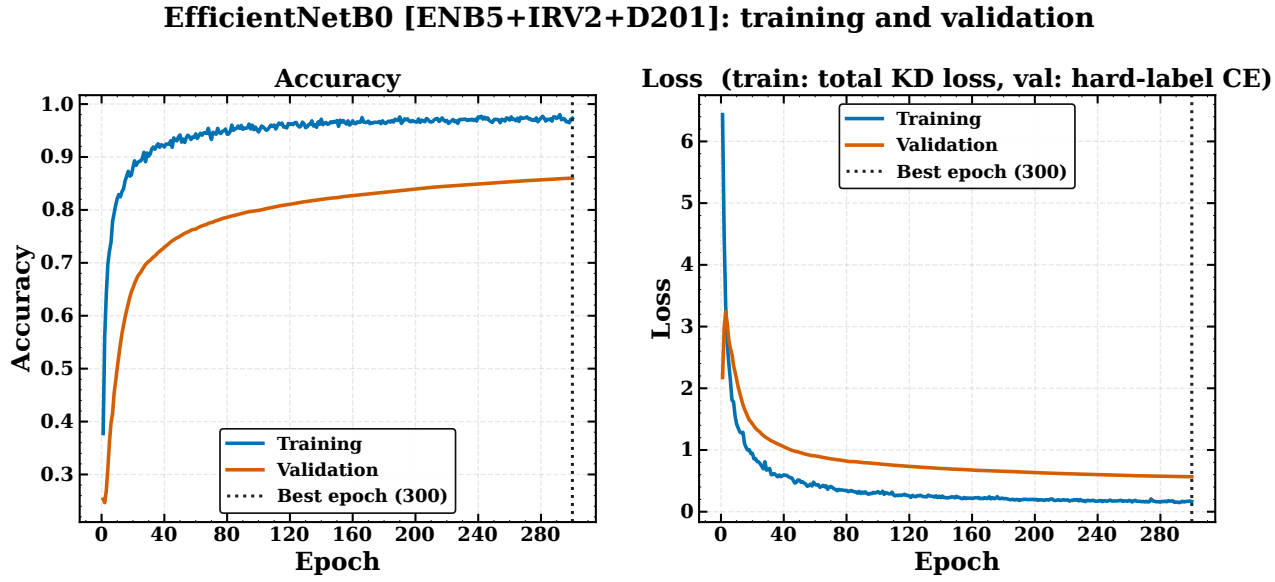
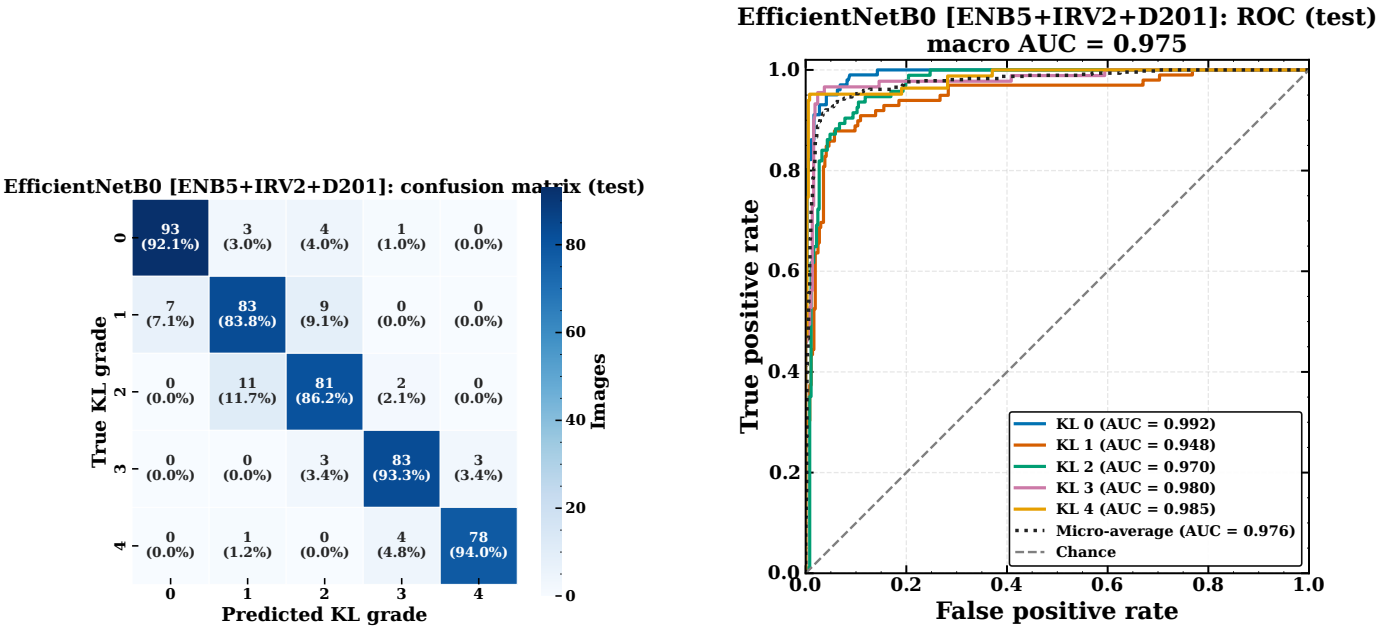


Figure 45: EfficientNetB0 [ENB5+IRV2+D201]: training and validation curves. Training loss is the combined objective; validation loss is hard-label cross-entropy.



(a) Test confusion matrix.

(b) One-vs-rest ROC curves.

Figure 46: EfficientNetB0 [ENB5+IRV2+D201]: test-set results.

9.2.7 EfficientNetB0 [ENB5+IRV2+R50]

Run summary

Teachers	Custom-ENB5, Inception-ResNetV2, ResNet50
Temperature / alpha	$T = 3$, $\alpha = 0.3$
Parameters / size	4.38 M / 17.5 MB
Epochs / best	300 / 299
Validation / test accuracy	0.8679 / 0.8798
Macro / weighted F1	0.8822 / 0.8802
Macro AUC	0.9735
Training time	53.8 min

Test-set classification report

Class	Precision	Recall	F1-score	Support
0	0.9451	0.8515	0.8958	101
1	0.7961	0.8283	0.8119	99
2	0.8469	0.8830	0.8646	94
3	0.9195	0.8989	0.9091	89
4	0.9080	0.9518	0.9294	83
Accuracy	–	–	0.8798	466
Macro average	0.8831	0.8827	0.8822	466
Weighted average	0.8822	0.8798	0.8802	466

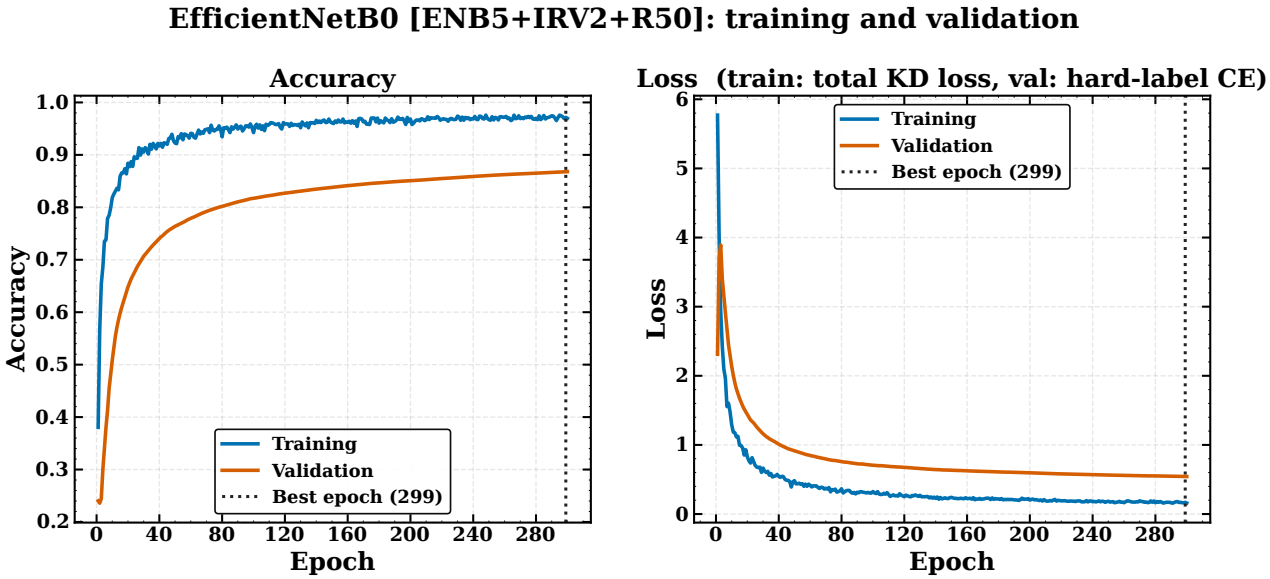
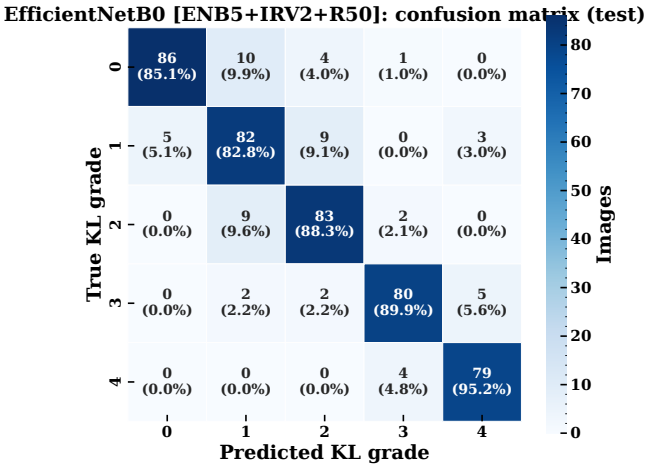
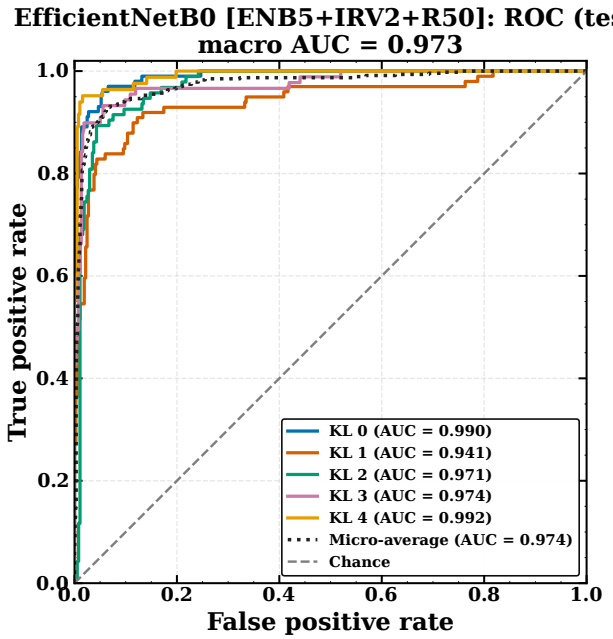


Figure 47: EfficientNetB0 [ENB5+IRV2+R50]: training and validation curves. Training loss is the combined objective; validation loss is hard-label cross-entropy.



(a) Test confusion matrix.



(b) One-vs-rest ROC curves.

Figure 48: EfficientNetB0 [ENB5+IRV2+R50]: test-set results.

9.2.8 EfficientNetB0 [ENB5+R50]

Run summary

Teachers	Custom-ENB5, ResNet50
Temperature / alpha	$T = 3$, $\alpha = 0.3$
Parameters / size	4.38 M / 17.5 MB
Epochs / best	300 / 300
Validation / test accuracy	0.8578 / 0.9013
Macro / weighted F1	0.9030 / 0.9008
Macro AUC	0.9769
Training time	54.1 min

Test-set classification report

Class	Precision	Recall	F1-score	Support
0	0.9118	0.9208	0.9163	101
1	0.8696	0.8081	0.8377	99
2	0.8485	0.8936	0.8705	94
3	0.9425	0.9213	0.9318	89
4	0.9419	0.9759	0.9586	83
Accuracy	–	–	0.9013	466
Macro average	0.9028	0.9039	0.9030	466
Weighted average	0.9013	0.9013	0.9008	466

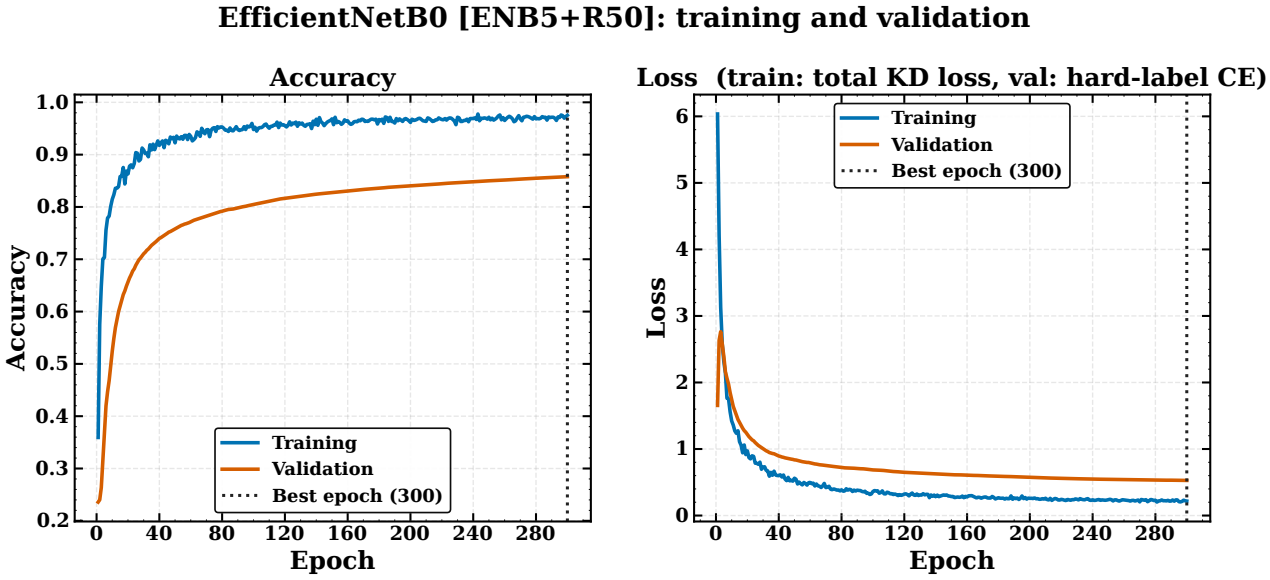
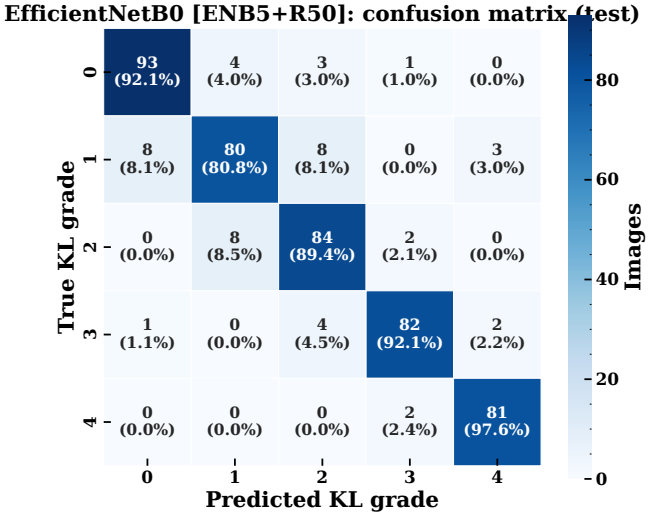
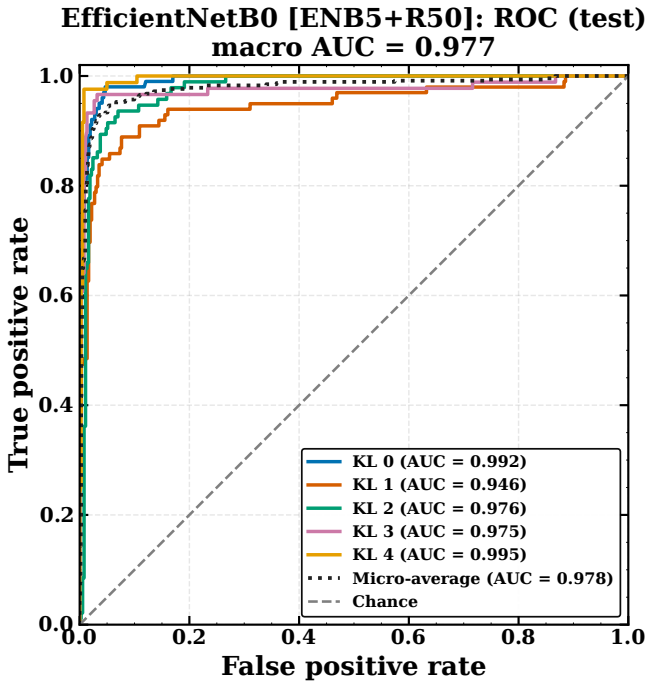


Figure 49: EfficientNetB0 [ENB5+R50]: training and validation curves. Training loss is the combined objective; validation loss is hard-label cross-entropy.



(a) Test confusion matrix.



(b) One-vs-rest ROC curves.

Figure 50: EfficientNetB0 [ENB5+R50]: test-set results.

9.2.9 EfficientNetB0 [IRV2]

Run summary

Teachers	InceptionResNetV2
Temperature / alpha	$T = 3$, $\alpha = 0.3$
Parameters / size	4.38 M / 17.5 MB
Epochs / best	300 / 300
Validation / test accuracy	0.8526 / 0.8712
Macro / weighted F1	0.8734 / 0.8718
Macro AUC	0.9765
Training time	49.3 min

Test-set classification report

Class	Precision	Recall	F1-score	Support
0	0.9247	0.8515	0.8866	101
1	0.8037	0.8687	0.8350	99
2	0.8247	0.8511	0.8377	94
3	0.8953	0.8652	0.8800	89
4	0.9277	0.9277	0.9277	83
Accuracy	–	–	0.8712	466
Macro average	0.8753	0.8728	0.8734	466
Weighted average	0.8738	0.8712	0.8718	466

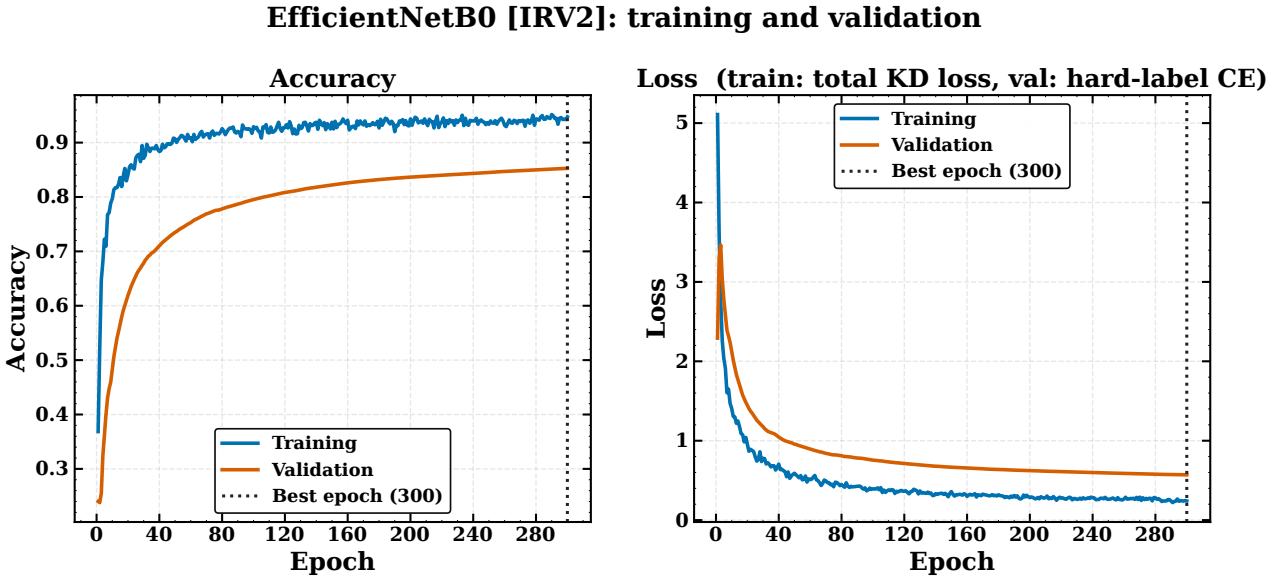
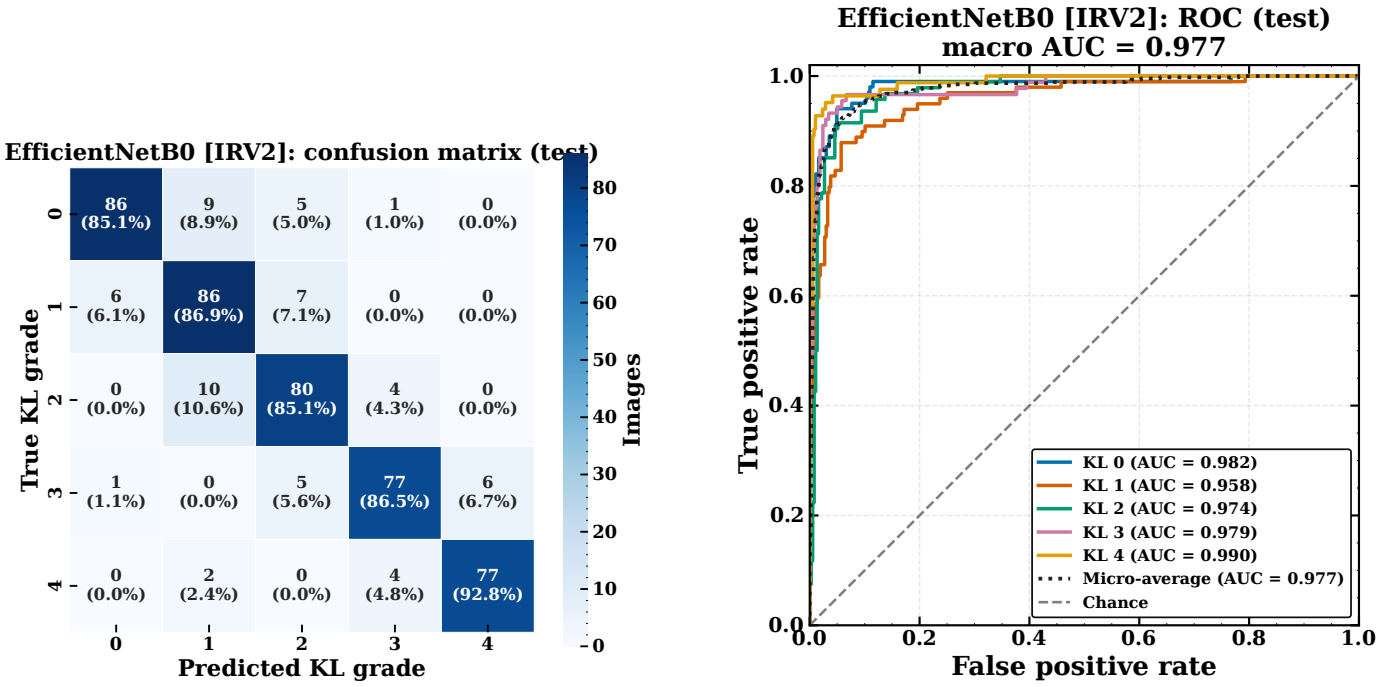


Figure 51: EfficientNetB0 [IRV2]: training and validation curves. Training loss is the combined objective; validation loss is hard-label cross-entropy.



(a) Test confusion matrix.

(b) One-vs-rest ROC curves.

Figure 52: EfficientNetB0 [IRV2]: test-set results.

9.2.10 EfficientNetB0 [IRV2+D201]

Run summary

Teachers	InceptionResNetV2, DenseNet201
Temperature / alpha	$T = 3$, $\alpha = 0.3$
Parameters / size	4.38 M / 17.5 MB
Epochs / best	300 / 300
Validation / test accuracy	0.8616 / 0.8670
Macro / weighted F1	0.8683 / 0.8654
Macro AUC	0.9743
Training time	51.0 min

Test-set classification report

Class	Precision	Recall	F1-score	Support
0	0.8319	0.9307	0.8785	101
1	0.8353	0.7172	0.7717	99
2	0.8421	0.8511	0.8466	94
3	0.8925	0.9326	0.9121	89
4	0.9500	0.9157	0.9325	83
Accuracy	–	–	0.8670	466
Macro average	0.8703	0.8694	0.8683	466
Weighted average	0.8673	0.8670	0.8654	466

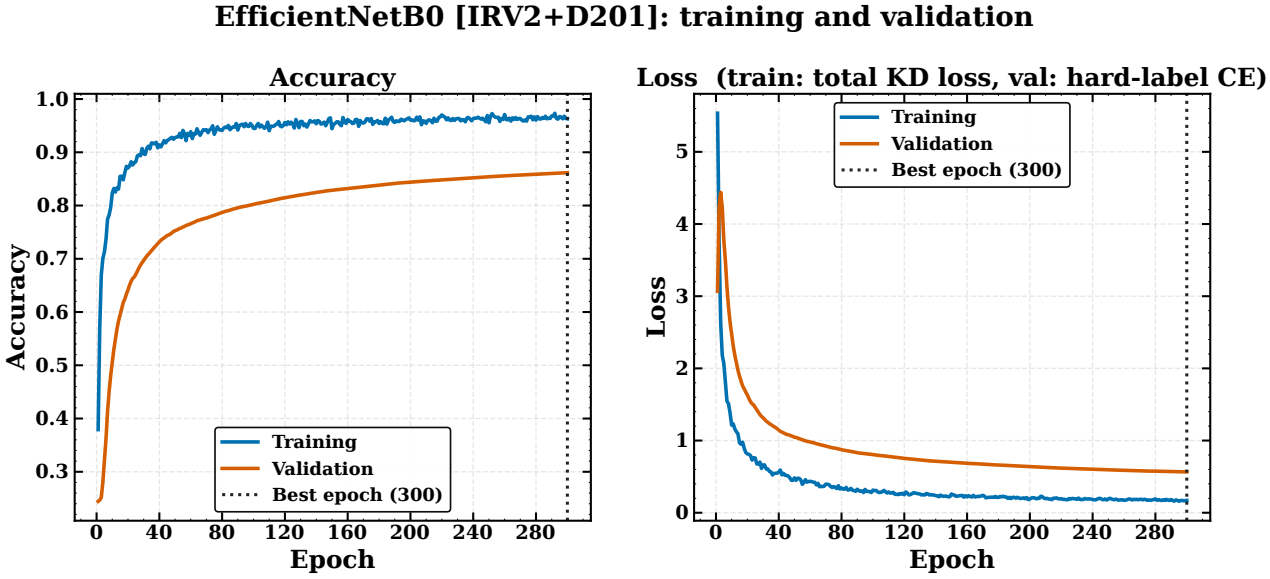
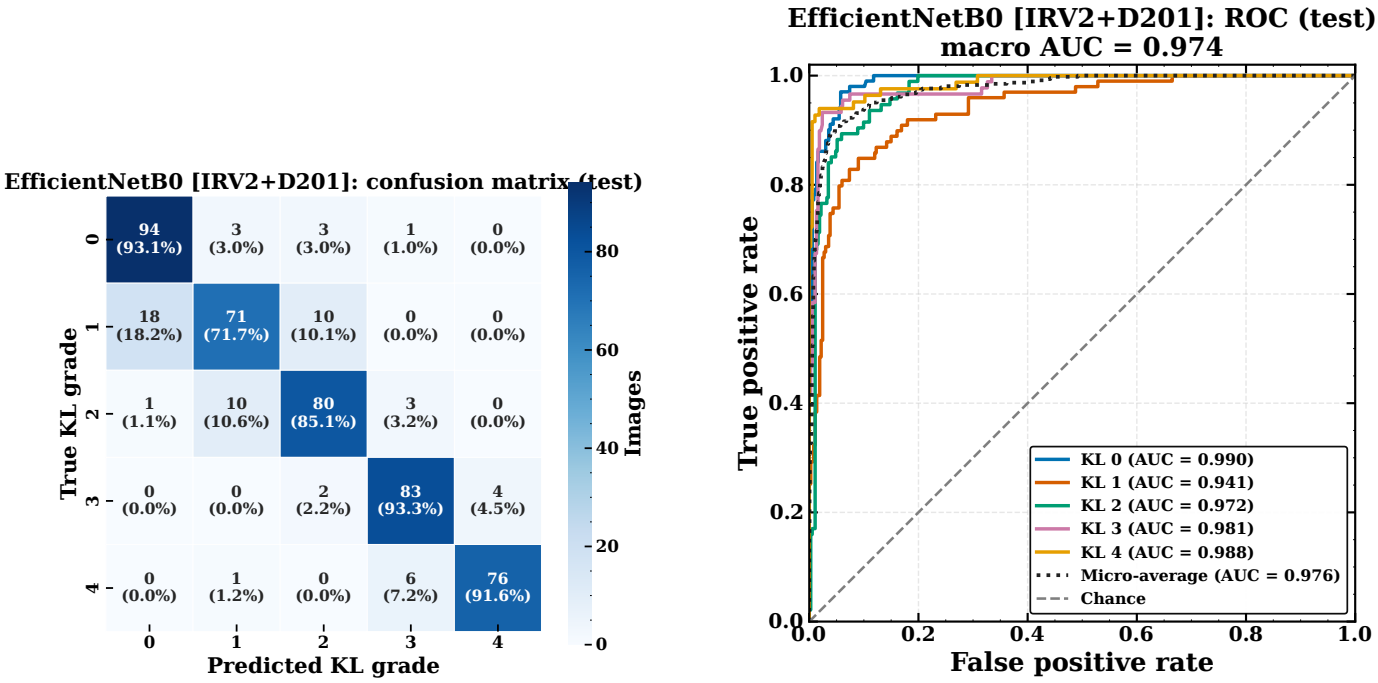


Figure 53: EfficientNetB0 [IRV2+D201]: training and validation curves. Training loss is the combined objective; validation loss is hard-label cross-entropy.



(a) Test confusion matrix.

(b) One-vs-rest ROC curves.

Figure 54: EfficientNetB0 [IRV2+D201]: test-set results.

9.2.11 EfficientNetB0 [IRV2+R50]

Run summary

Teachers	InceptionResNetV2, ResNet50
Temperature / alpha	$T = 3$, $\alpha = 0.3$
Parameters / size	4.38 M / 17.5 MB
Epochs / best	300 / 300
Validation / test accuracy	0.8518 / 0.8820
Macro / weighted F1	0.8844 / 0.8827
Macro AUC	0.9766
Training time	53.8 min

Test-set classification report

Class	Precision	Recall	F1-score	Support
0	0.9375	0.8911	0.9137	101
1	0.8454	0.8283	0.8367	99
2	0.7850	0.8936	0.8358	94
3	0.9157	0.8539	0.8837	89
4	0.9518	0.9518	0.9518	83
Accuracy	–	–	0.8820	466
Macro average	0.8871	0.8837	0.8844	466
Weighted average	0.8856	0.8820	0.8827	466

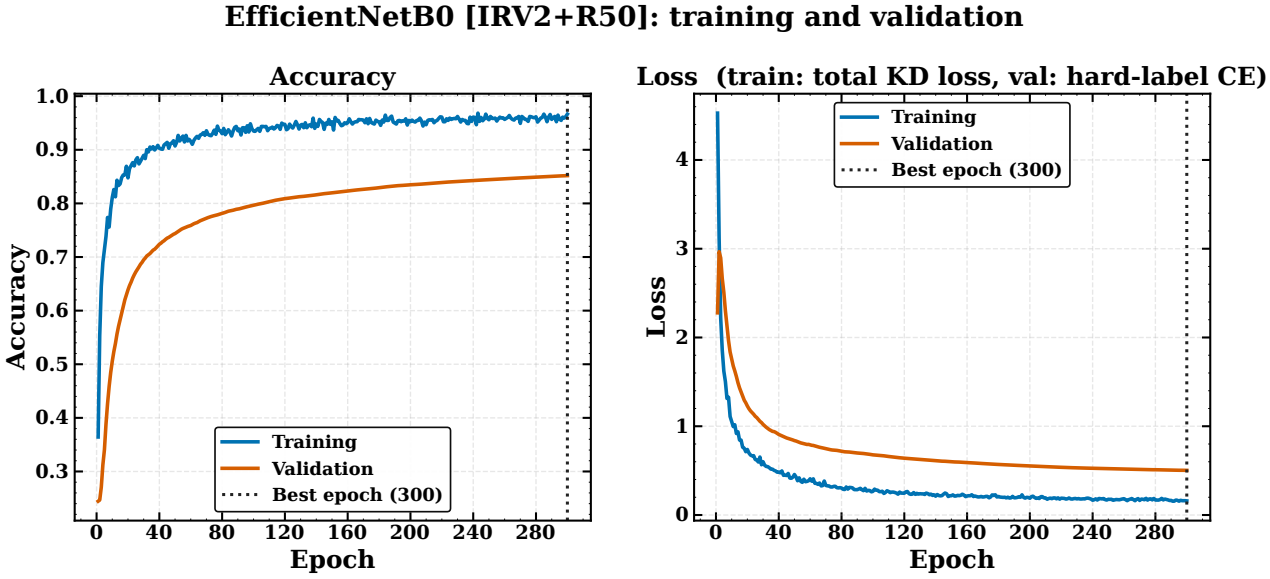
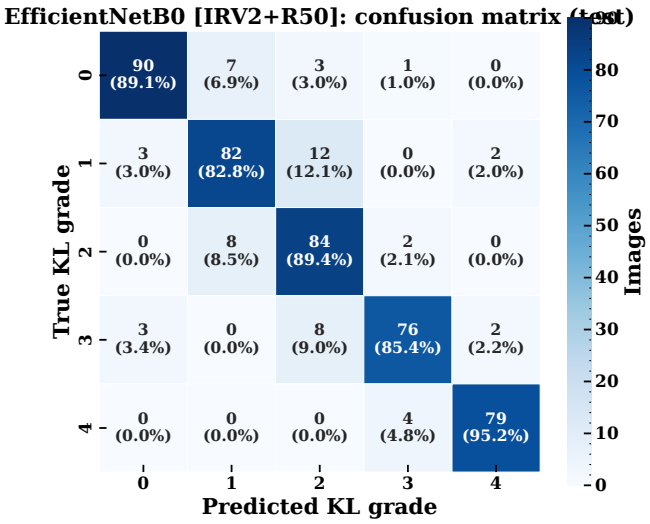
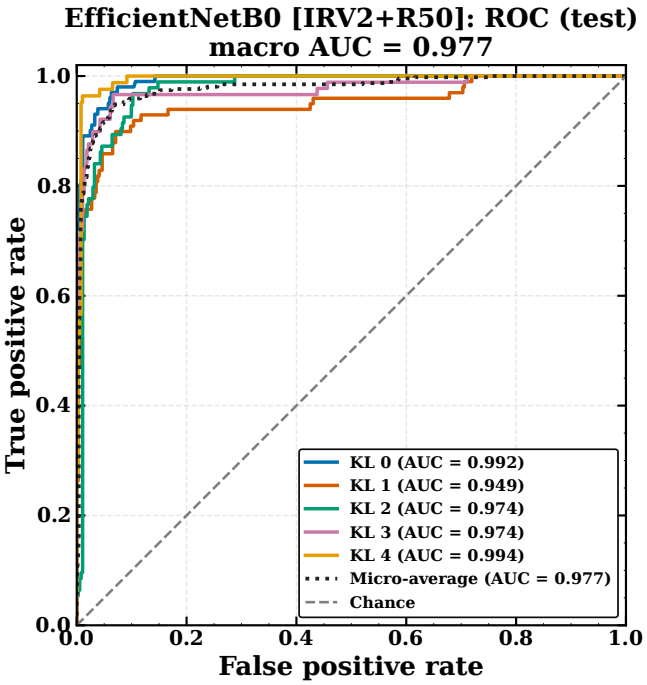


Figure 55: EfficientNetB0 [IRV2+R50]: training and validation curves. Training loss is the combined objective; validation loss is hard-label cross-entropy.



(a) Test confusion matrix.



(b) One-vs-rest ROC curves.

Figure 56: EfficientNetB0 [IRV2+R50]: test-set results.

9.2.12 EfficientNetB0 [R50]

Run summary

Teachers	ResNet50
Temperature / alpha	$T = 3$, $\alpha = 0.3$
Parameters / size	4.38 M / 17.5 MB
Epochs / best	300 / 300
Validation / test accuracy	0.8618 / 0.8691
Macro / weighted F1	0.8733 / 0.8703
Macro AUC	0.9722
Training time	52.8 min

Test-set classification report

Class	Precision	Recall	F1-score	Support
0	0.9263	0.8713	0.8980	101
1	0.7593	0.8283	0.7923	99
2	0.8000	0.8085	0.8042	94
3	0.9302	0.8989	0.9143	89
4	0.9634	0.9518	0.9576	83
Accuracy	–	–	0.8691	466
Macro average	0.8758	0.8718	0.8733	466
Weighted average	0.8727	0.8691	0.8703	466

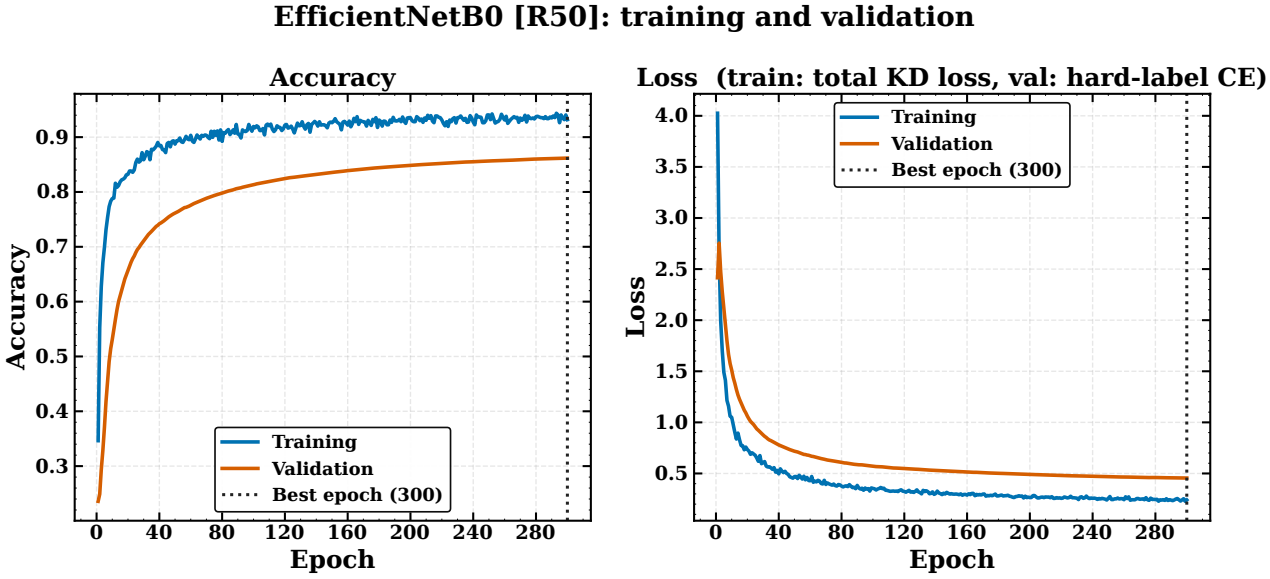
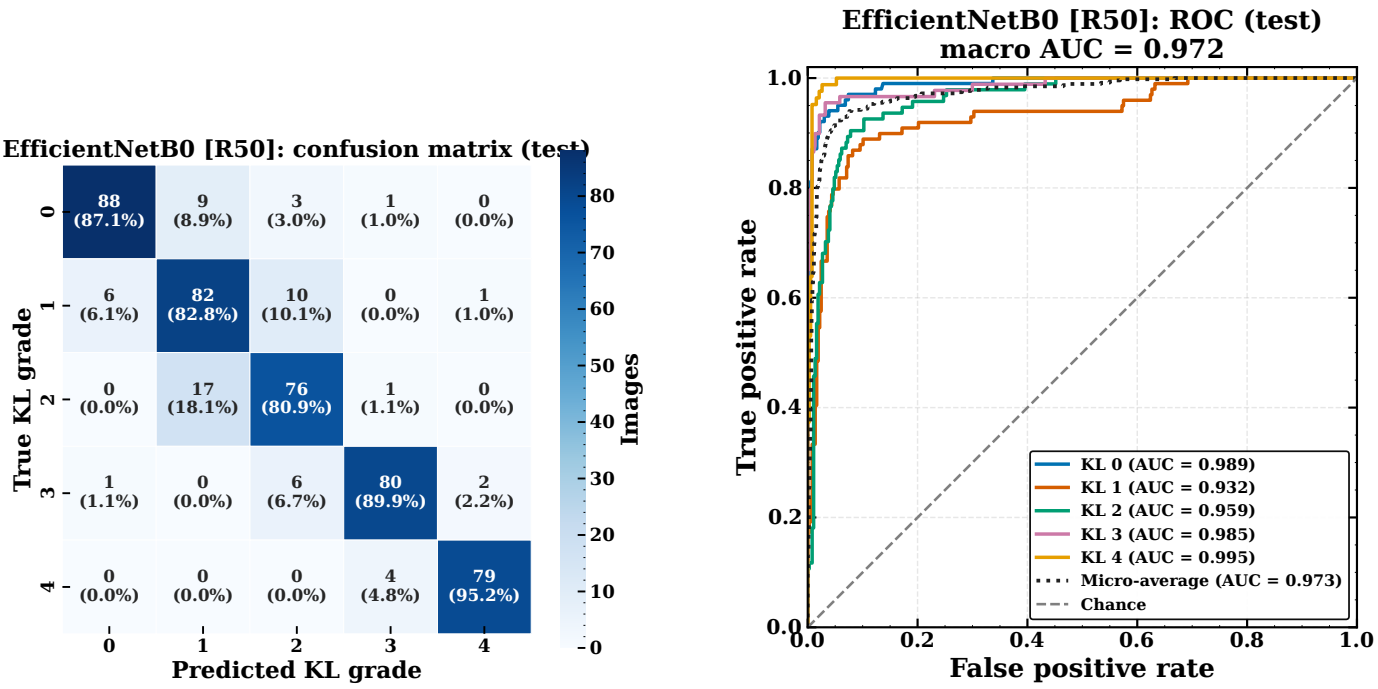


Figure 57: EfficientNetB0 [R50]: training and validation curves. Training loss is the combined objective; validation loss is hard-label cross-entropy.



(a) Test confusion matrix.

(b) One-vs-rest ROC curves.

Figure 58: EfficientNetB0 [R50]: test-set results.

9.3 EfficientNetB2

9.3.1 EfficientNetB2 [none]

Run summary

Teachers	none (control)
Temperature / alpha	–
Parameters / size	8.13 M / 32.2 MB
Epochs / best	84 / 74
Validation / test accuracy	0.8250 / 0.8541
Macro / weighted F1	0.8584 / 0.8530
Macro AUC	0.9676
Training time	15.6 min

Test-set classification report

Class	Precision	Recall	F1-score	Support
0	0.7603	0.9109	0.8288	101
1	0.7674	0.6667	0.7135	99
2	0.8587	0.8404	0.8495	94
3	0.9639	0.8989	0.9302	89
4	0.9643	0.9759	0.9701	83
Accuracy	–	–	0.8541	466
Macro average	0.8629	0.8586	0.8584	466
Weighted average	0.8569	0.8541	0.8530	466

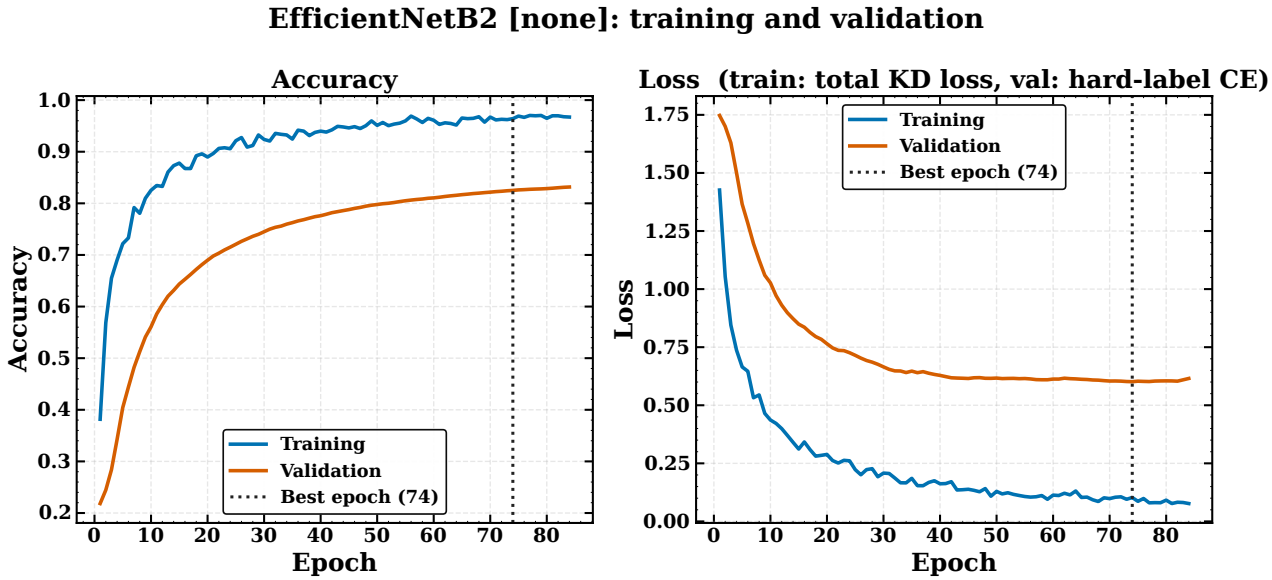
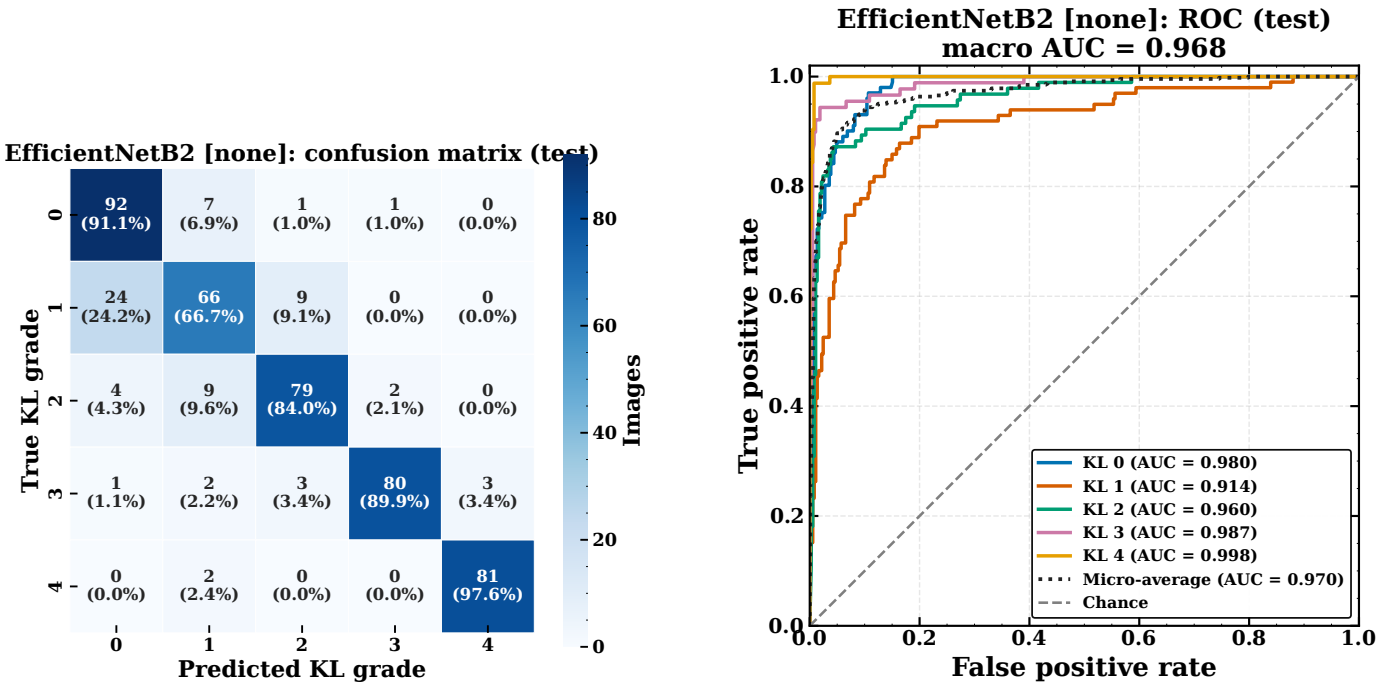


Figure 59: EfficientNetB2 [none]: training and validation curves. Training loss is the combined objective; validation loss is hard-label cross-entropy.



(a) Test confusion matrix.

(b) One-vs-rest ROC curves.

Figure 60: EfficientNetB2 [none]: test-set results.

9.3.2 EfficientNetB2 [D201]

Run summary

Teachers	DenseNet201
Temperature / alpha	$T = 3, \alpha = 0.3$
Parameters / size	8.13 M / 32.2 MB
Epochs / best	300 / 300
Validation / test accuracy	0.8758 / 0.8906
Macro / weighted F1	0.8926 / 0.8893
Macro AUC	0.9756
Training time	52.8 min

Test-set classification report

Class	Precision	Recall	F1-score	Support
0	0.8774	0.9208	0.8986	101
1	0.8409	0.7475	0.7914	99
2	0.8469	0.8830	0.8646	94
3	0.9432	0.9326	0.9379	89
4	0.9535	0.9880	0.9704	83
Accuracy	–	–	0.8906	466
Macro average	0.8924	0.8944	0.8926	466
Weighted average	0.8896	0.8906	0.8893	466

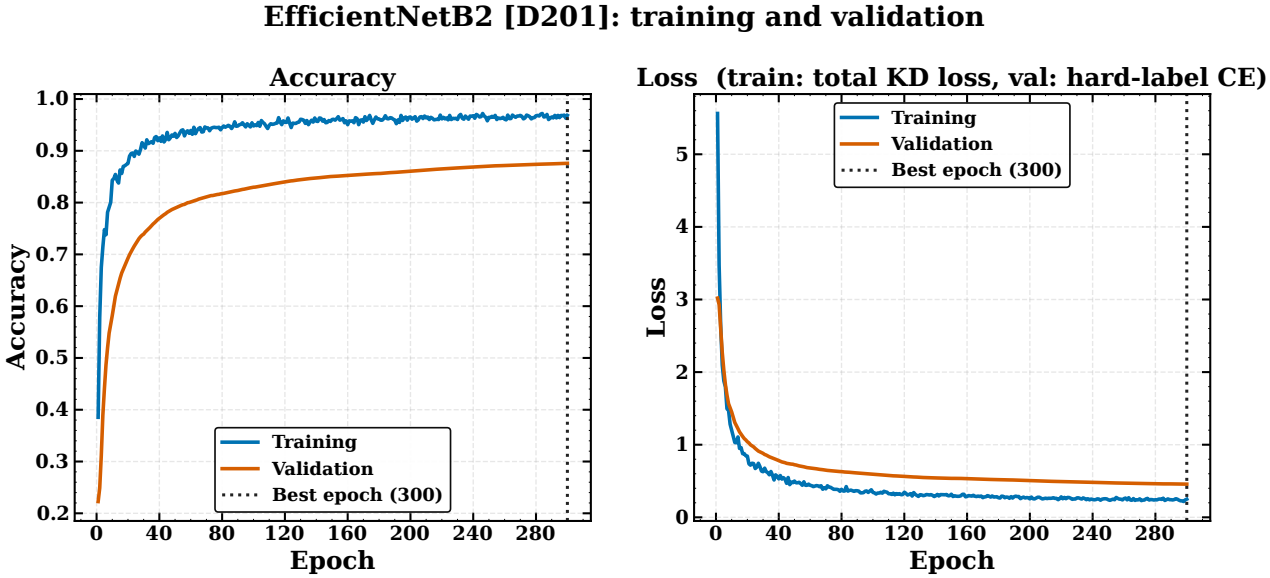
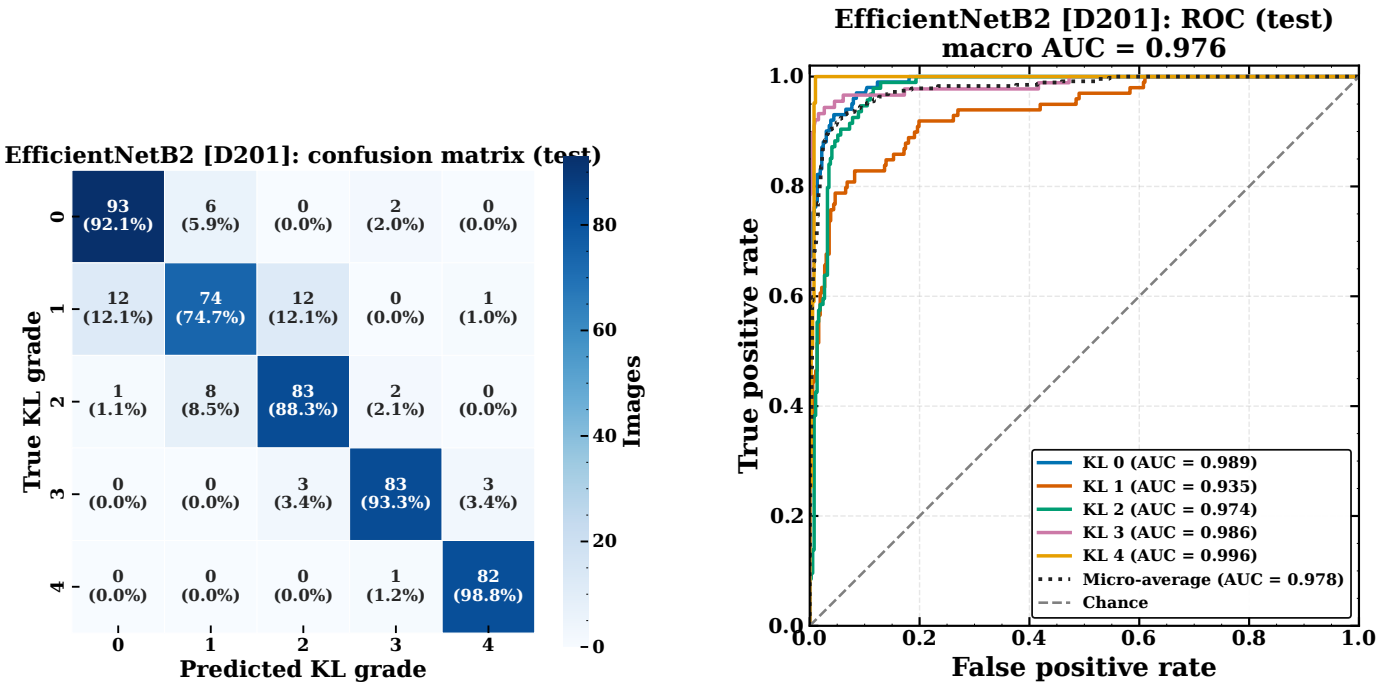


Figure 61: EfficientNetB2 [D201]: training and validation curves. Training loss is the combined objective; validation loss is hard-label cross-entropy.



(a) Test confusion matrix.

(b) One-vs-rest ROC curves.

Figure 62: EfficientNetB2 [D201]: test-set results.

9.3.3 EfficientNetB2 [ENB5]

Run summary

Teachers	Custom-ENB5
Temperature / alpha	$T = 3, \alpha = 0.3$
Parameters / size	8.13 M / 32.2 MB
Epochs / best	300 / 300
Validation / test accuracy	0.8797 / 0.8884
Macro / weighted F1	0.8917 / 0.8882
Macro AUC	0.9737
Training time	52.7 min

Test-set classification report

Class	Precision	Recall	F1-score	Support
0	0.8667	0.9010	0.8835	101
1	0.8021	0.7778	0.7897	99
2	0.8737	0.8830	0.8783	94
3	0.9647	0.9213	0.9425	89
4	0.9529	0.9759	0.9643	83
Accuracy	–	–	0.8884	466
Macro average	0.8920	0.8918	0.8917	466
Weighted average	0.8885	0.8884	0.8882	466

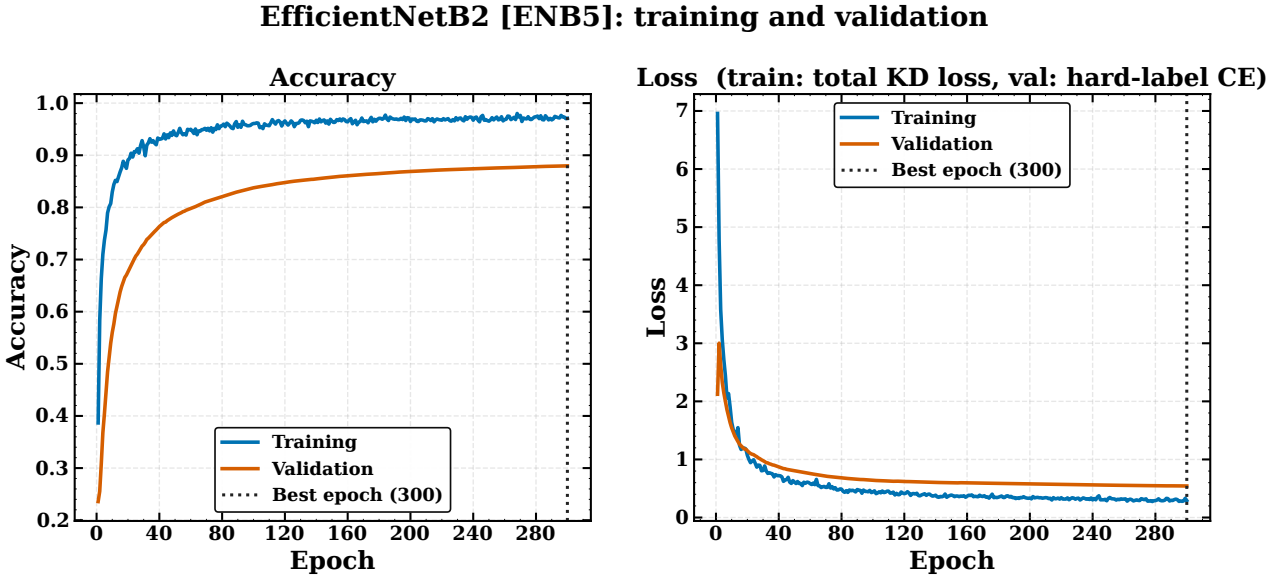
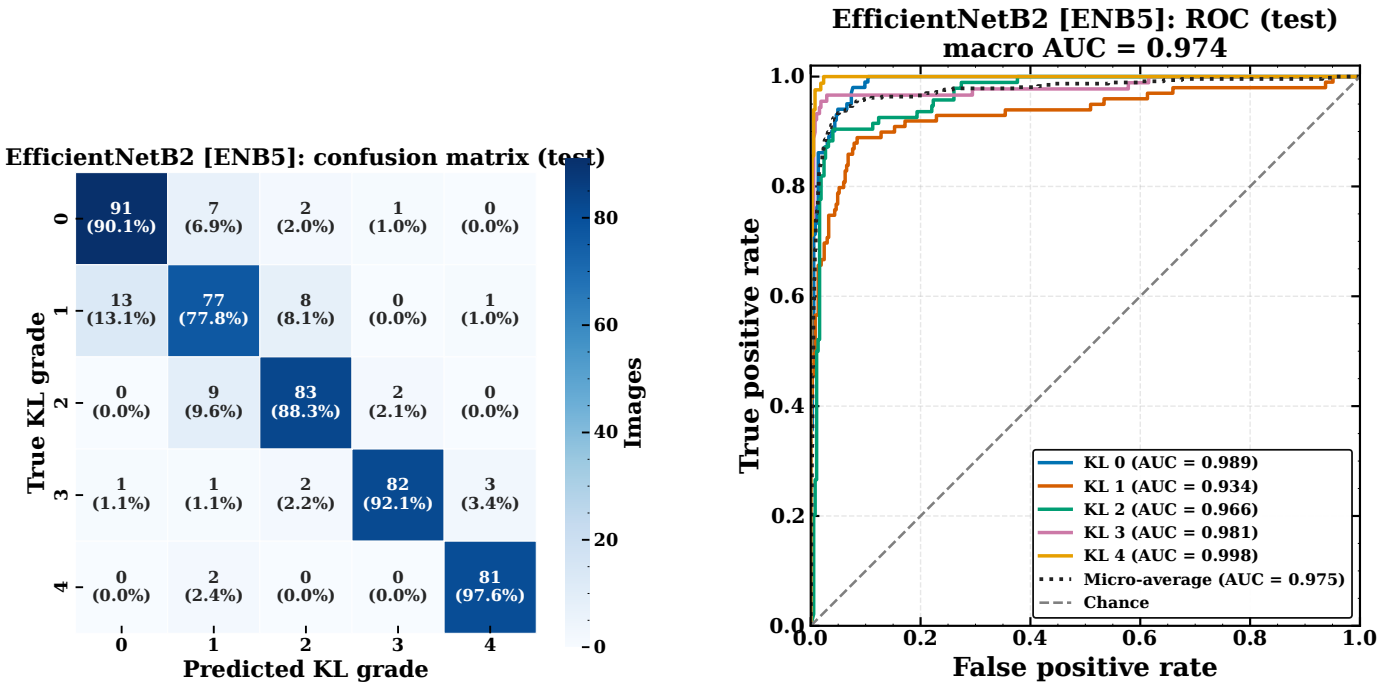


Figure 63: EfficientNetB2 [ENB5]: training and validation curves. Training loss is the combined objective; validation loss is hard-label cross-entropy.



(a) Test confusion matrix.

(b) One-vs-rest ROC curves.

Figure 64: EfficientNetB2 [ENB5]: test-set results.

9.3.4 EfficientNetB2 [ENB5+D201]

Run summary

Teachers	Custom-ENB5, DenseNet201
Temperature / alpha	$T = 3$, $\alpha = 0.3$
Parameters / size	8.13 M / 32.2 MB
Epochs / best	286 / 276
Validation / test accuracy	0.8763 / 0.8884
Macro / weighted F1	0.8901 / 0.8876
Macro AUC	0.9756
Training time	50.8 min

Test-set classification report

Class	Precision	Recall	F1-score	Support
0	0.8846	0.9109	0.8976	101
1	0.8539	0.7677	0.8085	99
2	0.8400	0.8936	0.8660	94
3	0.9326	0.9326	0.9326	89
4	0.9405	0.9518	0.9461	83
Accuracy	–	–	0.8884	466
Macro average	0.8903	0.8913	0.8901	466
Weighted average	0.8882	0.8884	0.8876	466

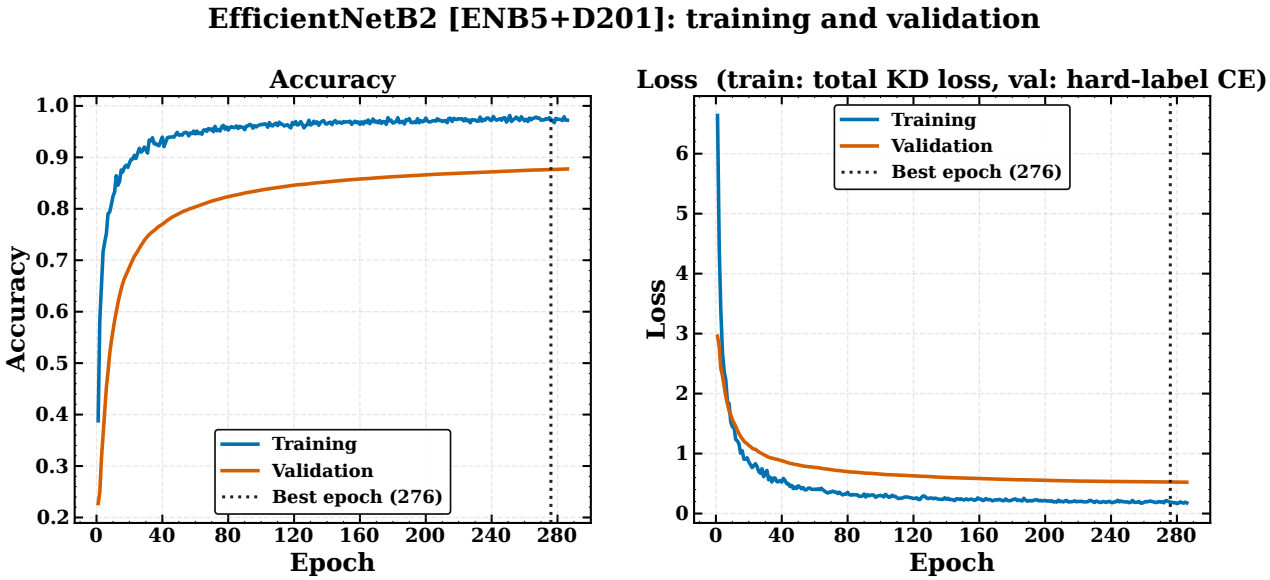
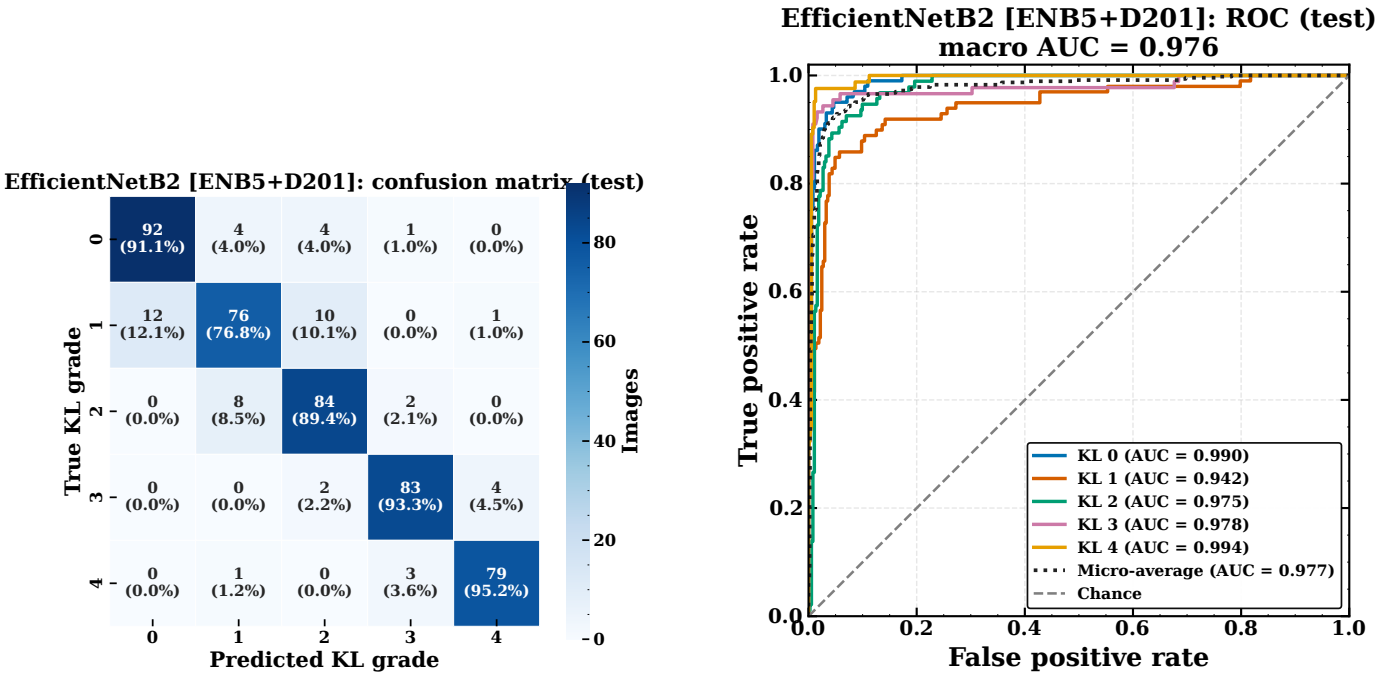


Figure 65: EfficientNetB2 [ENB5+D201]: training and validation curves. Training loss is the combined objective; validation loss is hard-label cross-entropy.



(a) Test confusion matrix.

(b) One-vs-rest ROC curves.

Figure 66: EfficientNetB2 [ENB5+D201]: test-set results.

9.3.5 EfficientNetB2 [ENB5+IRV2]

Run summary

Teachers	Custom-ENB5, Inception-ResNetV2
Temperature / alpha	$T = 3$, $\alpha = 0.3$
Parameters / size	8.13 M / 32.2 MB
Epochs / best	300 / 298
Validation / test accuracy	0.8751 / 0.8798
Macro / weighted F1	0.8817 / 0.8799
Macro AUC	0.9752
Training time	52.8 min

Test-set classification report

Class	Precision	Recall	F1-score	Support
0	0.9010	0.9010	0.9010	101
1	0.8081	0.8081	0.8081	99
2	0.8526	0.8617	0.8571	94
3	0.9121	0.9326	0.9222	89
4	0.9375	0.9036	0.9202	83
Accuracy	–	–	0.8798	466
Macro average	0.8823	0.8814	0.8817	466
Weighted average	0.8801	0.8798	0.8799	466

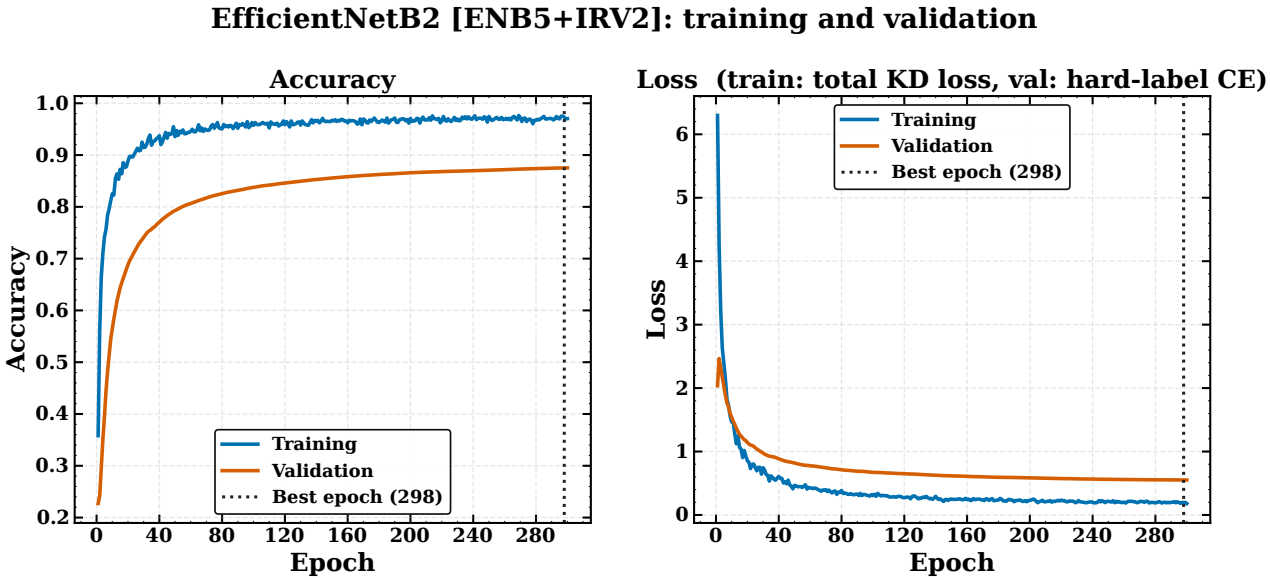
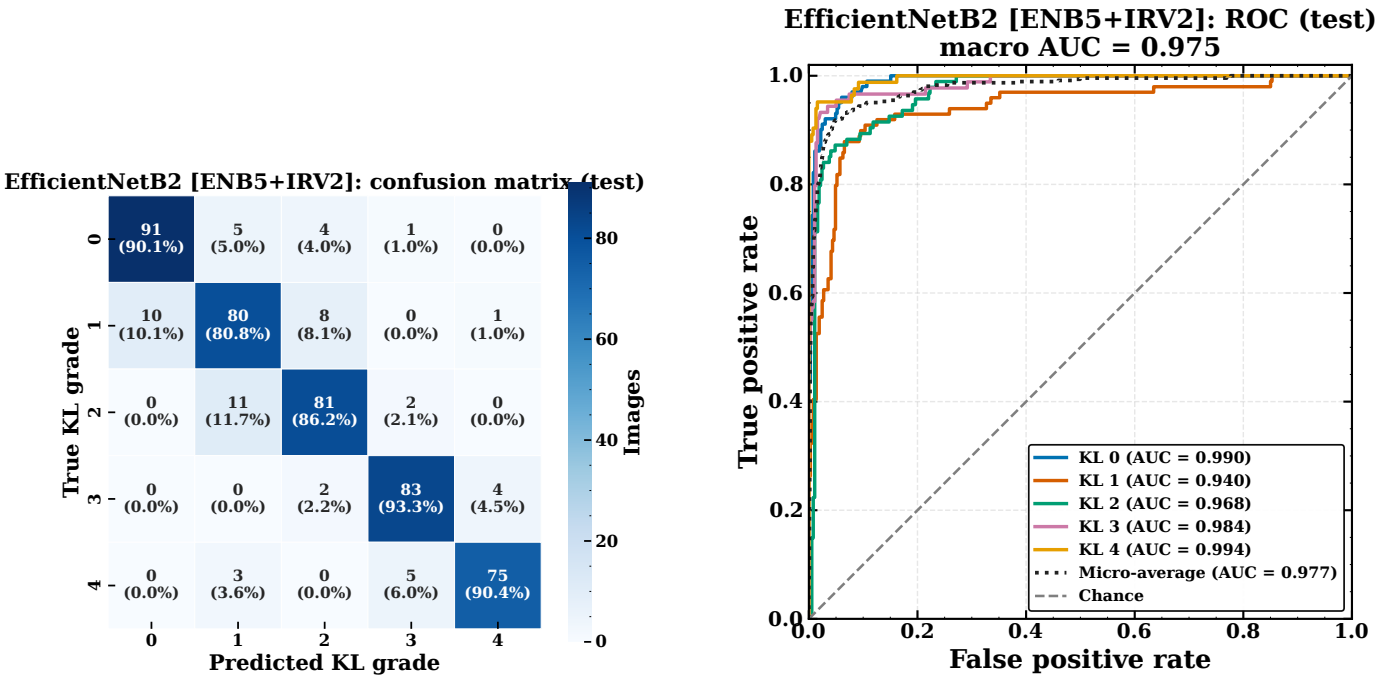


Figure 67: EfficientNetB2 [ENB5+IRV2]: training and validation curves. Training loss is the combined objective; validation loss is hard-label cross-entropy.



(a) Test confusion matrix.

(b) One-vs-rest ROC curves.

Figure 68: EfficientNetB2 [ENB5+IRV2]: test-set results.

9.3.6 EfficientNetB2 [ENB5+IRV2+D201]

Run summary

Teachers	Custom-ENB5, Inception-ResNetV2, DenseNet201
Temperature / alpha	$T = 3$, $\alpha = 0.3$
Parameters / size	8.13 M / 32.2 MB
Epochs / best	300 / 300
Validation / test accuracy	0.8808 / 0.8734
Macro / weighted F1	0.8754 / 0.8723
Macro AUC	0.9742
Training time	54.0 min

Test-set classification report

Class	Precision	Recall	F1-score	Support
0	0.8598	0.9109	0.8846	101
1	0.8202	0.7374	0.7766	99
2	0.8351	0.8617	0.8482	94
3	0.9121	0.9326	0.9222	89
4	0.9512	0.9398	0.9455	83
Accuracy	–	–	0.8734	466
Macro average	0.8757	0.8765	0.8754	466
Weighted average	0.8727	0.8734	0.8723	466

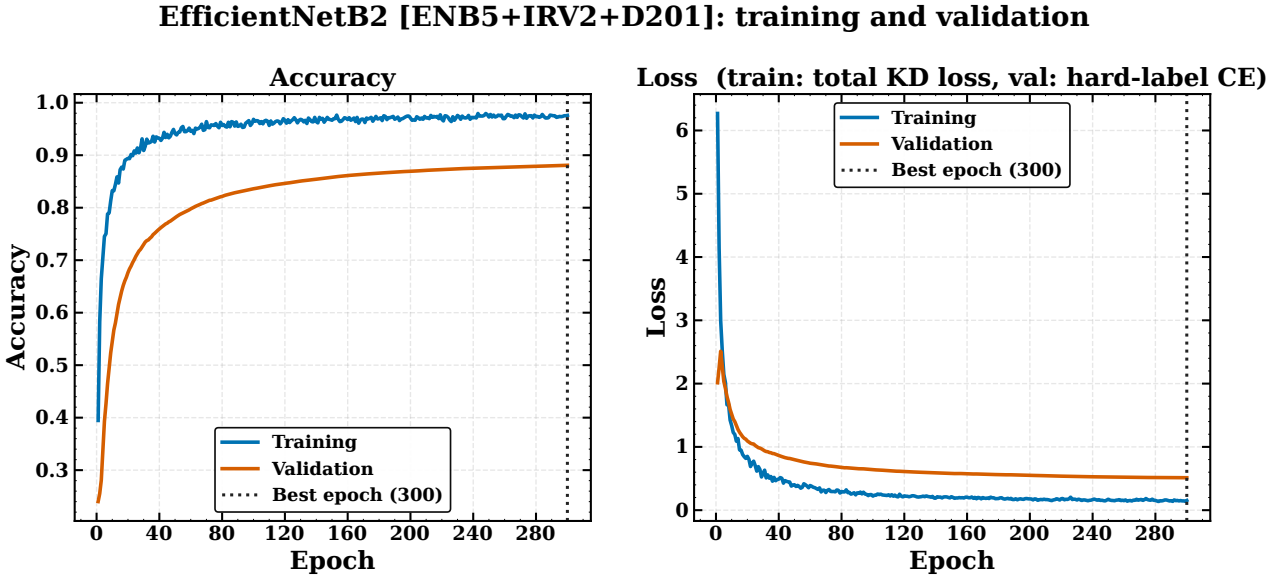
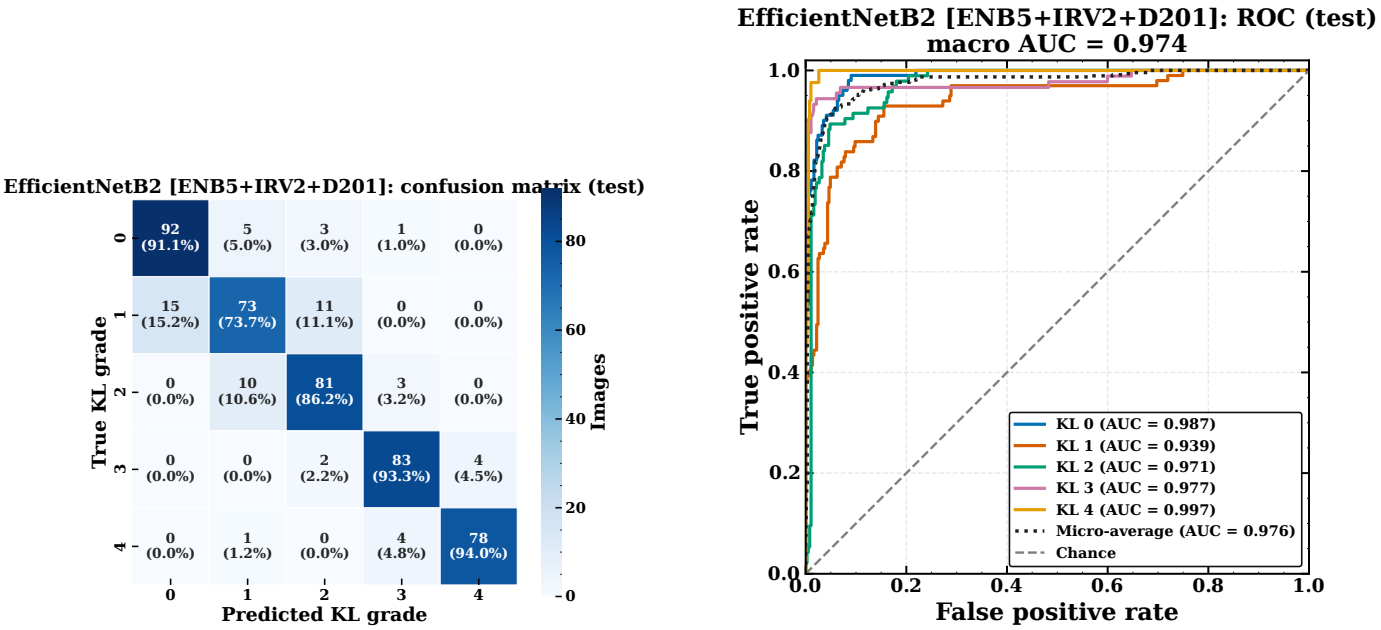


Figure 69: EfficientNetB2 [ENB5+IRV2+D201]: training and validation curves. Training loss is the combined objective; validation loss is hard-label cross-entropy.



(a) Test confusion matrix.

(b) One-vs-rest ROC curves.

Figure 70: EfficientNetB2 [ENB5+IRV2+D201]: test-set results.

9.3.7 EfficientNetB2 [ENB5+IRV2+R50]

Run summary

Teachers	Custom-ENB5, Inception-ResNetV2, ResNet50
Temperature / alpha	$T = 3$, $\alpha = 0.3$
Parameters / size	8.13 M / 32.2 MB
Epochs / best	300 / 295
Validation / test accuracy	0.8730 / 0.8906
Macro / weighted F1	0.8936 / 0.8908
Macro AUC	0.9763
Training time	52.0 min

Test-set classification report

Class	Precision	Recall	F1-score	Support
0	0.9167	0.8713	0.8934	101
1	0.8039	0.8283	0.8159	99
2	0.8542	0.8723	0.8632	94
3	0.9535	0.9213	0.9371	89
4	0.9419	0.9759	0.9586	83
Accuracy	–	–	0.8906	466
Macro average	0.8940	0.8938	0.8936	466
Weighted average	0.8916	0.8906	0.8908	466

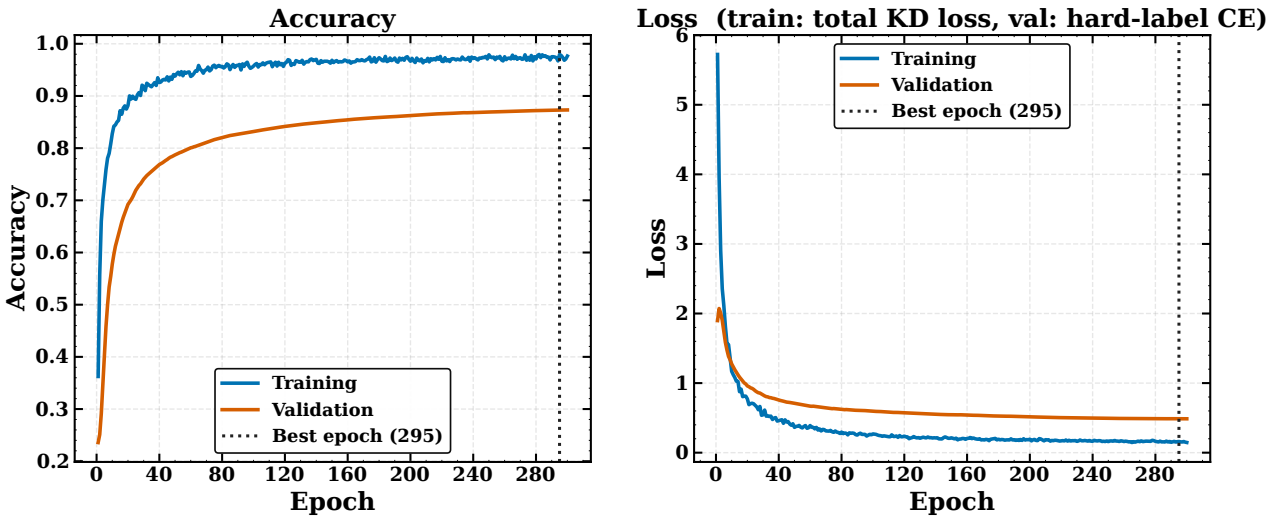
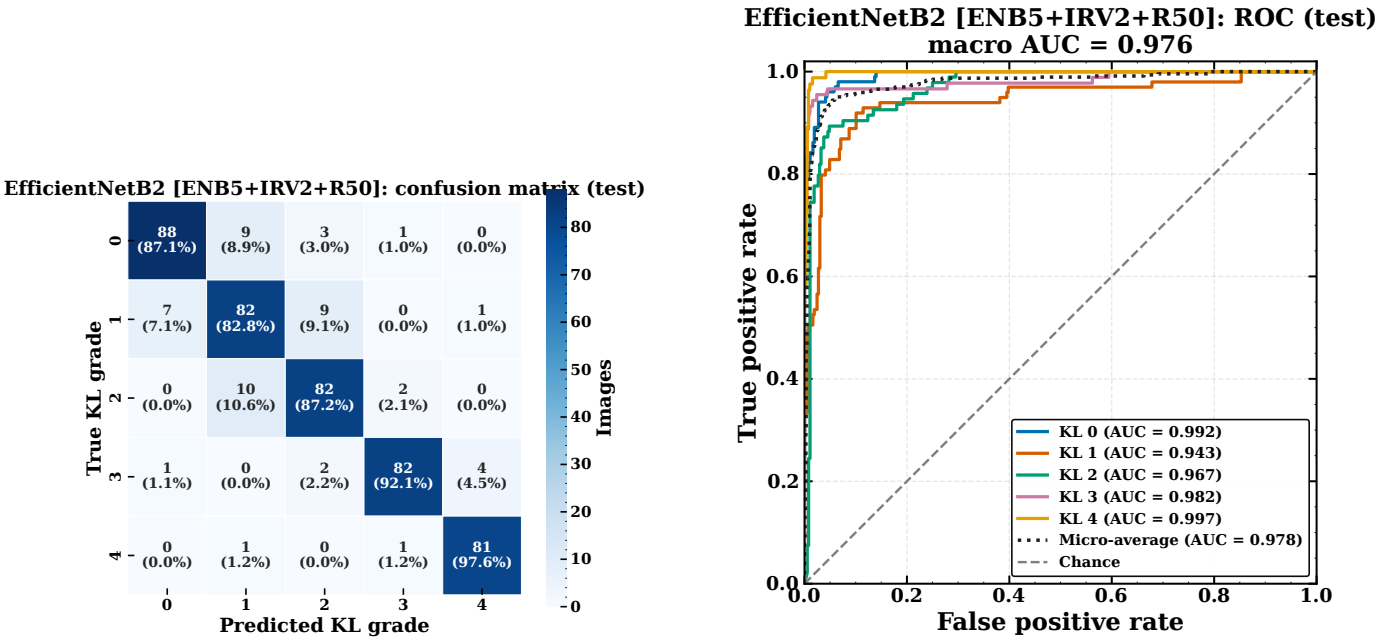


Figure 71: EfficientNetB2 [ENB5+IRV2+R50]: training and validation curves. Training loss is the combined objective; validation loss is hard-label cross-entropy.



(a) Test confusion matrix.

(b) One-vs-rest ROC curves.

Figure 72: EfficientNetB2 [ENB5+IRV2+R50]: test-set results.

9.3.8 EfficientNetB2 [ENB5+R50]

Run summary

Teachers	Custom-ENB5, ResNet50
Temperature / alpha	$T = 3$, $\alpha = 0.3$
Parameters / size	8.13 M / 32.2 MB
Epochs / best	300 / 300
Validation / test accuracy	0.8831 / 0.8820
Macro / weighted F1	0.8839 / 0.8803
Macro AUC	0.9747
Training time	51.1 min

Test-set classification report

Class	Precision	Recall	F1-score	Support
0	0.8440	0.9109	0.8762	101
1	0.8452	0.7172	0.7760	99
2	0.8416	0.9043	0.8718	94
3	0.9535	0.9213	0.9371	89
4	0.9419	0.9759	0.9586	83
Accuracy	–	–	0.8820	466
Macro average	0.8852	0.8859	0.8839	466
Weighted average	0.8821	0.8820	0.8803	466

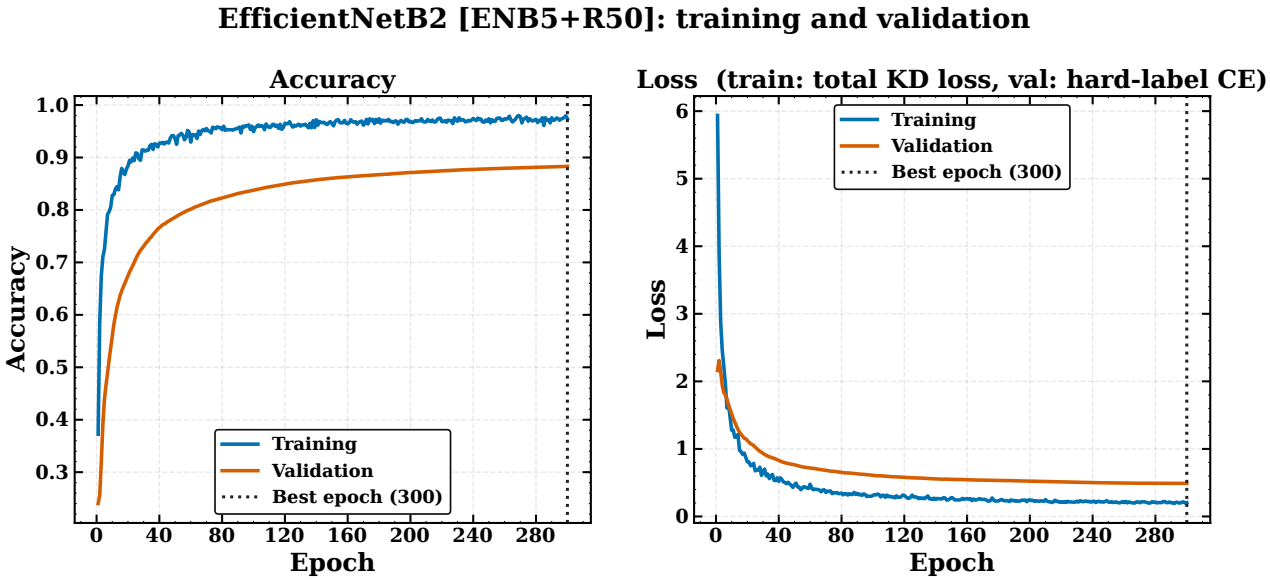
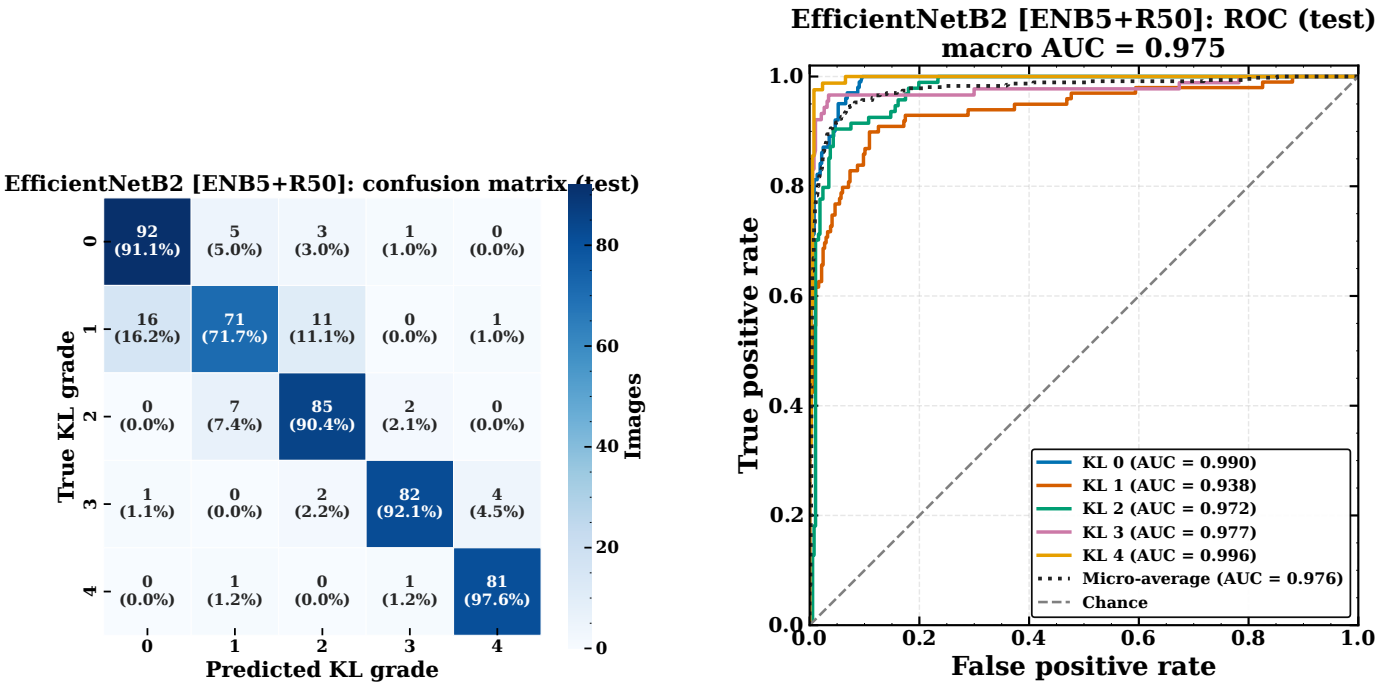


Figure 73: EfficientNetB2 [ENB5+R50]: training and validation curves. Training loss is the combined objective; validation loss is hard-label cross-entropy.



(a) Test confusion matrix.

(b) One-vs-rest ROC curves.

Figure 74: EfficientNetB2 [ENB5+R50]: test-set results.

9.3.9 EfficientNetB2 [IRV2]

Run summary

Teachers	InceptionResNetV2
Temperature / alpha	$T = 3$, $\alpha = 0.3$
Parameters / size	8.13 M / 32.2 MB
Epochs / best	286 / 276
Validation / test accuracy	0.8731 / 0.8820
Macro / weighted F1	0.8839 / 0.8816
Macro AUC	0.9743
Training time	50.4 min

Test-set classification report

Class	Precision	Recall	F1-score	Support
0	0.9082	0.8812	0.8945	101
1	0.8404	0.7980	0.8187	99
2	0.8283	0.8723	0.8497	94
3	0.9205	0.9101	0.9153	89
4	0.9195	0.9639	0.9412	83
Accuracy	–	–	0.8820	466
Macro average	0.8834	0.8851	0.8839	466
Weighted average	0.8820	0.8820	0.8816	466

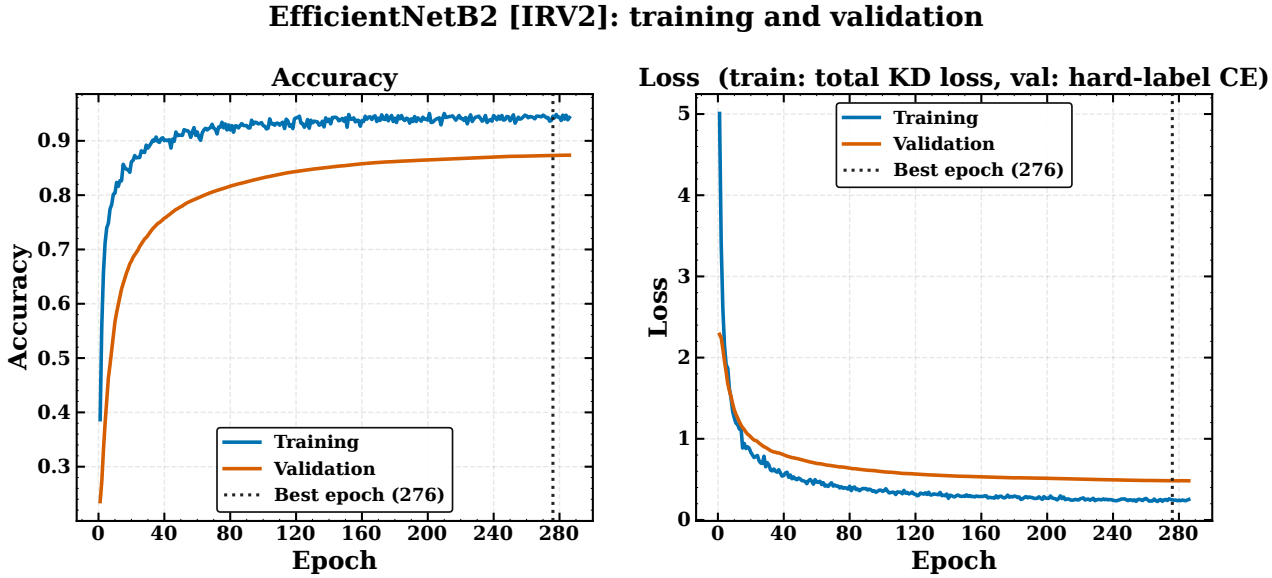
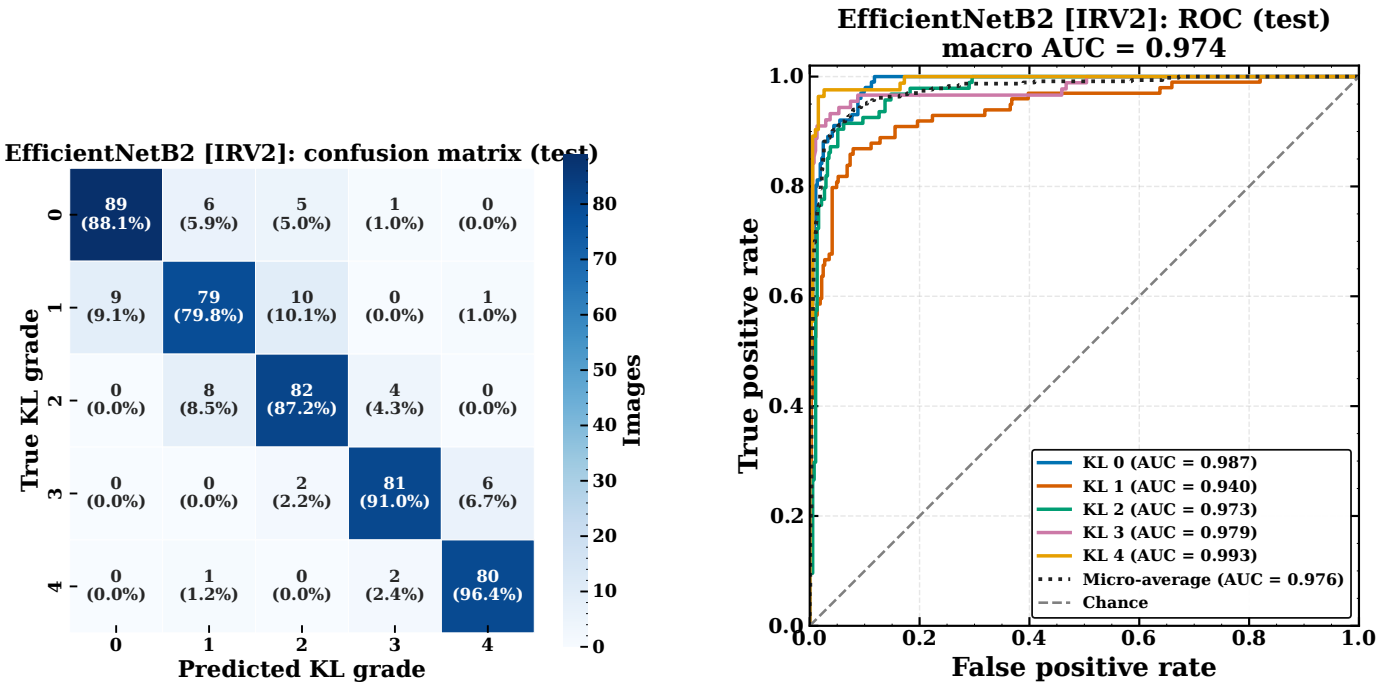


Figure 75: EfficientNetB2 [IRV2]: training and validation curves. Training loss is the combined objective; validation loss is hard-label cross-entropy.



(a) Test confusion matrix.

(b) One-vs-rest ROC curves.

Figure 76: EfficientNetB2 [IRV2]: test-set results.

9.3.10 EfficientNetB2 [IRV2+D201]

Run summary

Teachers	InceptionResNetV2, DenseNet201
Temperature / alpha	$T = 3$, $\alpha = 0.3$
Parameters / size	8.13 M / 32.2 MB
Epochs / best	300 / 299
Validation / test accuracy	0.8800 / 0.8798
Macro / weighted F1	0.8821 / 0.8800
Macro AUC	0.9781
Training time	52.4 min

Test-set classification report

Class	Precision	Recall	F1-score	Support
0	0.9167	0.8713	0.8934	101
1	0.8020	0.8182	0.8100	99
2	0.8469	0.8830	0.8646	94
3	0.9625	0.8652	0.9112	89
4	0.8901	0.9759	0.9310	83
Accuracy	–	–	0.8798	466
Macro average	0.8836	0.8827	0.8821	466
Weighted average	0.8823	0.8798	0.8800	466

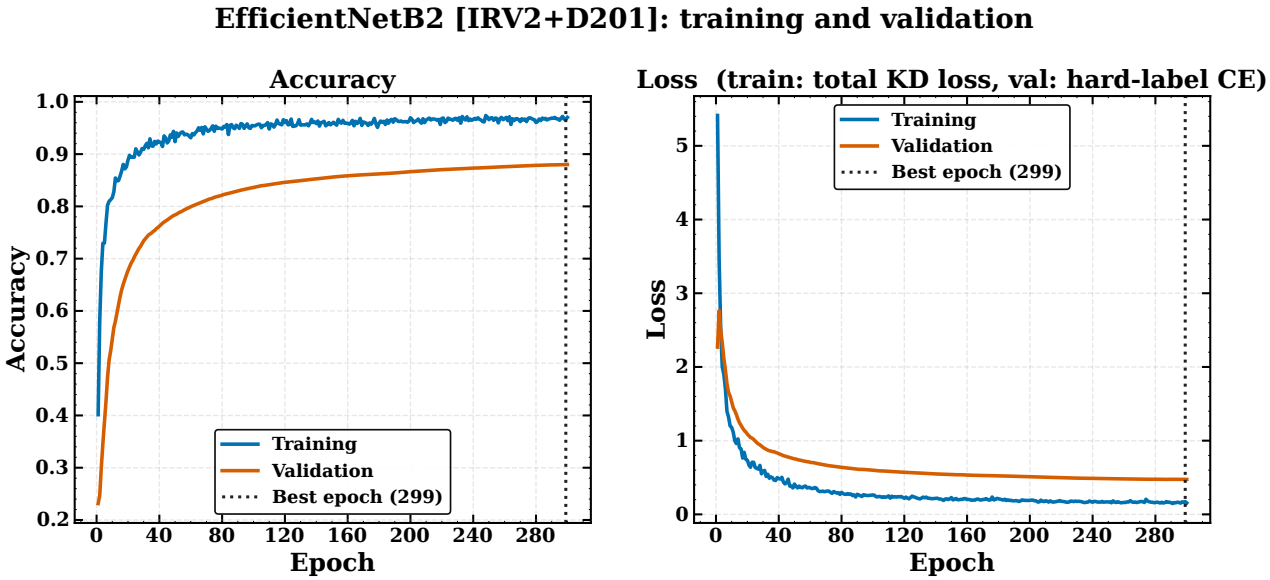
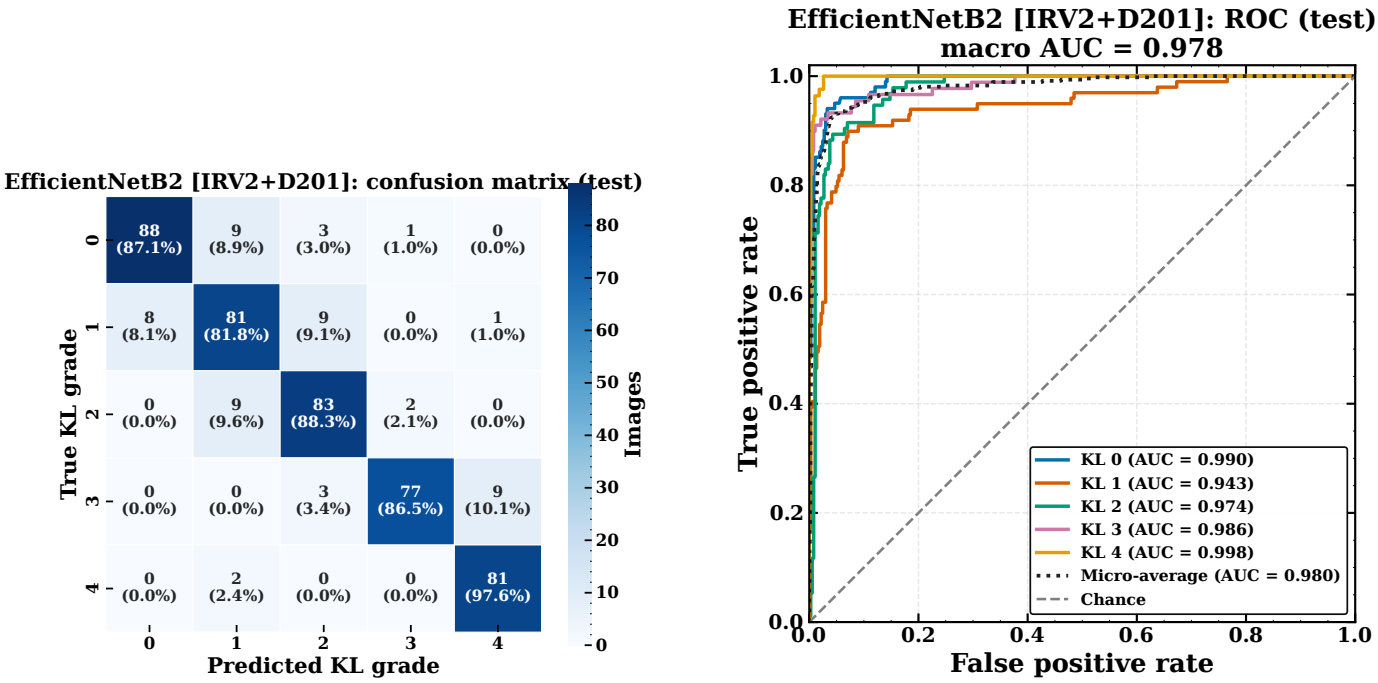


Figure 77: EfficientNetB2 [IRV2+D201]: training and validation curves. Training loss is the combined objective; validation loss is hard-label cross-entropy.



(a) Test confusion matrix.

(b) One-vs-rest ROC curves.

Figure 78: EfficientNetB2 [IRV2+D201]: test-set results.

9.3.11 EfficientNetB2 [IRV2+R50]

Run summary

Teachers	InceptionResNetV2, ResNet50
Temperature / alpha	$T = 3$, $\alpha = 0.3$
Parameters / size	8.13 M / 32.2 MB
Epochs / best	300 / 300
Validation / test accuracy	0.8772 / 0.8777
Macro / weighted F1	0.8805 / 0.8786
Macro AUC	0.9768
Training time	51.0 min

Test-set classification report

Class	Precision	Recall	F1-score	Support
0	0.9263	0.8713	0.8980	101
1	0.7736	0.8283	0.8000	99
2	0.8542	0.8723	0.8632	94
3	0.9121	0.9326	0.9222	89
4	0.9487	0.8916	0.9193	83
Accuracy	–	–	0.8777	466
Macro average	0.8830	0.8792	0.8805	466
Weighted average	0.8806	0.8777	0.8786	466

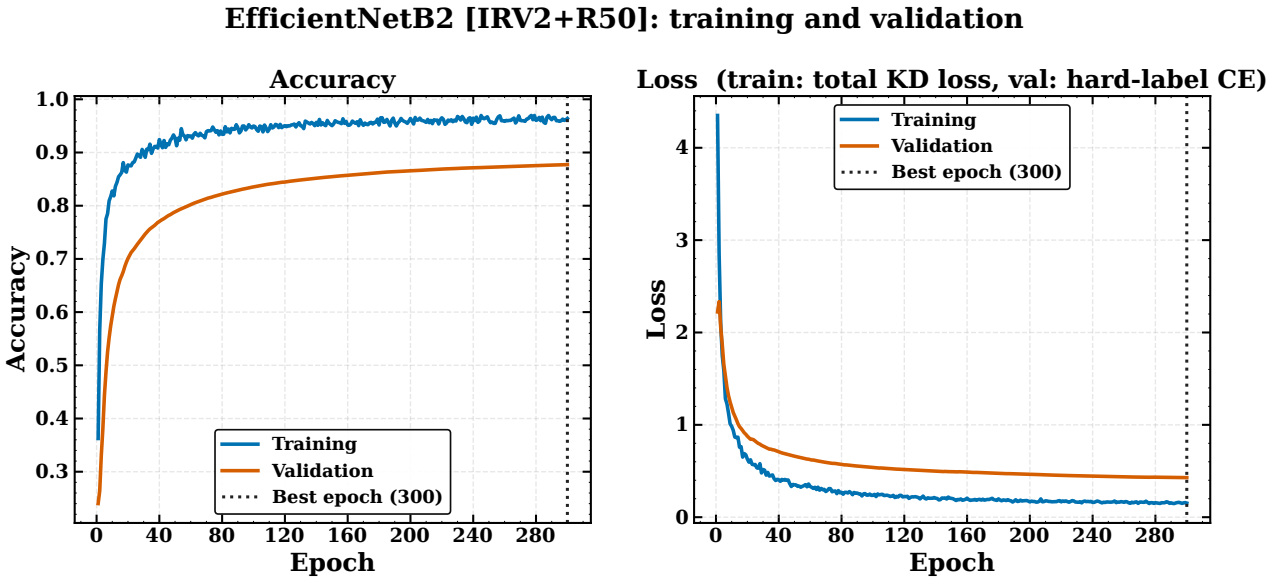
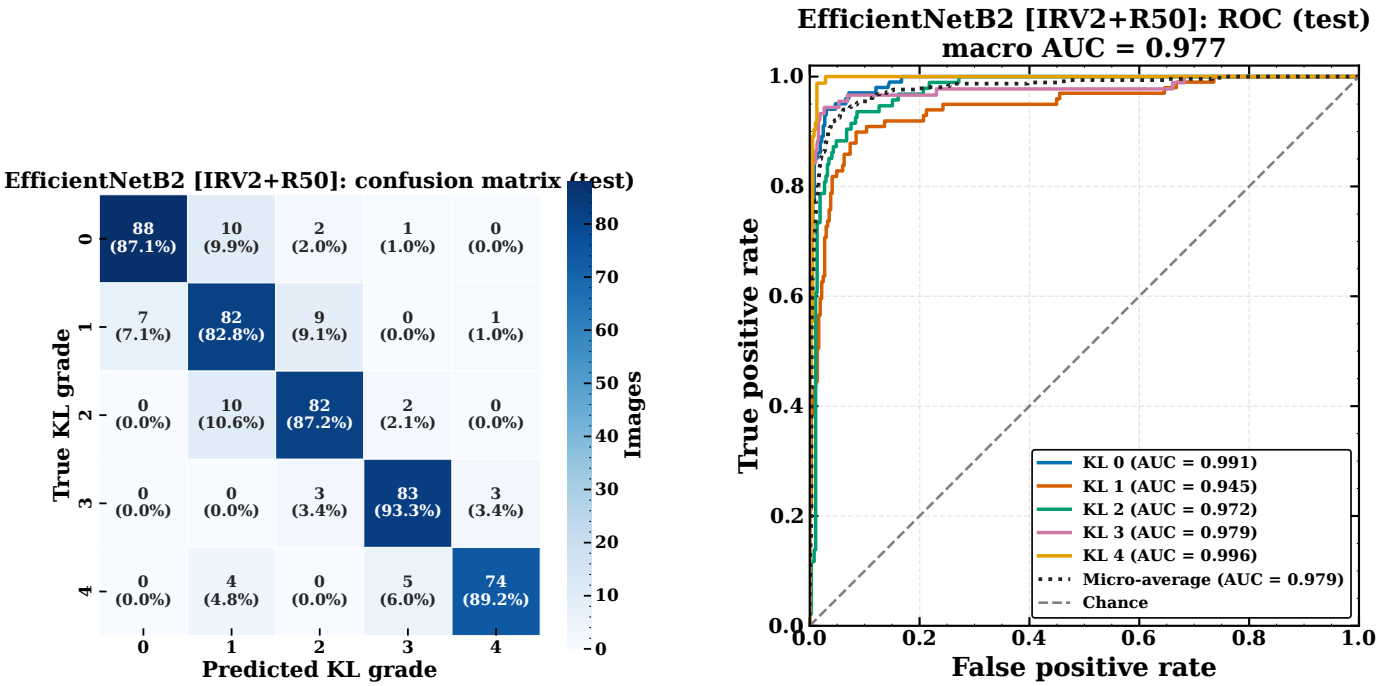


Figure 79: EfficientNetB2 [IRV2+R50]: training and validation curves. Training loss is the combined objective; validation loss is hard-label cross-entropy.



(a) Test confusion matrix.

(b) One-vs-rest ROC curves.

Figure 80: EfficientNetB2 [IRV2+R50]: test-set results.

9.3.12 EfficientNetB2 [R50]

Run summary

Teachers	ResNet50
Temperature / alpha	$T = 3, \alpha = 0.3$
Parameters / size	8.13 M / 32.2 MB
Epochs / best	300 / 300
Validation / test accuracy	0.8644 / 0.8970
Macro / weighted F1	0.8999 / 0.8971
Macro AUC	0.9734
Training time	51.0 min

Test-set classification report

Class	Precision	Recall	F1-score	Support
0	0.9000	0.8911	0.8955	101
1	0.8265	0.8182	0.8223	99
2	0.8515	0.9149	0.8821	94
3	0.9753	0.8876	0.9294	89
4	0.9535	0.9880	0.9704	83
Accuracy	–	–	0.8970	466
Macro average	0.9014	0.9000	0.8999	466
Weighted average	0.8985	0.8970	0.8971	466

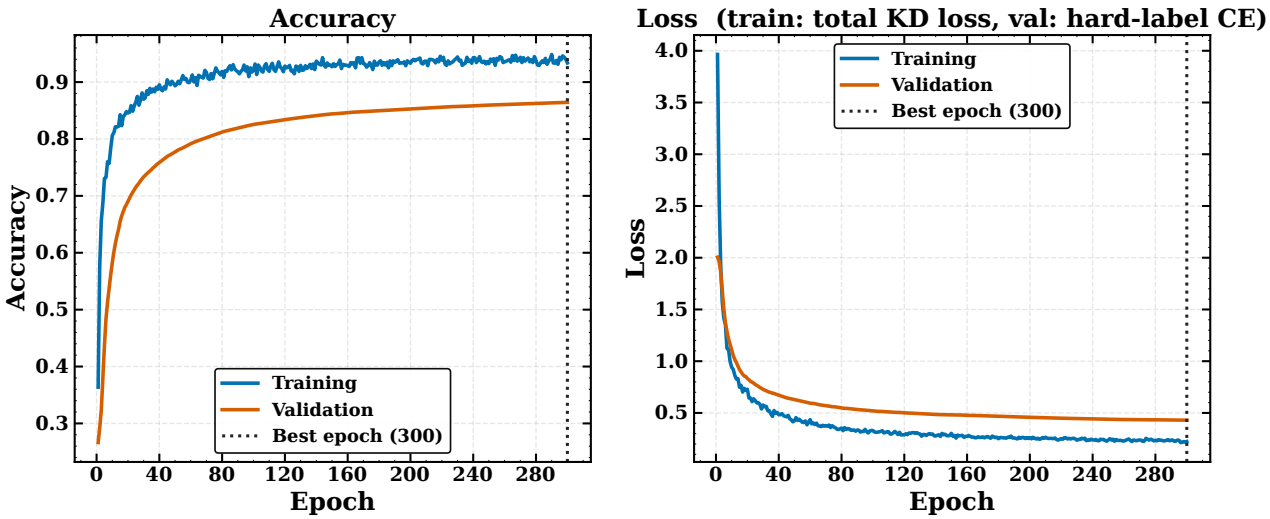
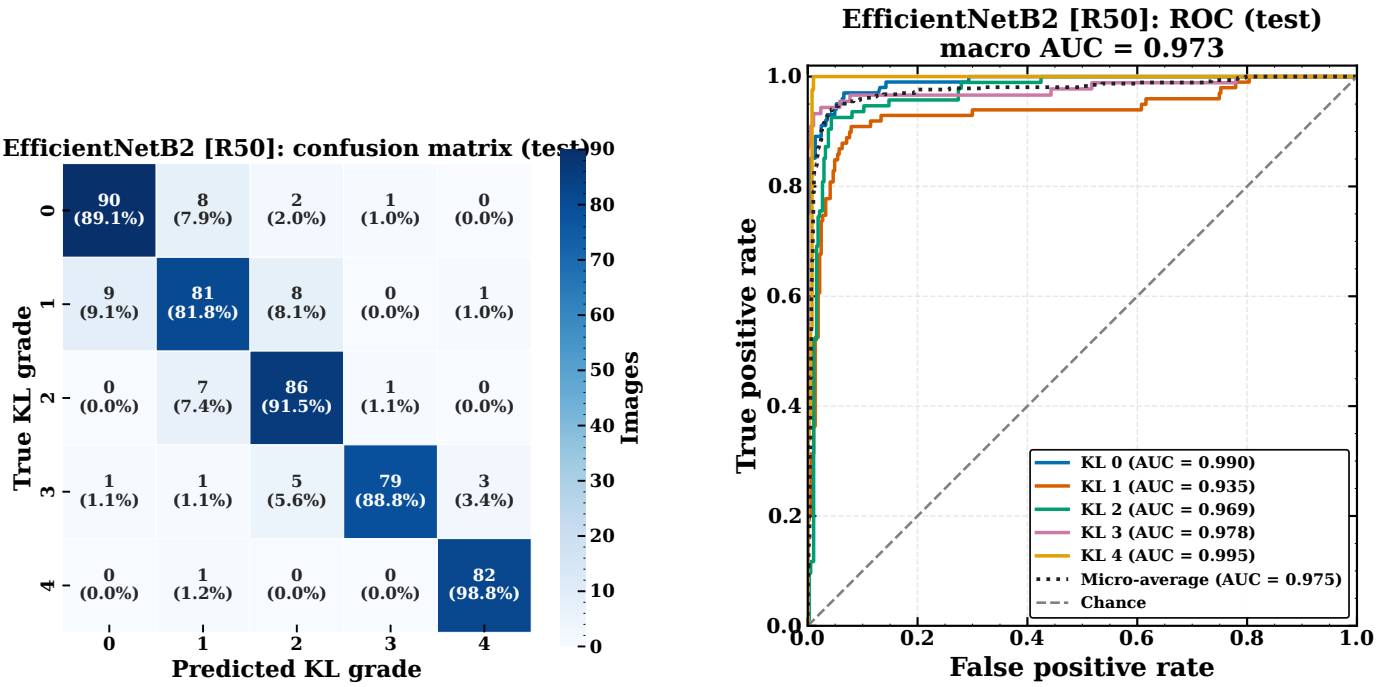


Figure 81: EfficientNetB2 [R50]: training and validation curves. Training loss is the combined objective; validation loss is hard-label cross-entropy.



(a) Test confusion matrix.

(b) One-vs-rest ROC curves.

Figure 82: EfficientNetB2 [R50]: test-set results.

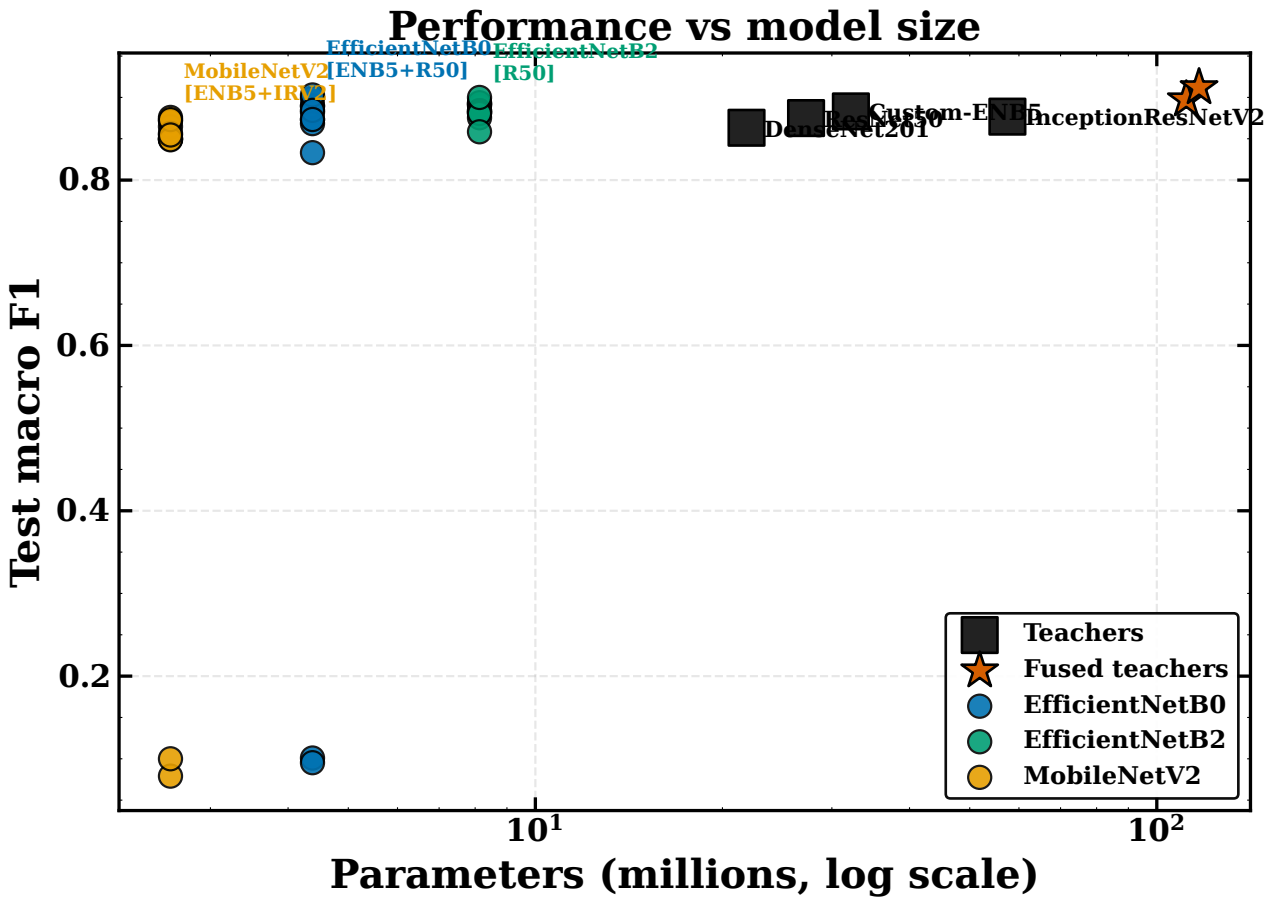


Figure 83: Test macro F1 against on-disk size. The distilled students occupy the upper-left corner: the accuracy of the ensemble at a fraction of its footprint.

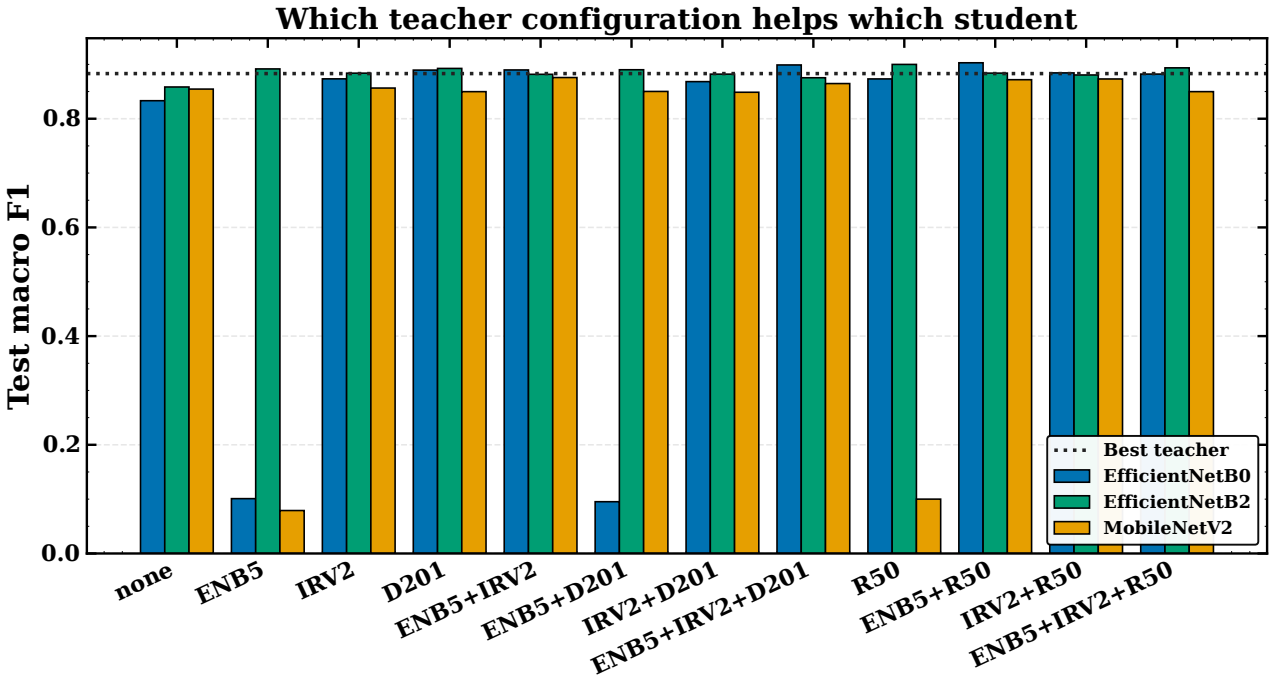


Figure 84: Teachers, fused ensembles, and the students distilled from them.

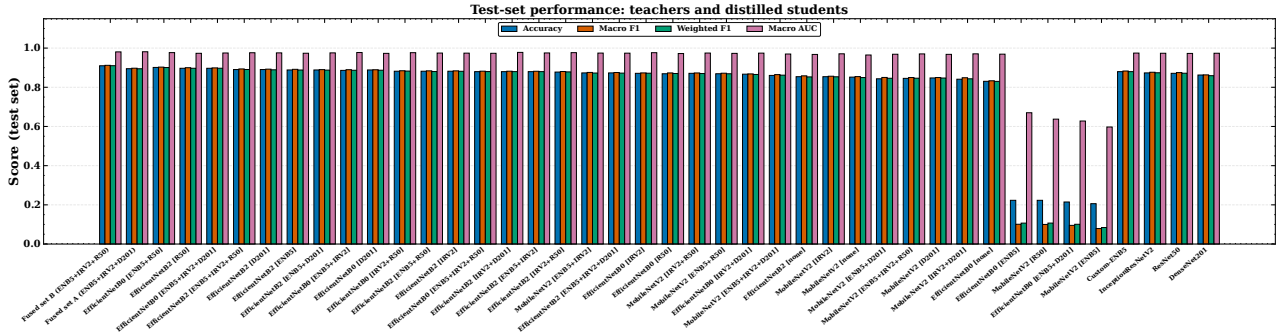


Figure 85: Test-set metrics across every model in the study.

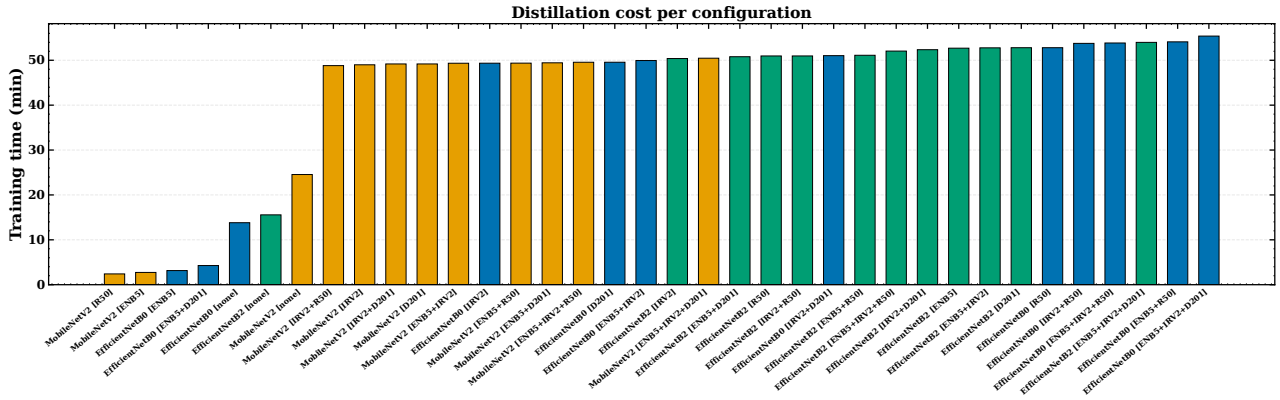


Figure 86: Training time per run. Distilled runs cost more per epoch, because every teacher runs a forward pass on each batch, and they also use far more epochs before early stopping fires.

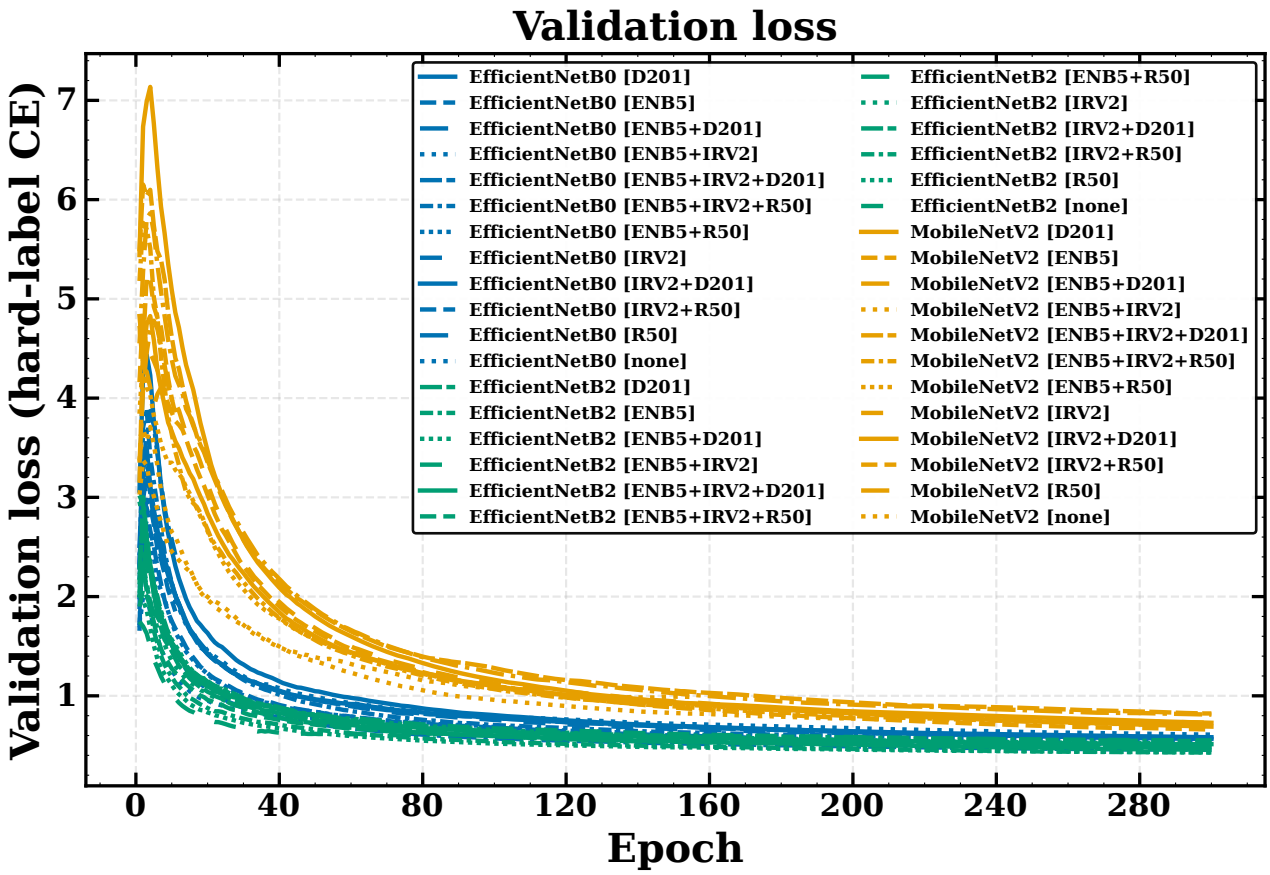


Figure 87: Hard-label validation loss for every run. The voided runs are the curves that stop at epoch 11.

11 Reproducing the Results

The notebook `kd/KD_MultiTeacher_Distillation.ipynb` produces every number and figure in this report, writing one folder per run under `kd/results/` with `metrics.json`, `history.csv`, the classification report, the confusion matrix, per-image predictions, the student weights, and vector plus 600 dpi figures. To re-train the eight voided runs with the warm-up fix, delete their folders and re-run the notebook: finished runs are reloaded from disk and only the deleted ones are trained again.